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Balancing Supply with Demand on Ride-hailing Platforms in Markets with Price Regulations: An Operational Approach

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Problem definition: Ride-hailing platforms face a supply-demand imbalance. During peak periods, the demand from passengers far exceeds the supply of drivers, while during off-peak periods, there is an abundance of supply but weak demand. The well-studied surge pricing can be challenging to implement in markets where prices are subject to regulation and are inflexible to adjust. We study an alternative operational approach that shifts the supply of drivers from off-peak to peak periods to address the supply-demand imbalance. **Methodology:** We propose two novel incentive schemes: the *qualification* scheme and the *prioritization* scheme. Specifically, the platform sets a work target for the peak period. Under the qualification scheme, the platform assigns off-peak service requests only to drivers who meet the peak period target. Under the prioritization scheme, the platform prioritizes off-peak requests for drivers who meet the peak period target. We analyze the effectiveness of these schemes, considering the openness of the platform's supply system. **Results:** For platforms with a closed system that only allows full-time drivers to provide service, the qualification scheme improves the total matching volume to a greater extent but hurts full-time drivers more than the prioritization scheme. For platforms with an open system that also allows the abundant part-time drivers to serve off-peak requests, the prioritization scheme outperforms the qualification scheme in improving total matching volume. Furthermore, the implementation of an incentive scheme in an open system may benefit both the platform and full-time drivers. **Managerial implications:** Our study suggests an alternative to surge pricing for on-demand platforms to address the supply-demand imbalance when prices are inflexible. It provides insights for platforms with varying levels of supply system openness in choosing appropriate incentive schemes. The welfare results offer guidance for platforms and policymakers regarding both matching volume and worker welfare.

Key words: Ride-hailing platforms; Supply-demand imbalance; Operational approach; Incentive schemes

1. Introduction

On-demand ride-hailing platforms, such as Didi, connect passengers with on-call drivers. A distinguishing feature of these two-sided markets is that the drivers are typically independent contractors rather than formal employees of the platform, giving them flexibility in choosing their work amount and schedule. However, as drivers' choices may not always align with the platform's interests, significant discrepancies between service supply and demand usually occur and bring great challenges to platforms. For example, Didi, the Chinese ride-hailing giant, constantly faces labor shortages during weekday mornings when requests for ride peak, and the resulting unfulfilled ride requests account for one fifth of the total demand (Shen 2019). In cities such as Beijing, the likelihood of securing a ride is lower during peak periods compared to normal periods.[1] One way to ease the supply-demand imbalance is surge pricing, which involves raising the price to attract more drivers during peak hours to meet the increased demand for rides. Nevertheless, in markets such as China where ride-hailing prices are regulated and less flexible,[2] implementing surge pricing can be challenging. Since 2017, Didi has faced multiple lawsuits for increasing prices by 20% during peak periods with accusations of abusing market power (Wang 2019).

Moreover, surge pricing itself has been criticized by passengers who feel exploited and view the practice as unfair and sometimes unethical. It may also hurt drivers. Using field data from one ride-hailing platform in China, Xu et al. (2024) shows that customers may misdirect complaints about surge pricing towards individual drivers, increasing the likelihood of drivers receiving complaints by a factor of 1.13.

In response to regulations on price control and surge pricing limitations, some Chinese platforms are turning to non-pricing strategies to address the supply-demand imbalances. For example, Licheng Zhuanche, a platform focusing on premium service, allocates off-peak ride requests only to drivers whose service levels during peak periods meet the platform's requirements. This approach aims to incentivize drivers to provide more services during peak periods.[3] Didi, on the other hand, implements the "Reputation Score" system in 2019, where each driver's score is positively correlated with his/her amount of service at peak hours. During the off-peak periods, those with high scores will be assigned ride requests with high priority.[4] Both Licheng Zhuanche and Didi's approaches do not rely on varying the price to adjust demand, but focus on changing service incentives to shift supply.

Motivated by the emerging practices, this paper studies two novel incentive schemes based on supply shifting and examines their effectiveness in solving the supply-demand imbalance and influence on platform profitability and users' welfare. In addition to the peak periods when the increase in demand causes a shortage of service supply, there are also off-peak periods when service supply exceeds demand. For example, drivers in Shanghai struggle to receive ride orders during off-peak times.[5] It is common for drivers to be idle during these periods (Feng et al. 2025). The two schemes, namely the *qualification* scheme resembling Licheng Zhuanche's approach, and the *prioritization* scheme similar to Didi's approach, aim to shift the supply of service from off-peak periods to peak periods without altering the transaction price. This is achieved by strategically assigning off-peak service requests to drivers based on their peak period work amount. Specifically, the platform sets a target level of service for the peak period. In the *qualification* scheme, during off-peak periods, the platform assigns service requests only to drivers who have provided service up to the work target during peak periods. This means that drivers who have provided less service during peak periods are not qualified to work during off-peak periods. In the *prioritization* scheme, the platform first assigns service requests in off-peak periods to drivers who have achieved the target during peak periods, and then assigns the remaining requests to other drivers. This prioritizes drivers who have met the target during peak periods for work during off-peak periods.

To examine the effectiveness of the incentive schemes, it is essential to understand the two main reasons of supply shortage during peak periods. The first reason is related to the composition of the service supply. For example, most Didi drivers are part time and half of them work fewer than 2 hours a day (Liao 2019, Liu and Lu 2021). Only 16.76% of Didi drivers work more than 6 hours a day (Liu and Lu 2021). As a result, many part-time drivers may not be able to work during peak periods due to their full-time jobs, causing a peak-period supply shortage.[6] The second reason stems from the worse traffic conditions, which lead even full-time drivers to be less motivated to work during peak periods despite the excessive demand. Multiple studies have consistently shown that traffic congestion (Stokols et al. 1978, Gulian et al. 1989, Hennessy and Wiesenthal 1997, Evans et al. 2002) causes increased driver stress, along with the emotional discomfort of working at rush hours (Morris and Hirsch 2016). In addition, drivers often receive unfair

low ratings from passengers due to factors beyond their control, such as traffic-related delays, further diminishing their motivation to provide service during the peak periods (Kapoor and Tucker 2017, Ashkrof et al. 2020). As a result, although full-time drivers can provide service during peak periods, they may reduce the service amount. In this study, we capture both causes of the peak period supply shortages and examine how qualification and prioritization schemes can shift service supply without changing the transaction price.

When studying the two incentive schemes, we also consider the openness of a platform's supply system. Some platforms adopt an *open* system, allowing both full-time and part-time workers to provide services. Other platforms adopt a *closed* system by excluding drivers who cannot provide service full time. For example, Caocao Chuxing, a new entrant to China's ride-hailing market, enforces strict work amount requirements which effectively deter part-time drivers from joining the platform (Tang 2018, Goh and Shirouzu 2018). Since January 2019, Didi has been transitioning from an open system to a closed system through implementing double licensing, requiring both drivers and their cars to be licensed for service provision. As registering a privately owned car for commercial use entails significant costs in China (Liao 2019, Nonninger 2019), the double licensing policy essentially excludes part-time drivers from supplying services. We show that the openness of a platform's supply system is crucial in determining the two incentive schemes' effectiveness.

To focus attention on the operational approach and reflect the price's inflexibility, we assume that both periods have the same price per unit of service time, which is exogenously given. We begin with a closed system in which part-time drivers are excluded from the service supply, and study a benchmark scenario in which full-time drivers freely allocate their total work time across peak and off-peak periods. Working during the peak period incurs an additional congestion cost, which reflects the psychological discomfort, high stress levels, and fatigue experienced by drivers. In deciding how to allocate their work hours, full-time drivers weigh the increased likelihood of receiving ride requests against the added congestion cost of working during peak periods. We find that the optimal allocation of work hours by full-time drivers may not align with the platform's best interests, as it does not lead to maximizing the total matching volume.

We then study the two incentive schemes and show that both schemes can improve the platform's total matching volume compared to the benchmark scenario. Furthermore,

in choosing the peak period target for both schemes, the platform chooses the smallest target which can induce all full-time drivers to participate, rather than choosing a larger target which only involves some of the drivers in the scheme. Given that both schemes engage all full-time drivers while excluding part-time drivers from the service supply, one may believe that the two schemes are equivalent for a platform with a closed system. Interestingly, we find that the qualification scheme leads to a higher total matching volume than the prioritization scheme. This is because, if not reaching the peak period work target, full-time drivers cannot provide service during the off-peak period under the qualification scheme, but can still receive ride requests under a prioritization scheme (although with a lower priority). This difference enables the platform to set a higher work target under a qualification scheme, shift more supply from the off-peak period to the peak period, and improve the total matching volume to a greater extent. Despite the increased matching volume, full-time drivers are worse off due to the longer work time during the peak period.

We next turn the focus to an open system where a significant number of part-time workers can provide service during the off-peak period. Compared to a closed system, full-time drivers under an open system allocate more of their service time to the peak period to avoid intense competition with part-time drivers in the off-peak period, resulting in a less severe peak period supply shortage. Similar to a closed system, for both schemes, the platform chooses an optimal peak period work target that involves all full-time drivers in the scheme. However, in contrast to a closed system, we find that a prioritization scheme outperforms a qualification scheme in improving the total matching volume. The prioritization scheme offers two key advantages over the qualification scheme. First, it effectively utilizes the service supply from part-time drivers, resulting in improved off-peak matching volume. Second, the prioritization scheme allows the platform to increase the peak period work target by leveraging the abundant off-peak supply from part-time drivers, thus improving the peak period matching volume. We also show that compared to the benchmark scenario without any scheme, the introduction of an incentive scheme in an open system may increase the platform's total matching volume and benefit full-time drivers simultaneously.

In extensions, we compare our proposed schemes with surge pricing in various performance measures, and discuss the situations where each approach can be effectively applied. We then relax the model assumption by allowing a finite number of part-time drivers in an open system and explore the robustness and boundary conditions of our main results.

The remainder of the paper proceeds as follows. After reviewing the literature in Section 2, we present the model in Section 3. Sections 4 and 5 analyze the benchmark scenario and the two schemes in a closed system and an open system, respectively. Section 6 extends the model to compare with surge pricing and to assess the results' robustness. Finally, Section 7 concludes the paper. All proofs are presented in Online Appendix and Supplementary Analysis.

2. Literature Review

The recent development of on-demand service platforms has drawn significant attention in academia. A growing body of literature has investigated various aspects of on-demand service platforms, especially operational issues on ride-hailing platforms. Examples of such issues include dynamic pricing and capacity management (Cachon et al. 2017, Taylor 2018, Bai et al. 2019, Yang et al. 2020, Hu et al. 2022), matching demand with supply (Bimpikis et al. 2019, Besbes et al. 2021, Hu and Zhou 2022), and labor welfare, government regulation, and gender-based issues (Yu et al. 2020, Tang et al. 2021, Besbes et al. 2024). We refer interested readers to Narasimhan et al. (2018), Wang and Yang (2019) and Benjaafar and Hu (2020) for reviews of studies on these topics.

Our paper contributes to the literature on using non-pricing instruments to facilitate service provision. Yang and Tang (2018) introduce a new fare-reward scheme for railway transit that aims to shift commuters' peak departures to off-peak periods. By incorporating the heterogeneity in commuters' scheduling flexibility, Tang et al. (2019) further propose a hybrid fare scheme that combines the fare-reward scheme with a non-rewarding scheme to shift peak traffic to off-peak periods. Yang et al. (2020) build a reward scheme integrated with a constrained surge pricing and show the novel approach can simultaneously benefit users, drivers, and the platform. Afèche et al. (2023) consider the interplay of passengers' admission control and drivers' re-positioning decisions over a spatial network, and find that the platform may reject requests originated from low-demand regions even when the supply is abundant, so that drivers will self relocate to high-demand locations. Different from the existing studies that largely focus on adjusting the demand for service, we fix the service demand but explore the shift of service supply through operational approaches.

Our paper also relates to the management of service providers' participation. Hu et al. (2022) study a platform's worker classifications. They show that compared with uniformly

classifying both long-term workers and ad hoc workers, discriminatorily classifying long-term workers as employees and ad hoc workers as contractors can mitigate undercutting and overjoining issues and lead to a Pareto improvement outcome. Castro et al. (2022) study a platform's supply prioritization over private agents, who are fully controlled by the platform and receive a fixed wage, and flexible agents, who freely determine to provide service and pay a commission to the platform. They show that depending on whether the market is over or under supplied, the platform should prioritize either the private or the flexible agents. Besbes et al. (2024) propose a novel prioritized-first-come-first-serve (PFCFS) policy, allowing drivers to reserve slots according to their temporal preferences to raise drivers' effective wage without hurting supply. Our study differs from existing research that relies on bonus incentives, information provision, worker classification, or supply prioritization to manage service providers. Instead, we consider the context when service providers are heterogeneous in their working schedules, and focus on shifting self-scheduling capacity from off-peak periods to peak periods through operational approaches to address the supply-demand imbalance. Our paper contributes to the literature by characterizing the optimal incentive scheme in the presence of heterogeneous service providers.

3. Model Setup

Consider an on-demand ride-hailing platform that receives ride requests from passengers and assigns them to available drivers. The transaction price of a trip typically depends on various factors such as the distance traveled, duration of the trip, and the number of baggage items. To keep the model focused, we assume the transaction price to be proportional to the service time, and denote the price per unit of service time as p. When a transaction occurs, the platform pays a proportion δ of the total transaction price to the driver, while retaining the remaining $1-\delta$ as its own revenue. To explore the use of an operational approach for balancing supply and demand, we treat the price per unit service time as exogenous. The assumption of exogenous pricing aligns with existing literature that explores non-pricing factors on two-sided platforms (Feng et al. 2021, Afèche et al. 2023), and is applicable for modeling markets with strong price regulation. In Section 6.1, we compare the performance of our operational approach with surge pricing.

Each passenger participates in the market at a length-less time point and sends her service request to the platform. The demand for service fluctuates significantly throughout

the day. During rush hours, which we refer to as the peak period, the service demand far exceeds the supply capacity from drivers, leading to significant supply shortage. Conversely, there are also hours such as early morning and late night when supply exceeds demand, which we refer to as the off-peak period.

3.1. Drivers

We consider two types of drivers. Some drivers can freely choose to work during both peak and off-peak periods, and are referred to as full-time drivers in our model. We use n to denote the size of the full-time drivers. Each full-time driver allocates his total work time, which is normalized to 1, between the peak and the off-peak periods. We use $\alpha \in [0, 1]$ and $1 - \alpha$ to denote the proportions of work time that a full-time driver allocates to the peak and the off-peak periods.[7] Evidenced by multiple studies (Hennessy and Wiesenthal 1999, Morris and Hirsch 2016, Huang et al. 2018) and as discussed in Section 1, drivers incur additional costs, such as psychological discomfort, high stress levels, and fatigue, when working during the congested peak periods. These additional costs could cause full-time drivers less willing to work during the peak period, especially when price is not adjusted. To reflect this, we normalize the cost of working during the off-peak period to zero, and introduce an additional congestion cost of $\frac{1}{2}k \cdot \alpha^2$ for working during the peak period. The quadratic cost function is used to capture the increasing marginal cost associated with the working time during the peak period. The parameter k measures the magnitude of the additional costs. It is important to emphasize that the additional costs are borne on drivers but not on the platform. As a result, drivers' decisions on how to allocate their working time may not align with the platform's optimal interests.

In addition to full-time drivers, some platforms also involve drivers who are only available to work in the off-peak period due to their primary jobs or other commitments during peak hours. We refer to the drivers who are constrained to work in the off-peak period as part-time drivers. To capture the supply surplus during the off-peak period in the simplest way, and to maintain analysis tractability and to focus attention on shifting full-time drivers' supply from off-peak period to peak period, we assume that the number of part-time drivers is infinite. The assumption is purely for technical tractability. In Section 6.2, we consider the case of finite part-time drivers and demonstrate the robustness of our main results.

In practice, there may also be drivers who are constrained to work during the peak period. As long as the additional service supply does not address the peak period's supply

shortage, the presence of these drivers would not affect a full-time driver's time allocation between peak and off-peak periods, and therefore does not qualitatively affect our main results on implementing the incentive schemes.[8] To maintain simplicity, we do not explicitly model this group of drivers.

3.2. The Supply-Demand Imbalance

For simplicity, we only consider the total demand of each period, which is understood as the integration of the number of passengers in each specific time point over the total duration of that period. We denote $D_p(p)$ as the peak period demand given price p and assume $D_p(p) > n$, reflecting the peak period supply shortage even when full-time drivers devote all their time to peak periods.[9] We use $D(\epsilon) = D_0 - p + \epsilon$ to denote the total service demand in the off-peak period, where ϵ is a mean-zero random factor to reflect the off-peak demand randomness and follows a distribution F, which is differentiable and has the support $[-s, s]$. We define $L \equiv D_0 - p$, and assume $s \in (0, \min\{n - L, L\})$. Here the constraint $s < n - L$ implies $D(\epsilon) = L + \epsilon < L + n - L = n$, indicating that service demand falls short of supply during the off-peak period when every full-time driver devotes all their time to the off-peak period. The constraint $s < L$ ensures that the realized off-peak demand $D(\epsilon)$ is always positive.

Lastly, if $k \leq \frac{np\delta}{n-L+s}$ (i.e., the congestion cost is very small), the service target might be lower than the amount a full-time driver would voluntarily work without any incentive scheme, making an incentive scheme irrelevant. Therefore, we constrain our analysis in the region $k > \frac{np\delta}{n-L+s}$. This region corresponds to a scenario where the driver congestion cost is not too small, which discourages them from working a long time in the peak under fixed pricing and presents an opportunity for our incentive schemes to effectively incentivize drivers to shift their availability from off-peak periods to peak hours and address the supply-demand imbalance.

3.3. Platform, Incentive Schemes, and System Openness

We start with a benchmark model in which the platform randomly assigns ride-hailing requests to all available drivers. We then analyze two incentive schemes, namely the *qualification* scheme and the *prioritization* scheme, with the goal of shifting service supply from the off-peak period to the peak period to solve the supply-demand imbalance. Specifically, the platform publicly sets a peak period work target α_0. Under the *qualification* scheme, the platform assigns off-peak period requests only to drivers who have fulfilled the peak

period target amount. Under the *prioritization* scheme, the platform first assigns off-peak requests to drivers who have fulfilled the peak period target, and then assigns the residual requests, if any, to other drivers.

The openness to service supply varies across platforms. Some platforms allow both full-time and part-time drivers to provide service, which we call an *open* system. Other platforms only allow full-time drivers to provide service, which we refer to a *closed* system. In practice, the service supply openness may depend on various factors, including concerns regarding transaction security, the platform's control over independent drivers, and the regulatory environment. For the purpose of this research, we take the system openness as exogenous, and study how the effectiveness of the two incentive schemes in improving matching volume depends on system openness. Table 1 summarizes our key notation.

Table 1 Summary of Key Notation

	Notation	Explanation
The exogenous parameters	p	Transaction price per unit of service time, $p \in (0, D_0)$
	n	The size of the group of full-time drivers
	D_0	The intercept of the off-peak demand function, $D_0 \in (0, n+p-s)$
	L	Expected demand during off-peak periods, $L \equiv D_0 - p$
	ϵ	Uncertain component of off-peak demand, $\epsilon \in [-s, s]$, $s \in (0, \min\{n-L, L\})$
	δ	The proportion of transaction price retained by the driver, $\delta \in (0,1)$
	k	Congestion cost coefficient, $k \in (\frac{np\delta}{n-L+s}, +\infty)$
The platform's decision	α_0	Peak period work target, $\alpha_0 \in (0,1)$
A full-time driver's decision	α	The proportion of time allocated to the peak period, $\alpha \in [0,1]$
The endogenous parameter	θ	The proportion of qualified/prioritized (full-time) drivers
Superscripts	B	Benchmark
	Q	Qualification scheme
	P	Prioritization scheme
Subscripts	q/uq	Qualified/unqualified (full-time) drivers
	p/up	Prioritized/unprioritized (full-time) drivers
	C	Closed system
	O	Open system

4. Closed System

A platform with a closed system allows only full-time drivers to provide services. For succinctness, we use drivers to refer to full-time drivers in this section. We start by examining a benchmark case in which no incentive scheme is implemented. Then, we show the impact of qualification and prioritization schemes on drivers' work time allocation, welfare, and the total matching volume.

4.1. Benchmark

We first derive the driver's equilibrium work time allocation when no incentive scheme is implemented. Recall that α is the proportion of time that a full-time driver allocates to the peak period. As demand exceeds supply in the peak period, a driver can always receive ride requests and earn $p \cdot \delta$ income for each unit of time. In the meanwhile, the driver also incurs the congestion cost $\frac{1}{2}k \cdot \alpha^2$. Therefore, the driver's net utility from working during peak periods is $p\delta \cdot \alpha - \frac{1}{2}k \cdot \alpha^2$.

We next consider the off-peak period and use $\hat{\alpha}$ to denote all drivers' average amount of work time allocated to the peak period in equilibrium. The total service supply during the off-peak period is $n(1-\hat{\alpha})$. Each driver can serve a passenger with certainty if $D(\epsilon) \geq n(1-\hat{\alpha})$, but otherwise serve a passenger with probability $\frac{D(\epsilon)}{n(1-\hat{\alpha})}$. Therefore, a driver's net utility from working during the off-peak period is $\min\{1, \frac{D(\epsilon)}{n(1-\hat{\alpha})}\} \cdot p\delta \cdot (1-\alpha)$. Summing up a driver's utility from both periods, we obtain the driver's utility function for a given realized ϵ as shown below, where the superscript B and subscript C refer to the benchmark and the closed system respectively:

$$U_C^B(\alpha, \epsilon) = p\delta \cdot \alpha - \frac{1}{2}k \cdot \alpha^2 + \min\{1, \frac{D(\epsilon)}{n(1-\hat{\alpha})}\} \cdot p\delta \cdot (1-\alpha). \tag{1}$$

A driver chooses α to maximize the expected utility $U_C^B(\alpha) = \int_{-s}^{s} U_C^B(\alpha, \epsilon) dF(\epsilon)$. Lemma 1 characterizes a full-time driver's optimal choice of α, the resulting utility, and the platform's expected total matching volume.

LEMMA 1 **(Closed System: Benchmark)**. *In the closed system and without an incentive scheme, the equilibrium is characterized as follows:*

- *Each full-time driver's optimal choice of α, denoted by α_C^B, is given as*

$$\alpha_C^B = \begin{cases} \alpha^b, & \text{if } k \leq k_1, \\ \frac{k+p\delta-\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{2k}, & \text{if } k > k_1, \end{cases}$$

  where α^b solves the implicit function $\alpha^b = \frac{p\delta}{k} \cdot \frac{\int_{-s}^{n(1-\alpha^b)-L} F(\epsilon)d\epsilon}{n(1-\alpha^b)}$ and $k_1 = \frac{p\delta s}{(L+s)(1-\frac{L+s}{n})}$.

- *Each full-time driver's utility, denoted by U_C^B, is given as*

$$U_C^B = \begin{cases} (1 - \frac{\int_{-s}^{n(1-\alpha^b)-L} F(\epsilon)d\epsilon}{n})p\delta - \frac{1}{2}k\alpha^{b2}, & \text{if } k \leq k_1, \\ \frac{-k^2+2\frac{kL}{n}p\delta+2\delta kp+\delta^2p^2+(k-p\delta)\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{4k}, & \text{if } k > k_1. \end{cases}$$

- *The platform's expected total matching volume across both periods, denoted by M_C^B, is*

$$M_C^B = \begin{cases} n - \int_{-s}^{n(1-\alpha^b)-L} F(\epsilon)d\epsilon, & \text{if } k \leq k_1, \\ L + n\frac{k+p\delta-\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{2k}, & \text{if } k > k_1. \end{cases}$$

Furthermore, we have $M_C^B < n$.

When allocating their work time between the two periods, a full-time driver weighs two opposing effects. On one hand, working during the off-peak period exposes them to increased competition and the risk of receiving fewer ride requests. On the other hand, working during the peak period means incurring the additional cost of congestion, despite the presence of ample demand. Lemma 1 shows that the total expected matching volume across both periods is always less than the total capacity (i.e., $M_C^B < n$). This is because full-time drivers, due to the high congestion costs in the peak period, tend to allocate more service to the off-peak period than what is demanded during that time. As a result, there is an excess of available capacity during the off-peak period, leading to idleness and under utilization of resources.

We next analyze the two proposed incentive schemes to study their effectiveness in increasing the matching volume.

4.2. Qualification Scheme

When adopting the *qualification* scheme, the platform assigns off-peak ride requests exclusively to drivers who have met the peak period work target. Therefore, for a given peak period work target α_0, each driver decides whether to participate in the scheme by allocating $\alpha \geq \alpha_0$ amount of time to the peak period. The platform, anticipating how each driver will respond to the target service amount, chooses the work target that maximizes the total matching volume. We first derive each driver's choice of work time allocation given the platform's choice of α_0, and then determine the platform's optimal choice of α_0.

To facilitate analysis, we define $\theta \in (0, 1]$ as the proportion of drivers qualified to provide service during the off-peak period. Given our model of a continuum of drivers, each driver's unilateral decision to participate or not in the scheme will not affect θ. We then specify a driver's utility of participating or not in the qualification scheme.

- If a driver does not participate in the qualification scheme, he can only provide service during the peak period. The unqualified driver, by working α_{uq} amount of time during the peak period, derives a utility of $U_{uq}(\alpha_{uq}) = p\delta \cdot \alpha_{uq} - \frac{1}{2}k \cdot \alpha_{uq}^2$. Using the first-order condition, we can solve for an unqualified driver's choice of α_{uq}, denoted by α_{uq}^Q, as $\alpha_{uq}^Q = \frac{p\delta}{k}$. The corresponding driver utility is $U_{uq}^Q = \frac{p^2\delta^2}{2k}$.

- If a driver participates in the qualification scheme, he needs to work at least α_0 during the peak period to be qualified to provide service in the off-peak period. We use α_q to represent the driver's work time during the peak period. Given the equilibrium average supply decision $\hat{\alpha}_q$ of all qualified drivers, the total service supply in the off-peak period becomes $n\theta(1-\hat{\alpha}_q)$. Thus, a qualified driver's utility for a realized ϵ is

$$U_q(\alpha_q, \epsilon) = \min\{1, \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_q)}\} \cdot p\delta \cdot (1-\alpha_q) + p\delta \cdot \alpha_q - \frac{1}{2}k \cdot \alpha_q^2, \tag{2}$$

and a qualified driver chooses α_q to maximize the expected utility

$$\max_{\alpha_q} U_q(\alpha) = \max_{\alpha_q} \Big(\int_{-s}^{s} U_q(\alpha_q, \epsilon) dF(\epsilon) \Big), \tag{3}$$

$$\text{s.t. } \alpha_0 \le \alpha_q \le 1. \tag{4}$$

The equilibrium condition requires $\alpha_q = \hat{\alpha}_q$. Our next result characterizes each qualified driver's equilibrium choice of α_q, denoted as α_q^Q.

LEMMA 2. *Given a closed system, under a qualification scheme with $\alpha_0 > \frac{p\delta}{k}$, each qualified driver chooses $\alpha_q^Q = \alpha_0$.*

To understand Lemma 2, note that a qualification scheme only becomes relevant if the work target exceeds what drivers would voluntarily choose when not participating in the scheme (i.e., $\frac{p\delta}{k}$). Drivers who participate in the scheme only work up to the platform's target in the peak period to qualify for off-peak work, as additional peak period work beyond the target would only incur congestion costs without providing extra benefits.

Anticipating each type of driver's choice of $(\alpha_q^Q, \alpha_{uq}^Q)$, the platform chooses the target α_0, which induces θ proportion of full-time drivers to voluntarily participate, to maximize the expected total matching quantity $M^Q(\theta(\alpha_0))$. The matching volume during the off-peak period equals the smaller of riding demand and service supply, that is, $\min\{D(\epsilon), n\theta(1-\alpha_0)\}$. The peak period matching volume is determined by the service supply (i.e., $n\theta\alpha_0 + n(1-\theta)\alpha_{uq}^Q$) due to the supply shortage. Thus,

$$M^Q(\theta(\alpha_0)) = \int_{-s}^{s} \min\{D(\epsilon), n\theta(1-\alpha_0)\} dF(\epsilon) + n\theta\alpha_0 + n(1-\theta)\alpha_{uq}^Q. \tag{5}$$

The platform's problem can be formulated as

$$\max_{\alpha_0} M^Q(\theta(\alpha_0)), \tag{6}$$

$$\text{s.t. } U_{uq}^Q \le U_q(\alpha_0) \text{ with the constraint binding if } \theta < 1. \tag{7}$$

Constraint (7) is the incentive compatibility condition. It suggests that each driver must be indifferent between being qualified and unqualified if there is a mix of both types of drivers. If all drivers voluntarily choose to be qualified, each driver derives weakly higher utility than not being qualified. Lemma 3 characterizes the equilibrium under the qualification scheme.

LEMMA 3 **(Closed System: Qualification Scheme)**. *In the closed system with a qualification scheme, the equilibrium is characterized as follows:*

- *The platform's optimal choice of work target, denoted by* α_0^Q*, is given as*

$$\alpha_0^Q = \begin{cases} 1 - \frac{L-s}{n}, & \text{if } k \le k_2, \\ \alpha_0^q, & \text{if } k_2 < k \le k_3, \\ \frac{p\delta}{k} + \sqrt{2\frac{L}{n}\frac{p\delta}{k}}, & \text{if } k > k_3, \end{cases}$$

  where α_0^q *is larger than* $\frac{p\delta}{k}$ *and solves the implicit function* $(1-\frac{1}{n}\int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon)p\delta - \frac{1}{2}k\alpha_0^{q2} = \frac{p^2\delta^2}{2k}$*,* $k_2 = \frac{p\delta\left(\sqrt{\frac{L-s}{n}\left(2-\frac{L-s}{n}\right)}+1\right)}{\left(1-\frac{L-s}{n}\right)^2}$*, and* $k_3 = \frac{p\delta\left(\sqrt{\frac{L}{n}\left(2-\frac{L+2s}{n}\right)}-\frac{s}{n}+1\right)}{\left(1-\frac{L+s}{n}\right)^2}$*.*
- *Given* α_0^Q*, all drivers participate in the scheme (i.e.,* $\theta(\alpha_0^Q) = 1$*). Each full-time driver's utility, denoted by* U_C^Q*, is given as*

$$U_C^Q = \begin{cases} p\delta - \frac{1}{2}\cdot k(1-\frac{L-s}{n})^2, & \text{if } k \le k_2, \\ \frac{p^2\delta^2}{2k}, & \text{if } k > k_2. \end{cases}$$

- *The platform's expected total matching volume across both periods, denoted by* M_C^Q*, is*

$$M_C^Q = \begin{cases} n, & \text{if } k \le k_2, \\ n - \int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon, & \text{if } k_2 < k \le k_3, \\ L + n\frac{p\delta}{k} + \sqrt{2nL\frac{p\delta}{k}}, & \text{if } k > k_3. \end{cases}$$

Lemma 3 shows that the platform maximizes the total matching volume by inducing all drivers to participate in the qualification scheme. That is, compared with setting a high target and only engaging some drivers in the scheme, the platform can improve the matching volume by lowering the target and engaging as many drivers as possible. Note that the capacity shifted to the peak period is fully utilized due to high demand, whereas the same capacity may be idle during the off-peak period because of the limited and stochastic demand. By having all drivers work a relatively longer time during the peak period, it not only increases the peak period matching volume but also alleviates competition during the off-peak period, thereby increasing overall capacity utilization.

The platform's choice of peak period work target critically hinges on the congestion coefficient k. Given that $k > \frac{np\delta}{n-L+s}$, if the congestion cost is small (i.e., $k \leq k_2$), the platform is able to raise the work target ($\alpha_0^Q = 1 - \frac{L-s}{n}$), which shifts large amount of service supply from the off-peak period to the peak period. The significant supply shift also fully eliminates the capacity idleness in the off-peak period, resulting in full capacity utilization ($M_C^Q = n$). However, as the congestion cost rises beyond a threshold (i.e., $k > k_2$), the platform has to lower the peak period work target to ensure that all drivers participate in the qualification scheme. As a result, the off-peak supply may still exceed the off-peak demand, which implies that the total matching volume, although improved compared to the benchmark case, could still be lower than total capacity.

4.3. Prioritization Scheme

We next turn attention to the *prioritization* scheme, under which the platform also announces a peak work target α_0. Unlike a qualification scheme, drivers who work less than α_0 during the peak period are still able to provide service during the off-peak period but are not given priority in being assigned ride requests. Specifically, the platform assigns off-peak period requests first to drivers who have worked up to α_0 during the peak period, and only assigns any remaining requests to the remaining drivers.

We follow the same steps as in Section 4.2 by first deriving each driver's work time allocation given a certain work target α_0, and then solving for the platform's optimal choice of α_0. We still use $\theta \in (0, 1]$ to denote the proportion of drivers who participate in the scheme and are prioritized in being assigned off-peak period requests. The remaining $1 - \theta$ proportion of drivers are unprioritized in being assigned off-peak period requests.

- If a driver does not participate in the prioritization scheme, he works $\alpha_{up} \in [0, \alpha_0)$ amount of time during the peak period and derives utility $p\delta \cdot \alpha_{up} - \frac{1}{2}k \cdot \alpha_{up}^2$ from the peak period. In the off-peak period, only if $D(\epsilon) - n\theta(1 - \hat{\alpha}_p) > 0$ (or equivalently, $\epsilon > n\theta(1 - \hat{\alpha}_p) - L$), unprioritized drivers can receive service requests as off-peak demand cannot be fully fulfilled by prioritized drivers. An unprioritized driver's chance of receiving a request is $\min\{1, \frac{D(\epsilon) - n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}\}$, where $\hat{\alpha}_p$ and $\hat{\alpha}_{up}$ are the average peak period supply of all prioritized and unprioritized drivers respectively. As a result, the unprioritized driver's utility from the off-peak period is $\min\{1, \frac{D(\epsilon) - n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}\} \cdot p\delta \cdot (1 - \alpha_{up})$. Combining the utility from both periods, an unprioritized driver's work time allocation problem is

$$\max_{\alpha_{up}<\alpha_0} U_{up}(\alpha_{up}) = \max_{\alpha_{up}} \Big(p\delta\alpha_{up} - \frac{1}{2}k\alpha_{up}^2 + \int_{n\theta(1-\hat{\alpha}_p)-L}^{s} \min\{1, \frac{D(\epsilon)-n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}\} p\delta(1-\alpha_{up})dF(\epsilon) \Big). \quad (8)$$

- If a driver participates in the prioritization scheme, he works $\alpha_p \in [\alpha_0, 1]$ in the peak period. We can similarly formulate his work time allocation problem as

$$\max_{\alpha_p \geq \alpha_0} U_p(\alpha_p) = \max_{\alpha_p} \Big(p\delta\alpha_p - \frac{1}{2}k\alpha_p^2 + \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)}\} dF(\epsilon) p\delta(1-\alpha_p) \Big). \quad (9)$$

Anticipating the equilibrium work time allocation of a prioritized driver (α_p^*) and that of an unprioritized driver (α_{up}^*), the platform chooses the target α_0 to maximize the total matching volume $M^P(\theta(\alpha_0))$. Note that the total service supply capacities during peak and off-peak periods are $n\theta\alpha_p^* + n(1-\theta)\alpha_{up}^*$ and $n\theta(1-\alpha_p^*) + n(1-\theta)(1-\alpha_{up}^*)$ respectively. Thus the total matching volume is

$$M^P(\theta(\alpha_0)) = \int_{-s}^{s} \min\{D(\epsilon), n\theta(1-\alpha_p^*) + n(1-\theta)(1-\alpha_{up}^*)\} dF(\epsilon) + n\theta\alpha_p^* + n(1-\theta)\alpha_{up}^*. \quad (10)$$

The platform's problem can be formulated as

$$\max_{\alpha_0} M^P(\theta(\alpha_0)), \quad (11)$$

$$\text{s.t. } U_{up}(\alpha_{up}^*) \leq U_p(\alpha_p^*) \text{ with the constraint binding if } \theta < 1. \quad (12)$$

Constraint (12) is the incentive compatibility condition that ensures that the proportions θ and $1-\theta$ of drivers self select to become prioritized and unprioritized drivers. Lemma 4 characterizes the equilibrium.

LEMMA 4 **(Closed System: Prioritization Scheme)**. *In the closed system with a prioritization scheme, the equilibrium is characterized as follows:*

- *The platform's optimal choice of work target, denoted by α_0^P, is given as*

$$\alpha_0^P = \begin{cases} \alpha_0^p, & \text{if } k \leq k_3, \\ \frac{p\delta}{k} + \sqrt{2\frac{L}{n}\frac{p\delta}{k}}, & \text{if } k > k_3, \end{cases}$$

where α_0^p solves the equation $(1 - \frac{1}{n}\int_{-s}^{n(1-\alpha_0^p)-L} F(\epsilon)d\epsilon)p\delta - \frac{1}{2}k\alpha_0^{p2} = \frac{p\delta(2k\int_{n(1-\alpha_0^p)-L}^{s} dF(\epsilon) + p\delta(1-\int_{n(1-\alpha_0^p)-L}^{s} dF(\epsilon))^2)}{2k}$.

- *Given α_0^P, all drivers participate (i.e., $\theta(\alpha_0^P) = 1$). Each full-time driver's utility is*

$$U_C^P = \begin{cases} (1 - \frac{1}{n}\int_{-s}^{n(1-\alpha_0^p)-L} F(\epsilon)d\epsilon)p\delta - \frac{1}{2}k\alpha_0^{p2}, & \text{if } k \leq k_3, \\ \frac{p^2\delta^2}{2k}, & \text{if } k > k_3. \end{cases}$$

- *The platform's expected total matching volume across both periods, denoted by M_C^P, is*

$$M_C^P = \begin{cases} n - \int_{-s}^{n(1-\alpha_0^p)-L} F(\epsilon)d\epsilon, & \text{if } k \le k_3, \\ L + n\frac{p\delta}{k} + \sqrt{2nL\frac{p\delta}{k}}, & \text{if } k > k_3. \end{cases}$$

Furthermore, we have $M_C^P < n$.

Similar to the result of a qualification scheme, with a prioritization scheme the platform also chooses a target α_0^P that induces all drivers to participate. But unlike a qualification scheme, the platform is unable to achieve a full capacity utilization (i.e., $M_C^P < n$) even when the congestion cost is small. To appreciate the intuition, it is important to note that with a prioritization scheme, drivers still have the opportunity to work during the off-peak period, despite a lower priority for receiving service requests if they choose not to participate in the scheme. This distinction from a qualification scheme limits the platform's ability to set a high peak work target when engaging all drivers. When the congestion cost is small, a qualification scheme allows the platform to shift enough supply to the peak period and fully eliminate the capacity idleness in the off-peak period, while a prioritization scheme cannot achieve this due to the lower work target set by the platform.

Interestingly, when the congestion cost is large (i.e., $k > k_3$), both the prioritization and qualification schemes achieve the same outcomes for the platform's work target, each driver's utility, and the total matching volume (see Lemma 4 and Lemma 3). This convergence occurs because, in the presence of high congestion costs, the platform faces limitations in shifting sufficient supply to the peak period to eliminate off-peak capacity idleness, regardless of the scheme employed. Consequently, drivers who do not participate in either scheme are unable to receive service requests during the off-peak period, leading to a driver facing the same trade-off between participating or not in both of the schemes. As a result, the platform's optimal choice of the peak work target, and the corresponding driver welfare and transaction volumes are the same across both schemes.

4.4. Comparison of the Benchmark and Two Schemes in a Closed System

We now compare the benchmark scenario with the two incentive schemes to examine their effects on the total matching volume and drivers' welfare.

PROPOSITION 1. *In the closed system, given $k > \frac{np\delta}{n-L+s}$ and that the price per unit of service time is fixed, we have the following results.*

- *Implementing a prioritization scheme improves the platform's total matching volume compared to the benchmark. However, it lowers each driver's welfare.*
- *Implementing a qualification scheme further increases the total matching volume compared to the prioritization scheme. However, it further lowers each driver's welfare.*
- *The platform sets a (weakly) higher peak period work target when employing a qualification scheme as opposed to a prioritization scheme.*

The reason that a driver becomes worse off under a prioritization scheme, despite a higher total matching volume, is that the peak work target set by the platform is higher than that a driver would voluntarily choose (in the absence of any scheme). As the service supply can always be utilized during the peak period, but can be wasted during the off-peak period if demand is insufficient, the prioritization scheme increases the total matching volume by shifting supply from the off-peak period to the peak period. However, each individual driver is worse off because of the additional congestion cost caused by the prolonged work time during the peak period.

Recall that under either scheme, the platform uses a peak period work target that induces all drivers to participate in the scheme. One may expect that the two schemes are equally effective in shifting off-peak supply to the peak period. However, as readily indicated by Lemma 3 and Lemma 4, the qualification scheme proves to be more effective in improving the total match volume. If a driver does not meet the peak period work target, he can still work during the off-peak period under the prioritization scheme, but is excluded from the service supply under the qualification scheme. As a result, the platform is able to set a higher target under the qualification scheme to shift more service supply from the off-peak period to the peak period and improve the matching volume to a greater extent. However, drivers are further hurt as a result of the longer work time during the peak period.

5. Open System

A platform with an open system allows both full-time and part-time drivers to provide service. In this section, we analyze the benchmark case and the two incentive schemes in an open system and compare their performances.

5.1. Benchmark

When no scheme is implemented and with the presence of the large supply capacity from part-time drivers, a full-time driver's chance of serving a passenger during the off-peak

period is negligible. With the simplifying assumption of infinite part-time drivers, a full-time driver derives zero utility from working in the off-peak period. Therefore, a full-time driver's utility when allocating α time to the peak period is

$$U_O^B(\alpha) = p\delta \cdot \alpha - \frac{1}{2}k \cdot \alpha^2. \tag{13}$$

Lemma 5 summarizes the benchmark equilibrium.

LEMMA 5 **(Open System: Benchmark)**. *In the open system without an incentive scheme, the equilibrium is characterized as follows:*

- *Each full-time driver's choice of α, denoted by α_O^B, is $\alpha_O^B = \frac{p\delta}{k}$.*
- *Each full-time driver's utility is given as $U_O^B = \frac{p^2\delta^2}{2k}$.*
- *The platform's expected total matching volume across both periods is $M_O^B = L + n\frac{p\delta}{k}$.*

Comparing the matching volume and driver welfare in an open system with those in a closed system, we have following result.

PROPOSITION 2. *In the absence of an incentive scheme, a platform achieves a higher total matching volume in an open system than in a closed system. However, full-time drivers are worse off.*

An open system improves the number of matching volume in two ways. Directly, it expands the service supply during the off-peak period by involving part-time drivers. Indirectly, it shifts the full-time drivers' work time more toward the peak period so that the peak period matching volume also increases. However, the full-time drivers are worse off as they have to work more during the peak period and bear additional congestion costs.

5.2. Qualification Scheme

When the platform adopts a qualification scheme, part-time drivers will not be assigned any service requests as they do not serve passengers during the peak period. Therefore, the qualification scheme in an open system is effectively the same as that in a closed system. The equilibrium has the same characterization as that in Lemma 3.

5.3. Prioritization Scheme

Following the similar procedure as that for a closed system, we obtain the equilibrium outcomes of using a prioritization scheme in an open system, as shown in Lemma 6.

LEMMA 6 **(Open System: Prioritization Scheme)**. *In the open system with a prioritization scheme, the equilibrium is characterized as follows:*

- *The platform's choice of peak period work target, denoted by* $\alpha_{0_O}^P$*, is*

$$\alpha_{0_O}^P = \begin{cases} \sqrt{\frac{p\delta}{k}(2-\frac{p\delta}{k})}, & \text{if } k \le k_2, \\ \alpha_0^q, & \text{if } k_2 < k \le k_3, \\ \frac{p\delta}{k} + \sqrt{2\frac{L}{n}\frac{p\delta}{k}}, & \text{if } k > k_3. \end{cases}$$

- *All full-time drivers participate in the scheme. Each driver's utility is* $U_O^P = \frac{p^2\delta^2}{2k}$.
- *The platform's expected total matching volume, denoted by* M_O^P*, is*

$$M_O^P = \begin{cases} L + n\sqrt{\frac{p\delta}{k}(2-\frac{p\delta}{k})}, & \text{if } k \le k_2, \\ L + n\alpha_0^q, & \text{if } k_2 < k \le k_3, \\ L + n\frac{p\delta}{k} + \sqrt{2nL\frac{p\delta}{k}}, & \text{if } k > k_3. \end{cases}$$

When a prioritization scheme is implemented, full-time drivers who do not participate in the scheme face competition from part-time drivers during off-peak periods. With an abundance of part-time drivers, the chances of an unprioritized driver receiving a service request during off-peak hours become extremely low, resulting in negligible utility. This similarity to a qualification scheme, where not participating also leads to zero off-peak utility, may lead one to believe that the platform sets the same peak period work target for both schemes. However, a comparison between Lemma 6 and Lemma 3 suggests that this intuition holds true only when the congestion cost is significant (i.e., $k > k_2$). In scenarios with a small congestion cost (i.e., $k \le k_2$), the platform sets a higher work target under the prioritization scheme compared to a qualification scheme.

To understand the reason, first note that to increase the matching volume under a qualification scheme, the platform seeks to maximize the utilization of full-time drivers' service. To achieve this, the platform shifts full-time drivers' work time to the peak period to minimize the likelihood of drivers experiencing idleness during the off-peak periods as the supply of drivers may surpass the demand. With a small congestion cost, the platform using a qualification scheme is able to set a high peak work target which shifts enough full-time drivers' service to eliminate the off-peak supply idleness. However, under a prioritization scheme, all off-peak period service requests are fulfilled due to the abundant part-time drivers. Therefore, in contrast to that under a qualification system, the platform seeks to shift full-time drivers' work time to the peak period as much as possible. As a result, the platform is able to further increase the work target under a prioritization scheme due to the small congestion cost.

5.4. Comparison of the Benchmark and Two Schemes in an Open System

On comparing the benchmark scenario and the two incentive schemes in an open system, we have the following results.

PROPOSITION 3. *In the open system, given $k > \frac{np\delta}{n-L+s}$ and that the price per unit of service time is fixed, we have the following results.*

- *Compared to the benchmark scenario, implementing a qualification scheme increases the platform's total matching volume if $k > k_4$ but decreases the total matching volume otherwise. Here*

$$k_4 = \begin{cases} \frac{p\delta}{1-L/n}, & \text{if } s \le \sqrt{n^2-L^2}+L-n, \\ \tilde{k}, & \text{if } s > \sqrt{n^2-L^2}+L-n, \end{cases}$$

where $\tilde{k}$ satisfies $n - \int_{-s}^{n(1-\tilde{\alpha}_0^q)-L} F(\epsilon)d\epsilon - L - n\frac{p\delta}{\tilde{k}} = 0$, and $\tilde{\alpha}_0^q$ solves the implicit equation $(1 - \frac{1}{n}\int_{-s}^{n(1-\tilde{\alpha}_0^q)-L} F(\epsilon)d\epsilon)p\delta - \frac{1}{2}\tilde{k}\tilde{\alpha}_0^{q2} = \frac{p^2\delta^2}{2\tilde{k}}$.

  In addition, it increases full-time drivers' welfare if $k \le k_2$ but has no impact on their welfare otherwise. Here k_2 is defined in Lemma 3.

- *Compared to the benchmark scenario, implementing a prioritization scheme increases the platform's total matching volume without affecting full-time drivers' welfare.*

- *Compared to the qualification scheme, implementing a prioritization scheme (weakly) increases platform's total matching volume. It lowers full-time drivers' welfare if $k \le k_2$, but otherwise does not affect their welfare.*

Recall that in a closed system, a qualification scheme always improves the total matching volume compared to the benchmark. Interestingly, Proposition 3 shows that in an open system, a qualification scheme may decrease the total matching volume. To see the reason, note that a qualification scheme, although increases the peak period matching volume by shifting full-time drivers' services to the peak period, may also reduce the off-peak matching volume by rejecting part-time drivers from providing services. The total matching volume increases only when the congestion cost for full-time drivers is high. If the congestion cost is low, there is already an abundant supply of full-time drivers during the peak period when no qualification scheme is implemented. Consequently, the increase in peak period matching volume resulting from the supply shift in the scheme is outweighed by the loss of matching volume during the off-peak period due to the exclusion of part-time drivers.

Unlike a qualification scheme, a prioritization scheme improves the total matching volume (compared to the benchmark scenario) by shifting the supply of full-time drivers to

the peak period while still allowing part-time drivers to work during the off-peak period. It also leads to a higher matching volume than a qualification scheme, in contrast to what we observe in a closed system where the prioritization scheme is inferior to the qualification scheme in increasing matching volume (see Proposition 1). This is because the prioritization scheme not only better utilizes the supply from part-time drivers, but also enables the platform to set a higher peak period work target (as explained in Section 5.3), thereby effectively shifting more of the full-time drivers' supply to the peak period.

Proposition 3 further demonstrates the impacts of implementing incentive schemes on a full-time driver's welfare. In contrast to a closed system where the implementation of incentive schemes always hurt full-time drivers, a qualification scheme in an open system can benefit full-time drivers compared to not having any scheme. This happens when the congestion cost is small (i.e., $k \leq k_2$). In such a case, full-time drivers are not significantly affected by longer working hours during the peak period, but they greatly benefit from reduced competition during off-peak periods due to the exclusion of part-time drivers under the qualification scheme. We also notice that threshold k_4 could be smaller than k_2. In this case, it implies a range of the congestion cost $k \in (k_4, k_2]$, under which a platform's implementing a qualification scheme benefits both the platform and full-time drivers (compared to not implementing any scheme).

We also observe that under certain situations the welfare of full-time drivers is not affected by the implementation of either a qualification or a prioritization scheme. Note that in the absence of any scheme, a full-time driver's chances of receiving ride requests during off-peak periods are negligible. Consequently, the driver only derives positive utility during the peak period. When the platform uses either scheme to optimally shift full-time drivers' work time, it may set the work target such that the full-time drivers' gain from working in the off-peak period exactly offsets their loss due to the longer work time in the peak period, making the full-time drivers' welfare unaffected.

6. Model Extensions

In this section, we first compare the performance of the proposed incentive schemes to surge pricing in Section 6.1. We then relax our simplifying assumption by considering finite number of part-time drivers in Section 6.2.[10]

6.1. Incentive Schemes vs. Surge Pricing

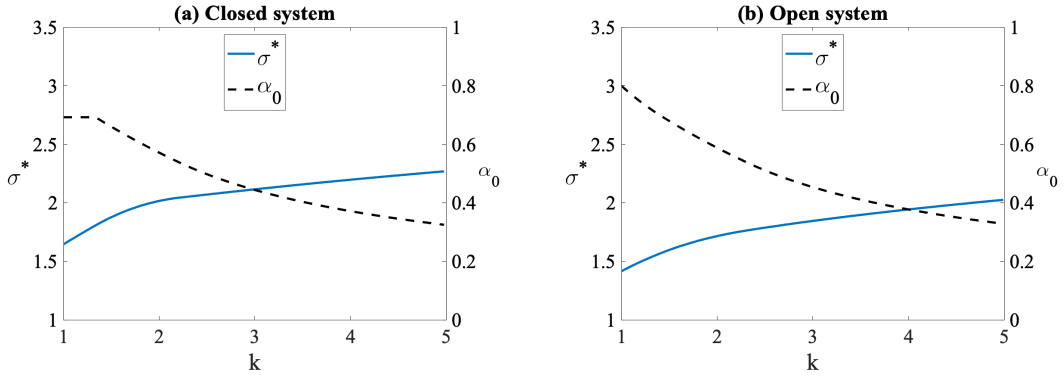
A key motivation of our proposed incentive schemes is that prices in many markets can be subject to regulations and have limited usefulness in addressing the supply-demand imbalance. Nevertheless, it would be helpful to compare the performance of our proposed schemes with surge pricing, the practice of raising the transaction price during the peak period to address supply shortages. To model surge pricing, we let the peak period demand be $D_p - p_s$, where D_p denotes the base demand in the peak period (i.e., the demand for ride service when transaction price is zero) and p_s denotes the surging price in the peak period. Same as in the main model, p represents the regular price at the off-peak period. We define $\sigma = \frac{p_s}{p} > 1$ as the platform's peak period price multiplier, which reflects the level of price surging.

Given an exogenous price multiplier σ, we can analytically derive the total matching volume, the platform's profit, and drivers' utility when the platform uses surge pricing under both a closed system and an open system. The detailed analysis and results are provided in Appendix B of the Online Appendix. To compare with the incentive schemes, we calculate the minimum price multiplier needed for a surge pricing scheme to achieve the same profit as that by the optimal incentive scheme.[11] As the analytical comparison losses tractability, we resort to numerical analysis. The parameter values are carefully chosen based on several empirical studies including Bai et al. (2019), Xiong et al. (2021), Bimpikis et al. (2019), Wang and Zhang (2022). We provide the detailed methodologies in SA.D.1 of the Supplementary Analysis. We also carry out a series of robustness checks, varying one parameter at a time across ten sets of parameter values. The detailed results can be found in SA.D.2 of the Supplementary Analysis.

Recall from Propositions 1 and 2 that the qualification scheme generates a higher matching volume in a closed system, whereas the prioritization scheme generates a higher matching volume in an open system. Thus in the closed system, we analyze the price multiplier needed for surge pricing to match the same profit as the qualification scheme, while in the open system, we assess the price multiplier needed for surge pricing to match the prioritization scheme. Figure 1 shows that in both a closed system and an open system, given the parameter values and with a relatively small congestion cost (i.e., $k = 1$), the platform needs to raise its base price by approximately 50% for surge pricing to match the

profit of the optimal incentive scheme. For this scenario (i.e., $k = 1$), the optimal scheme requires a work target around 0.7 (i.e., drivers need to allocate about 70% of their total daily work time to the peak period) in a closed system and around 0.8 (i.e., drivers need to allocate about 80% of their total daily work time to the peak period) in an open system.[12] As drivers' congestion cost grows larger, the multiplier required by surge pricing further increases, whereas the optimal incentive scheme's work target decreases. In the case of higher congestion cost like $k = 5$, surge pricing in both systems requires almost doubling the price (i.e., $\sigma^* \approx 2$), whereas the optimal incentive scheme only requires the work target to be slightly above 0.3 in both systems.

Figure 1 **The minimum price multiplier needed for surge pricing to achieve the same profit of an incentive scheme. ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

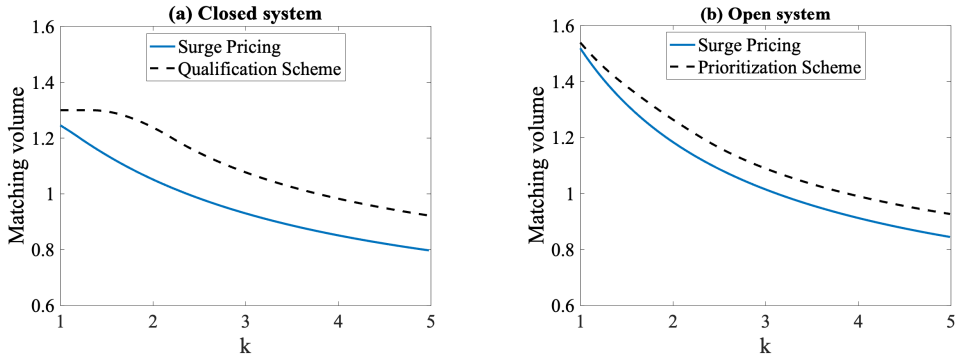


As previously discussed, the proposed incentive schemes can serve as alternatives to surge pricing in markets where prices are regulated. Even in less regulated markets, implementing surge pricing with a higher price multiplier may face challenges. For example, a 50% price increase could draw criticism for unethical practices. The comparison suggests that as the congestion cost increases, implementing surge pricing becomes more challenging, while using the incentive scheme to achieve the same profits may become easier as the required work target decreases.

When deriving the optimal incentive schemes, the goal is to maximize the total matching volume since the price is exogenous. We now align the objective of surge pricing to also maximize the total matching volume and compare it with the optimal incentive scheme in each system. Interestingly, we find the possibility that surge pricing does not achieve the same matching volume as the incentive scheme. To understand the reason, note that while surge pricing increases the peak period service supply, it also decreases the peak period

service demand, as a result reducing the overall matching volume. When the peak period demand is not significantly higher than the off-peak period demand, we could observe a lower total matching volume from surge pricing. Figure 2 illustrates this result in both a closed system and an open system.

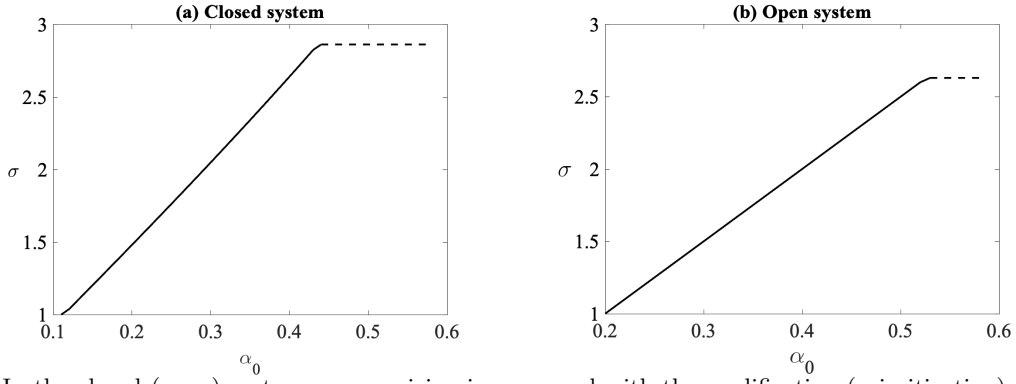
Figure 2 **The comparison of maximum matching volume between surge pricing and incentive schemes (**$D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$**)**



We further examine the mapping between the service target α_0 and the price multiplier σ such that the incentive scheme and surge pricing yield the same matching volume. In this analysis, the service target α_0 and the price multiplier σ are not necessarily the optimal ones but only enable the incentive scheme (i.e., the qualification scheme given a closed system, and the prioritization scheme given an open system) and surge pricing to reach the same matching volume. Figure 3 presents the outcomes in both a closed system and an open system. For example, in a closed system and given the chosen parameter values, as the work target of the qualification scheme changes from 0.2 to 0.3, the price multiplier of surge pricing needs to change from around 1.4 (i.e., a 40% price increase) to 2 (i.e., a 100% price surge). If a high price multiplier becomes the obstacle in implementing surge pricing, the mapping result suggests that the proposed incentive schemes with reasonable work target may be feasible alternatives to achieve the same matching volume. Figure 3 also shows the region where surge pricing cannot achieve the same matching volume as in the incentive scheme, consistent with our observations in Figure 2.

In summary, the comparison suggests that our proposed schemes are viable alternatives to surge pricing, not only when prices are regulated but also when a high price multiplier is required but poses challenges in implementing surge pricing. Furthermore, as drivers' congestion cost grows, implementing surge pricing could become more difficult because of a higher price multiplier needed, while implementing an incentive scheme may become easier

Figure 3 **The minimum price multiplier needed for surge pricing to achieve/approach the matching volume of an incentive scheme. ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, k = 2, \epsilon \sim U[-0.1, 0.1]$)**



Note: (1) In the closed (open) system, surge pricing is compared with the qualification (prioritization) scheme. We use the "dashed" line to represent the case that the surge pricing fails to achieve the same matching volume as the incentive scheme. (2) In creating each figure, α_0 begins at the level that each full-time driver would choose voluntarily in the absence of an incentive scheme, corresponding to the case $\sigma = 1$ (i.e., no surge pricing).

as the work target required decreases. The analysis also shows that the incentive schemes could have an advantage in improving the total matching volume.

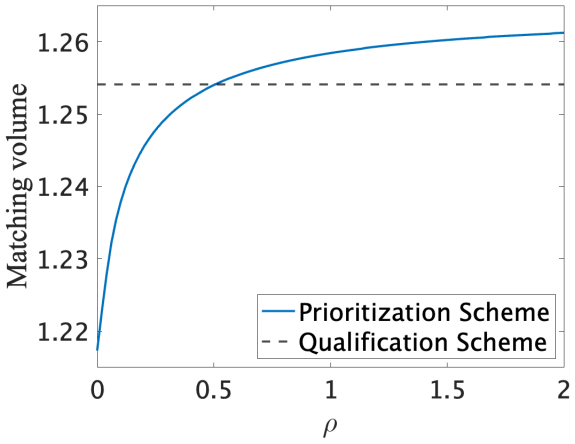
6.2. Finite Part-time Drivers

In the main model, we assumed that in an open system, an infinite number of part-time drivers can work during the off-peak period. The purpose of this assumption is to capture the supply surplus in the off-peak period due to the involvement of part-time drivers in the simplest way and to study how this surplus enables an incentive scheme, particularly a prioritization scheme, to effectively incentivize full-time drivers to participate. We now relax this assumption by considering the case where the number of part-time drivers is finite, and explore to what extent our result—that the prioritization scheme outperforms the qualification scheme in an open system (see Proposition 3)—continues to hold.

To explicitly incorporate a finite number of part-time drivers, we denote the ratio between part-time drivers and full-time drivers in an open system as ρ ($\rho > 0$). The total number of part-time drivers is then ρn. We denote the fraction of time per day that a part-time driver works as γ (note that we normalize the daily working hours of a full-time driver to 1, so $\gamma < 1$). Therefore, each part-time driver provides a supply of γ per day, and the total daily supply from part-time drivers is $\rho n \gamma$.

Since part-time drivers are not allowed to provide service under a qualification scheme, the equilibrium characterization under this scheme remains unaffected when the number of part-time drivers is finite. However, the equilibrium characterization under a prioritization

Figure 4 **Comparison of total matching volume in an open system with finite part-time drivers. (**$D_0 = 1, p = 0.5, \delta = 0.8, n = 1.3, k = 2, \gamma = 0.5, \epsilon \sim U[-0.1, 0.1]$**)**



scheme depends on the number of available part-time drivers. Recall that in our main model, the advantage of a prioritization scheme over a qualification scheme stems from the inclusion of abundant part-time drivers. This inclusion allows the platform to benefit from further improving the work target under a prioritization scheme, even when the likelihood of full-time drivers experiencing idleness is eliminated. The next proposition shows that this result still holds as long as the number of part-time drivers exceeds a certain threshold.

PROPOSITION 4. *In an open system with a finite number of part-time drivers, given that the price per unit of service time is fixed, a prioritization scheme leads to a higher matching volume than a qualification scheme as long as* ρ *is above a threshold (i.e.,* $\rho > \rho^*$*). The threshold* ρ^* *is obtained by solving* $M_O^{P\dagger}|_{\rho=\rho^*} = M_C^Q$*, where* M_C^Q *is given in Lemma 3 and* $M_O^{P\dagger}|_{\rho=\rho^*}$ *is provided in SA.B.1 of the Supplementary Analysis.*

We further visualize the comparison result in Figure 4. As the number of part-time drivers decreases, the benefit of allowing these part-time drivers to serve as backup supply to meet off-peak demand under the prioritization scheme diminishes. In the scenario where the number of part-time drivers becomes zero, the open system degenerates into a closed system. In such a case, the qualification scheme outperforms the prioritization scheme, consistent with our prediction in Proposition 1.

7. Conclusion

This study examines the challenge of intertemporal supply-demand imbalance faced by on-demand ride-hailing platforms. In contrast to existing literature that seeks demand-side

solutions such as surge pricing, this study explores an operational approach and focuses on supply-side solutions. We propose two novel incentive schemes: the qualification scheme and the prioritization scheme. Under the qualification scheme, the platform sets a peak period work target and assigns off-peak requests only to drivers who meet the peak period work target. Under the prioritization scheme, the platform also sets a peak period work target but prioritizes the off-peak requests for drivers who meet the target. Our research finds that for platforms with a closed system that only allows full-time drivers, the qualification scheme is more effective in improving the total matching volume. However, it hurts full-time drivers more than the prioritization scheme. On the other hand, for platforms with an open system that also includes abundant part-time drivers to serve off-peak requests, the prioritization scheme outperforms the qualification scheme in improving total matching volume. Moreover, we find that the implementation of an incentive scheme in an open system may benefit both the platform and full-time drivers.

Managerially, our study offers an alternative to surge pricing for on-demand platforms to alleviate the intertemporal supply-demand imbalance. The findings are particularly useful for markets where prices are subject to regulation and inflexible to adjust. Besides China, India also regulates the price in the ride-hailing industry by capping surge pricing during peak demand at a maximum of 1.5 times the base fare (Dash 2020). In Karnataka, an Indian state, surge pricing is entirely prohibited (Siddiqui 2024). Similarly, the Philippine government limits surge multipliers to a maximum of 2 (Ilagan and Rainis 2024). Especially when a large price multiplier poses challenges in implementing surge pricing, we find that the proposed incentive schemes could have an implementation advantage. Furthermore, we show that the incentive schemes could potentially lead to a total matching volume that cannot be achieved by surge pricing. Our results provide useful insights for platforms with varying degrees of supply system openness on the choice of an incentive scheme. The welfare results offer guidance to platforms and policymakers who care about not only the number of matches made but also the welfare of independent contractors.

Our model can serve as a starting point for exploring other important issues in the broader gig economy. First, for the purpose of this study, we assume the composition of full-time and part-time workers is exogenous. Future research could explore the impact of implementing an incentive scheme on workers' strategic decisions to work full-time or

part-time, and thus the worker composition on the supply side. Second, one of our goals is to understand the effect of supply system openness on the performance of incentive schemes. Further work could incorporate relevant factors to explore the platform's choice of system openness. Third, although our paper examines two extremes of a platform's openness—which in practice usually exists on a continuous spectrum—the results of the closed system analysis apply to platforms that primarily rely on full-time drivers to provide service, while the results of the open system analysis apply to platforms that involve a large number of part-time drivers. For example, Proposition 4 indicates that the effectiveness of the two schemes hinges on the ratio of part-time drivers to full-time drivers, which could be used to measure the openness of a platform's supply system. Investigating a platform's optimal choice of its supply system openness could be a promising direction for future research. Fourth, it would be beneficial to expand on the current work by examining the competition between platforms that can adopt different incentive schemes. Finally, empirical validation of our model predictions using field data would provide valuable insights.

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Notes

[1]See https://www.chinadaily.com.cn/a/201907/10/WS5d257efaa3105895c2e7cb93.html (Accessed April 2025).

[2]For example, city governments in China can set government-guided prices for ride-hailing services and enforce them when they deem it necessary. See http://www.gov.cn/zhengce/2022-12/06/content_5730384.htm (Accessed April 2025).

[3]See http://www.sohu.com/a/458602481_442726 (Accessed April 2025).

[4]See http://cloud.tencent.com/developer/news/572397 (Accessed April 2025).

[5]See https://iot.ofweek.com/2023-06/ART-132215-8500-30600985.html (Accessed April 2025).

[6]In Section SA.B.3 of the Supplementary Analysis, we demonstrate the robustness of our results when allowing a proportion of part-time drivers to provide service during peak periods.

[7]In our main model, we do not put constraint on the length of peak and off-peak periods. That is, a full-time driver can allocate all his working time to either the peak or the off-peak period. Sometimes the peak period may be shorter than a full time driver's total working time. We consider this situation by allowing the length of the peak period to be capped at $\bar{\alpha} \in (0,1)$. In this case, a full time driver's choice needs to satisfy $\alpha \in [0, \bar{\alpha}]$. We show the robustness of our main results and present the analysis in SA.B.5 of the Supplementary Analysis.

[8]The detailed analysis can be found in the SA.B.3 of the Supplementary Analysis.

[9]Although not explicitly modeled, the demand during peak periods can also be volatile. Our results, however, remain unaffected as long as the peak demand always exceeds the service supply ($D_p(p) > n$). In the Supplementary Analysis SA.B.4, we analyze the scenario in which the peak period demand exceeds n with certain probability and demonstrate the robustness of our results.

[10]In Appendix C of the Online Appendix, we further incorporate drivers' heterogeneity in their congestion costs, and discuss the robustness and boundary conditions of our results. In SA.D.5 of the Supplementary Analysis, we conducted additional robustness checks by considering drivers' heterogeneous congestion costs.

[11]In SA.D.4 of the Supplementary Analysis, we use an alternative comparison approach by comparing surge pricing and the incentive schemes when they are all maximizing the platform's profits. With its full flexibility in adjusting prices, it is not surprising that surge pricing yields the highest platform profit. However, we find that the incentive scheme generates a higher matching volume and increases consumer surplus.

[12]In reality, the length of the peak period may be limited to less than 1 so that a worker cannot allocate its full time to the peak period. In SA.D.3 of the Supplementary Analysis, we analyze cases where the service target has upper bounds of 0.5, 0.4 and 0.3, respectively. We show that the result of comparison with surge pricing is robust as long as drivers' congestion costs are high.

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Online Appendix for "Balancing Supply with Demand on Ride-hailing Platforms in Markets with Price Regulations: An Operational Approach"

This file provides the proofs of the lemmas and propositions of the main model, two lemmas related to surge pricing, and the analysis of the case with heterogeneous congestion costs. Proofs related to model extensions and numerical analyses are delegated to the Supplementary Analysis.

In the following, the equilibrium results are indicated with superscript $*$. We use subscripts C, O to represent *Closed system* and *Open system* respectively, subscripts $q/uq, p/up$ to represent *qualified/unqualified drivers* and *prioritized/unprioritized drivers* respectively, and use superscripts B, Q, P to represent *Benchmark, Qualification scheme*, and *Prioritization scheme*, respectively.

Appendix A. Proofs of Lemmas and Propositions of the Main Model

Proof of Lemma 1

In a closed system and without any incentive scheme, an individual driver's problem is to maximize his utility as follows:

$$\max_{\alpha\in[0,1]} U^B(\alpha) = \max_{\alpha\in[0,1]} \left(Pr^B p\delta(1-\alpha) + p\delta\alpha - \frac{1}{2}k\alpha^2 \right),$$

where $Pr^B = \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n(1-\hat{\alpha})}\} dF(\epsilon)$ represents the driver's probability of receiving ride request during the off-peak period and $\hat{\alpha} \in [0,1]$ denotes the average working time of all other full-time drivers at the peak period. Depending on the relative size of the demand $D(\epsilon) = L + \epsilon$ and the supply $n(1-\hat{\alpha})$ during the off-peak period, there are three cases.

- *Case 1: $n(1-\hat{\alpha}) > max\{D(\epsilon)\} = L + s$*

  In this case, the off-peak market is over-supplied. The expected driver utility is $E_\epsilon(U^B(\alpha)) = \left(\frac{L}{n(1-\hat{\alpha})}(1-\alpha) + \alpha\right) p\delta - \frac{1}{2}k\alpha^2$. Since it is concave in α, we use the first-order condition (FOC) to solve the optimal α, and obtain $\alpha(\hat{\alpha}) = \frac{p\delta}{k}(1 - \frac{L}{n(1-\hat{\alpha})})$. The equilibrium condition requires $\alpha(\hat{\alpha}) = \hat{\alpha}$, which yields the optimal $\alpha = \frac{k+p\delta-\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{2k} \equiv \alpha_1^\dagger$ (the other solution exceeds 1 and thus is infeasible). By solving $n(1-\alpha_1^\dagger) > L + s$, we obtain $k > \frac{p\delta s}{(L+s)(1-\frac{L+s}{n})} \equiv k_1$. Therefore, when $k > k_1$, $\alpha_1^\dagger$ is the optimal solution in this case.

- *Case 2: $min\{D(\epsilon)\} \le n(1-\hat{\alpha}) \le max\{D(\epsilon)\}$*

  In this case, the demand can be larger or smaller than the supply during the off-peak period. Driver utility $U^B(\alpha) = Pr^B p\delta(1-\alpha) + p\delta\alpha - \frac{1}{2}k\alpha^2$, where

$$\begin{aligned}
Pr^B =& \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n(1-\hat{\alpha})}\} dF(\epsilon) = \int_{-s}^{n(1-\hat{\alpha})-L} \frac{L+\epsilon}{n(1-\hat{\alpha})} dF(\epsilon) + \int_{n(1-\hat{\alpha})-L}^{s} dF(\epsilon) \\
=& \left(\frac{L}{n(1-\hat{\alpha})} \int_{-s}^{n(1-\hat{\alpha})-L} dF(\epsilon) + \frac{1}{n(1-\hat{\alpha})} \int_{-s}^{n(1-\hat{\alpha})-L} \epsilon dF(\epsilon) \right) + (F(s) - F(n(1-\hat{\alpha})-L)) \\
=& \left(\frac{L}{n(1-\hat{\alpha})} F(n(1-\hat{\alpha})-L) + \frac{1}{n(1-\hat{\alpha})} \left(\epsilon F(\epsilon)|_{-s}^{n(1-\hat{\alpha})-L} - \int_{-s}^{n(1-\hat{\alpha})-L} F(\epsilon) d\epsilon \right) \right) + 1 - F(n(1-\hat{\alpha})-L) \\
=& \left(\frac{L}{n(1-\hat{\alpha})} F(n(1-\hat{\alpha})-L) + \frac{1}{n(1-\hat{\alpha})} \left((n(1-\hat{\alpha})-L) F(n(1-\hat{\alpha})-L) - \int_{-s}^{n(1-\hat{\alpha})-L} F(\epsilon) d\epsilon \right) \right) + \\
& 1 - F(n(1-\hat{\alpha})-L) \\
=& 1 - \frac{1}{n(1-\hat{\alpha})} \int_{-s}^{n(1-\hat{\alpha})-L} F(\epsilon) d\epsilon
\end{aligned}$$

Therefore, we have $U^B(\alpha) = \Big(1-(1-\alpha)\frac{\int_{-s}^{n(1-\hat{\alpha})-L} F(\epsilon)d\epsilon}{n(1-\hat{\alpha})}\Big)p\delta - \frac{1}{2}k\alpha^2$. Solving FOC, we have $\alpha(\hat{\alpha}) = \frac{p\delta}{k}A(\hat{\alpha})$, where $A(\hat{\alpha}) = \frac{\int_{-s}^{n(1-\hat{\alpha})-L} F(\epsilon)d\epsilon}{n(1-\hat{\alpha})}$. The equilibrium condition requires $\alpha(\hat{\alpha}) = \hat{\alpha}$, whose solution is denoted as $\alpha_2^\dagger$. Next, we can show that $\frac{dA(\hat{\alpha})}{d\hat{\alpha}} = -nF[n(1-\hat{\alpha})-L] < 0$. Thus, for $\hat{\alpha} = \frac{p\delta}{k}A(\hat{\alpha})$, LHS (RHS) increases (decreases) in $\hat{\alpha}$, which indicates there is at most one solution. The equilibrium occurs in this case only if the following two conditions are satisfied: (i) $\hat{\alpha} < \frac{p\delta}{k}A(\hat{\alpha})$ when $\hat{\alpha} = 1 - \frac{L+s}{n}$ and (ii) $\hat{\alpha} > \frac{p\delta}{k}A(\hat{\alpha})$ when $\hat{\alpha} = 1 - \frac{L-s}{n}$, which lead to $k \le \frac{p\delta s}{(L+s)(1-\frac{L+s}{n})} \equiv k_1$. Therefore, when $k \le k_1$, $\alpha_2^\dagger$, which satisfies $\alpha_2^\dagger = \frac{p\delta}{k}\frac{\int_{-s}^{n(1-\alpha_2^\dagger)-L} F(\epsilon)d\epsilon}{n(1-\alpha_2^\dagger)}$, is the optimal solution in this case.

- *Case 3:* $n(1-\hat{\alpha}) < min\{D(\epsilon)\} = L - s$

  In this case, the off-peak market is always under-supplied. Driver utility $U^B(\alpha) = p\delta - \frac{1}{2}k\alpha^2$. Note that $U^B(\alpha)$ decreases in α, so the optimal solution is $\alpha(\hat{\alpha}) = 0$. The equilibrium condition requires $\alpha(\hat{\alpha}) = \hat{\alpha}$ and thus $\hat{\alpha} = 0$, the condition in *Case 3* becomes $n < L - s$, which violates the parameter space ($s < n - L$) in main model. Therefore, *Case 3* will not occur in equilibrium.

By combining the results from the above three cases, we have the equilibrium results as follows. If $k > k_1$, the optimal α equals $\alpha_1^\dagger$; if $k \le k_1$, the optimal α equals $\alpha_2^\dagger$. The total matching volume is $M^B(\alpha) = \int_{-s}^{s} \min\{D(\epsilon), n(1-\alpha)\}dF(\epsilon) + n\alpha$. We plug the values of $\alpha_1^\dagger$ and $\alpha_2^\dagger$ from *Case 1* and *Case 2*, into $U^B(\alpha)$ and $M^B(\alpha)$, and obtain the expressions of U_C^B, M_C^B presented in Lemma 1. We can further prove $M_C^B < n$ as shown below. When $k > k_1$, $L + n\frac{k+p\delta-\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{2k} < n$ always holds. When $k \le k_1$, $\int_{-s}^{n(1-\alpha^b)-L} F(\epsilon)d\epsilon > 0$ if $n(1-\alpha^b) - L \ge -s$ ($\alpha^b = 1 - \frac{L-s}{n}$ could not be the solution, since it fails to make $\alpha(\hat{\alpha}) = \frac{p\delta}{k}A(\hat{\alpha})$ hold). Therefore, we have proved $M_C^B < n$. ■

Proof of Lemma 2.

In a closed system, for an unqualified driver, the problem is $\underset{\alpha_{uq}\in(0,1)}{max}\ U_{uq}(\alpha_{uq}) = p\delta\alpha_{uq} - \frac{1}{2}k\alpha_{uq}^2$. Solving FOC yields the optimal solution, denoted as $\alpha_{uq}^Q = \frac{p\delta}{k}$. The qualification scheme is ineffective if the service target during peak period falls below this threshold, so we have the condition that $\alpha_0 > \frac{p\delta}{k}$.

Suppose the proportion of qualified drivers is θ and their average time allocated to the peak period is $\hat{\alpha}_q$, the optimization problem of an individual qualified driver is $\underset{\alpha_q\in[\alpha_0,1]}{max}\ U_q(\alpha_q) = Pr^Q(\hat{\alpha}_q)\cdot p\delta(1-\alpha_q) + p\delta\alpha_q - \frac{1}{2}k\alpha_q^2$, where $Pr^Q(\hat{\alpha}_q) = \int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_q)}\}dF(\epsilon)$. Since $\frac{dU_q(\alpha_q)}{d\alpha_q} = (1-Pr^Q)p\delta - k\alpha_q \le (1-Pr^Q)p\delta - k\alpha_0 < (1-Pr^Q)p\delta - k\frac{p\delta}{k} = -Pr^Q p\delta < 0$, we have the optimal solution, denoted as $\alpha_q^Q = \alpha_0$.■

Proof of Lemma 3.

Based on the results from Lemma 2, the service time allocated to the peak period by qualified and unqualified drivers are given by $\alpha_q^Q = \alpha_0$ and $\alpha_{uq}^Q = \frac{p\delta}{k}$, respectively. The utility of qualified and unqualified drivers are $U_q(\alpha_0) = Pr^Q(\alpha_0)\cdot p\delta(1-\alpha_0) + p\delta\alpha_0 - \frac{1}{2}k\alpha_0^2$ and $U_{uq}^Q = \frac{p^2\delta^2}{2k}$, where $Pr^Q(\alpha_0) = \int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n\theta(1-\alpha_0)}\}dF(\epsilon)$. Using the above information, the ride-hailing platform sets the service target during peak period α_0 to maximize the total matching volume, as shown below:

$$\underset{\alpha_0\in(\alpha_{uq}^Q,1]}{max}\ M^Q(\alpha_0) = \int_{-s}^{s}\min\{D(\epsilon), n\theta(1-\alpha_0)\}dF(\epsilon) + n\theta\alpha_0 + n(1-\theta)\alpha_{uq}^Q \tag{A.1}$$

$$\text{s.t. } U_q(\alpha_0) \geq U_{uq}^Q, \text{ with the constraint binding if } \theta < 1. \tag{A.2}$$

In the following, we first prove that given the service target α_0, all drivers participate the qualification scheme, i.e., $\theta^*(\alpha_0) = 1$. Then, we substitute $\theta^*(\alpha_0) = 1$ and characterize the optimal decision on the service target α_0.

1). *Proof of $\theta^*(\alpha_0) = 1$*

For simplicity, we will use θ^* to represent $\theta^*(\alpha_0)$ hereafter. Suppose (α_0^*, θ^*) is the optimal outcome, where $\theta^* \in (0,1)$ and $\alpha_0^* \in (\frac{p\delta}{k}, 1]$. Then, the total matching volume is shown in Equation (A.3) and subject to $U_q^* = U_{uq}^Q$, i.e., $Pr_q^* p\delta(1-\alpha_0^*) + p\delta\alpha_0^* - \frac{1}{2}k{\alpha_0^*}^2 = \frac{p^2\delta^2}{2k}$, where $Pr_q^* = \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n\theta^*(1-\alpha_0^*)}\} dF(\epsilon)$.

$$M^* = \begin{cases} n\theta^* + n(1-\theta^*)\frac{p\delta}{k}, & \text{if } n\theta^*(1-\alpha_0^*) - L < -s, \\ n\theta^* + n(1-\theta^*)\frac{p\delta}{k} - \int_{-s}^{n\theta^*(1-\alpha_0^*)-L} F(\epsilon)d\epsilon, & \text{if } -s < n\theta^*(1-\alpha_0^*) - L < s, \\ L + n\theta^*\alpha_0^* + n(1-\theta^*)\frac{p\delta}{k}, & \text{if } n\theta^*(1-\alpha_0^*) - L > s, \end{cases} \tag{A.3}$$

Let $\alpha_0' = \alpha_0^* - (1-\theta^*)(\alpha_0^* - \frac{p\delta}{k})$. We first prove $\theta' = \theta|_{\alpha_0' = \alpha_0^* - (1-\theta^*)(\alpha_0^* - \frac{p\delta}{k})} = 1$. That is, all drivers participate in the scheme given the peak target α_0'. In the following, we will prove $\theta' = 1$ by showing $U_q' > U_{uq}^Q$. To prove $U_q' > U_{uq}^Q$ (i.e., $Pr_q' p\delta(1-\alpha_0') + p\delta\alpha_0' - \frac{1}{2}k{\alpha_0'}^2 > \frac{p^2\delta^2}{2k}$), it is equivalent to prove $(\alpha_0' - \frac{p\delta}{k})^2 < 2\frac{p\delta}{k} Pr_q'(1-\alpha_0')$. For the left-hand side of the inequality, $(\alpha_0' - \frac{p\delta}{k})^2 = (\theta^*)^2(\alpha_0^* - \frac{p\delta}{k})^2 = (\theta^*)^2(2\frac{p\delta}{k} Pr_q^*(1-\alpha_0^*))$, where the second equality is derived from $U_q^* = U_{uq}^Q$ (i.e., $Pr_q^* p\delta(1-\alpha_0^*) + p\delta\alpha_0^* - \frac{1}{2}k{\alpha_0^*}^2 = \frac{p^2\delta^2}{2k}$). Therefore, proving $U_q' > U_{uq}^Q$ is equivalent to prove $Pr_q^*(1-\alpha_0^*)\theta^{*2} < Pr_q'(1-\alpha_0')$. Note that $Pr_q^* n\theta^*(1-\alpha_0^*) = (\int_{-s}^{n\theta^*(1-\alpha_0^*)-L} \frac{D(\epsilon)}{n\theta^*(1-\alpha_0^*)} dF(\epsilon) + \int_{n\theta^*(1-\alpha_0^*)-L}^{s} dF(\epsilon)) n\theta^*(1-\alpha_0^*) = n\theta^*(1-\alpha_0^*) - \int_{-s}^{n\theta^*(1-\alpha_0^*)-L} F(\epsilon)d\epsilon$ and $Pr_q' n(1-\alpha_0') = (\int_{-s}^{n(1-\alpha_0')-L} D(\epsilon)dF(\epsilon) + \int_{n(1-\alpha_0')-L}^{s} dF(\epsilon)) n(1-\alpha_0') = n(1-\alpha_0') - \int_{-s}^{n(1-\alpha_0')-L} F(\epsilon)d\epsilon$. Because the function $(x - \int_{-s}^{x-L} F(\epsilon)d\epsilon)$ increases in x (> 0) and $n(1-\alpha_0') > n\theta^*(1-\alpha_0^*)$, we have $Pr_q' n(1-\alpha_0') > Pr_q^* n\theta^*(1-\alpha_0^*)$. Thus, $Pr_q'(1-\alpha_0') > Pr_q^*\theta^*(1-\alpha_0^*) > Pr_q^*\theta^{*2}(1-\alpha_0^*)$. Therefore, we have proved $U_q' > U_{uq}^Q$. Since the utility of participating in the scheme is strictly greater than the utility of not participating, given the peak target α_0', all drivers will choose to participate. That is, $\theta' = 1$.

Given the peak target $\alpha_0' = \alpha_0^* - (1-\theta^*)(\alpha_0^* - \frac{p\delta}{k})$ and $\theta' = 1$, the total matching volume of M' is shown in Equation (A.4), and subject to $U_q' \geq U_{uq}^Q$, i.e., $Pr_q' p\delta(1-\alpha_0') + p\delta\alpha_0' - \frac{1}{2}k{\alpha_0'}^2 \geq \frac{p^2\delta^2}{2k}$, where $Pr_q' = \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n(1-\alpha_0')}\} dF(\epsilon)$. We next show that $M' \geq M^*$. If $n(1-\alpha_0') - L \leq -s$, $M' = n > M^*$. If $-s < n(1-\alpha_0') - L \leq s$ but $n\theta^*(1-\alpha_0^*) - L \leq -s$, $M' - M^* = n - \int_{-s}^{n(1-\alpha_0')-L} F(\epsilon)d\epsilon - (n\theta^* + n(1-\theta^*)\frac{p\delta}{k}) = n(1-\theta^*)(1-\frac{p\delta}{k}) - \int_{-s}^{n\theta^*(1-\alpha_0^*)+n(1-\theta^*)(1-\frac{p\delta}{k})-L} F(\epsilon)d\epsilon$. Let $x \equiv n(1-\theta^*)(1-\frac{p\delta}{k})$ and $m \equiv n\theta^*(1-\alpha_0^*) - L$, we can show that $H(x) = x - \int_{-s}^{x+m} F(\epsilon)d\epsilon > 0$ for any $x \in [0, s-a]$, where $m \in (0, -s]$, thus $M' > M^*$. If $-s < n(1-\alpha_0') - L \leq s$ but $-s < n\theta^*(1-\alpha_0^*) - L \leq s$, $M' - M^* = n - \int_{-s}^{n(1-\alpha_0')-L} F(\epsilon)d\epsilon - (n\theta^* + n(1-\theta^*)\frac{p\delta}{k} - \int_{-s}^{n\theta^*(1-\alpha_0^*)-L} F(\epsilon)d\epsilon) = n(1-\theta^*)(1-\frac{p\delta}{k}) - \int_{n\theta^*(1-\alpha_0^*)-L}^{n\theta^*(1-\alpha_0^*)+n(1-\theta^*)(1-\frac{p\delta}{k})-L} F(\epsilon)d\epsilon > 0$. If $n(1-\alpha_0') - L > s$ but $n\theta^*(1-\alpha_0^*) - L \leq -s$, $M' - M^* = L + n\alpha_0' - (n\theta^* + n(1-\theta^*)\frac{p\delta}{k}) = L - n\theta^*(1-\alpha_0^*) > 0$. If $n(1-\alpha_0') - L > s$ but $-s < n\theta^*(1-\alpha_0^*) - L \leq s$, $M' - M^* = L + n\alpha_0' - (n\theta^* + n(1-\theta^*)\frac{p\delta}{k} - \int_{-s}^{n\theta^*(1-\alpha_0^*)} F(\epsilon)d\epsilon) = L - n\theta^*(1-\alpha_0^*) + \int_{-s}^{n\theta^*(1-\alpha_0^*)-L} F(\epsilon)d\epsilon$. Because $-x + \int_{-s}^{x} F(\epsilon)d\epsilon > 0$ for any $x \in [-s, s]$, we have $M' > M^*$. If $n(1-\alpha_0') - L > n\theta^*(1-\alpha_0^*) - L > s$, $M' - M^* = L + n\alpha_0' - (L + n\theta^*\alpha_0^* + n(1-\theta^*)\frac{p\delta}{k}) = 0$. Therefore, $M' \geq M^*$.

$$M' = \begin{cases} n, & \text{if } n(1-\alpha_0') - L < -s, \\ n - \int_{-s}^{n(1-\alpha_0')-L} F(\epsilon)d\epsilon, & \text{if } -s < n(1-\alpha_0') - L < s, \\ L + n\alpha_0', & \text{if } n(1-\alpha_0') - L > s, \end{cases} \tag{A.4}$$

Given $\alpha_0' = \alpha_0^* - (1-\theta^*)(\alpha_0^* - \frac{p\delta}{k})$, we have shown $\theta' = 1$ and $M' \geq M^*$. Since $\frac{dM'}{d\alpha_0'} > 0$, the platform can increase the total matching volume by choosing a peak target greater than α_0', in which case the total matching volume exceeds M^*. This contradicts with that (α_0^*, θ^*) is the optimal outcome. Therefore, $\theta^* = 1$ in equilibrium.

2). *Characterization of* α_0^*

Next, we plug in $\theta^*(\alpha_0) = 1$ and characterize the optimal α_0 by considering the following three cases, which are determined by the relative sizes of demand $(L+\epsilon)$ and supply $(n\theta(1-\alpha_0))$ during the off-peak period.

- *Case 1:* $n(1-\alpha_0) \leq \min\{D(\epsilon)\} = L - s$.

  In this case, the off-peak market is under-supplied, so a qualified driver's utilization during off-peak periods is $Pr^Q(\alpha_0) = 1$. The utility for a qualified driver is $U_q(\alpha_0) = p\delta - \frac{1}{2}k\alpha_0^2$ for any given α_0, while the utility for an unqualified driver is $U_{uq}^Q = \frac{p^2\delta^2}{2k}$. The scheme is effective only if $U_q(\alpha_0) \geq U_{uq}^Q$, that is, $\alpha_0 \leq \sqrt{\frac{p\delta}{k}(2 - \frac{p\delta}{k})}$. Given the total matching volume $M^Q(\alpha_0) = n\theta + n(1-\theta)\alpha_{uq}^Q$, plugging $\theta^*(\alpha_0) = 1$ into the supply constraint during off-peak period, we obtain $\alpha_0 \geq 1 - \frac{L-s}{n}$ in this case. As α_0 needs to satisfy both $\alpha_0 \leq \sqrt{\frac{p\delta}{k}(2 - \frac{p\delta}{k})}$ and $\alpha_0 \geq 1 - \frac{L-s}{n}$, we must have $1 - \frac{L-s}{n} \leq \sqrt{\frac{p\delta}{k}(2 - \frac{p\delta}{k})}$, which is equivalent to $k \leq \frac{p\delta\left(\sqrt{\frac{L-s}{n}\left(2 - \frac{L-s}{n}\right)} + 1\right)}{\left(1 - \frac{L-s}{n}\right)^2} \equiv k_2$. Since $M^Q(\alpha_0) = n$, raising α_0 cannot further improve the matching volume. Therefore, the optimal service target should be set as $1 - \frac{L-s}{n}$, denoted by α_0^Q in the main text. We substitute the values of α_0^Q and $\theta^*(\alpha_0^Q)$ into $U_q(\alpha_0), M^Q(\alpha_0)$, yielding $U_C^Q = p\delta - \frac{1}{2}k(1 - \frac{L-s}{n})^2, M_C^Q = n$ as presented in Lemma 3.

- *Case 2:* $L - s = \min\{D(\epsilon)\} < n(1-\alpha_0) \leq \max\{D(\epsilon)\} = L + s$.

  In this case, the total matching volume is $M^Q(\alpha_0) = n\theta - \int_{-s}^{n\theta(1-\alpha_0)-L} F(\epsilon)d\epsilon + n(1-\theta)\frac{p\delta}{k}$ and the utility of a qualified driver is given by $U_q(\alpha_0) = (1 - \frac{1}{n\theta}\int_{-s}^{n\theta(1-\alpha_0)-L} F(\epsilon)d\epsilon)p\delta - \frac{1}{2}k\alpha_0^2$. By plugging in $\theta^*(\alpha_0) = 1$, we obtain $M^Q(\alpha_0) = n - \int_{-s}^{n(1-\alpha_0)-L} F(\epsilon)d\epsilon$ and $U_q(\alpha_0) = (1 - \frac{1}{n}\int_{-s}^{n(1-\alpha_0)-L} F(\epsilon)d\epsilon)p\delta - \frac{1}{2}k{\alpha_0}^2$. In equilibrium, $U_q(\alpha_0) = U_{uq}^Q$ should hold, so the optimal service target is the solution of the function $(1 - \frac{1}{n}\int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon)p\delta - \frac{1}{2}k{\alpha_0^q}^2 = \frac{p^2\delta^2}{2k}$, which is denoted as α_0^q in the main text. We can further prove that $\frac{dU_q(\alpha_0)}{d\alpha_0} = F[n(1-\alpha_0) - L]p\delta - k\alpha_0 < 0$. Thus, to guarantee the existence of the solution in this case where $1 - \frac{L+s}{n} \leq \alpha_0^q < 1 - \frac{L-s}{n}$, $U_q(1 - \frac{L+s}{n}) \geq U_{uq}^Q > U_q(1 - \frac{L-s}{n})$ must hold. Solving the inequality, we have $k_2 < k \leq \frac{p\delta\left(\sqrt{\frac{L}{n}\left(2 - \frac{L+2s}{n}\right)} - \frac{s}{n} + 1\right)}{\left(1 - \frac{L+s}{n}\right)^2} \equiv k_3$. We then plug α_0^q into $U_q(\alpha_0)$ and $M^Q(\alpha_0)$, yielding $U_C^Q = \frac{p^2\delta^2}{2k}, M_C^Q = n - \int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon$ as presented in Lemma 3.

- *Case 3:* $n(1-\alpha_0) > \max\{D(\epsilon)\} = L + s$.

  In this case, the off-peak market is over-supplied. The total matching volume is $M^Q(\alpha_0) = L + n\theta\alpha_0 + n(1-\theta)\frac{p\delta}{k}$ and the utility for a qualified driver is $U_q(\alpha_0) = (\frac{L}{n\theta} + \alpha_0)p\delta - \frac{1}{2}k\alpha_0^2$, plugging $\theta^*(\alpha_0) = 1$ into which yields $M^Q(\alpha_0) = L + n\alpha_0$ and $U_q(\alpha_0) = (\frac{L}{n} + \alpha_0)p\delta - \frac{1}{2}k{\alpha_0}^2$. Solving $U_q(\alpha_0) \geq U_{uq}^Q$ yields

$\alpha_0 \le \frac{p\delta}{k} + \sqrt{2\frac{L}{n}\frac{p\delta}{k}}$. Since $M^Q(\alpha_0)$ increases in α_0, the optimal service target is $\frac{p\delta}{k} + \sqrt{2\frac{L}{n}\frac{p\delta}{k}}$, denoted by α_0^Q in the main text. To guarantee that α_0^Q falls within the range of this case, α_0^Q must satisfy $n(1-\alpha_0^Q) > L+s$, i.e., $\alpha_0^Q < 1 - \frac{L+s}{n}$, which yields $k > k_3$. Substituting α_0^Q into $U_q(\alpha_0)$ and $M^Q(\alpha_0)$ yields $U_C^Q = \frac{p^2\delta^2}{2k}$ and $M_C^Q = L + n\frac{p\delta}{k} + \sqrt{2nL\frac{p\delta}{k}}$ as presented in Lemma 3. ■

Proof of Lemma 4.

For a prioritized driver, given α_0, θ, and $\hat{\alpha}_p$, the problem is

$$\max_{\alpha_p \in [\alpha_0, 1)} U_p(\alpha_p) = Pr_p(\hat{\alpha}_p) \cdot p\delta(1-\alpha_p) + p\delta\alpha_p - \frac{1}{2}k\alpha_p^2, \tag{A.5}$$

where $Pr_p(\hat{\alpha}_p) = \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)}\} dF(\epsilon)$.

For unprioritized drivers, they can take residual service requests during the off-peak period. Given α_0, $\theta, \hat{\alpha}_p, \hat{\alpha}_{up}$, an unprioritized driver's problem is

$$\max_{\alpha_{up} \in [0, \alpha_0)} U_{up}(\alpha_{up}) = Pr_{up}(\hat{\alpha}_p, \hat{\alpha}_{up}) \cdot p\delta(1-\alpha_{up}) + p\delta\alpha_{up} - \frac{1}{2}k\alpha_{up}^2, \tag{A.6}$$

where $Pr_{up}(\hat{\alpha}_p, \hat{\alpha}_{up}) = \int_{n\theta(1-\hat{\alpha}_p)-L}^{s} \min\{1, \frac{D(\epsilon)-n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}\} dF(\epsilon)$.

We first derive α_{up}^* and prove $\alpha_p^* = \alpha_0$ as shown below. In Equation (A.6), solving FOC yields $\alpha_{up}(\hat{\alpha}_p, \hat{\alpha}_{up}) = \frac{p\delta}{k}(1 - Pr_{up}(\hat{\alpha}_p, \hat{\alpha}_{up}))$. The equilibrium condition requires $\alpha_{up}(\hat{\alpha}_p, \hat{\alpha}_{up}) = \hat{\alpha}_{up}$, yielding $\alpha_{up}^\dagger$, which satisfies $\alpha_{up}^\dagger = \frac{p\delta}{k}(1 - Pr_{up}(\hat{\alpha}_p, \alpha_{up}^\dagger))$. If $\alpha_0 \le \alpha_{up}^\dagger$, the peak target is so small that all drivers will be equally prioritized, and the prioritization scheme becomes ineffective. Thus, the service target $\alpha_0 > \alpha_{up}^\dagger$ should hold, and the unprioritized drivers' optimal working amount in peak period is $\alpha_{up}^* = \alpha_{up}^\dagger$. In the following, we restrict our focus on the case that the unprioritized driver's problem has an interior solution $\alpha_{up}^* = \alpha_{up}^\dagger$ $(< \alpha_0)$. Then, we show that $\alpha_p^* = \alpha_0$. Note that $Pr_p(\hat{\alpha}_p) = \int_{-s}^{n\theta(1-\hat{\alpha}_p)-L} \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)} dF(\epsilon) + \int_{n\theta(1-\hat{\alpha}_p)-L}^{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})-L} dF(\epsilon) + \int_{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})-L}^{s} dF(\epsilon) > \int_{n\theta(1-\hat{\alpha}_p)-L}^{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})-L} \frac{D(\epsilon)-n\theta(1-\hat{\alpha}_0)}{n(1-\theta)(1-\hat{\alpha}_{up})} dF(\epsilon) + \int_{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})-L}^{s} dF(\epsilon) = Pr_{up}(\hat{\alpha}_p, \hat{\alpha}_{up})$. From (A.5), since $\frac{dU_p(\alpha_p)}{d\alpha_p} = \Big(1 - Pr_p(\hat{\alpha}_p)\Big)p\delta - k\alpha_p \le \Big(1 - Pr_p(\hat{\alpha}_p)\Big)p\delta - k\alpha_0 < \Big(1 - Pr_p(\hat{\alpha}_p)\Big)p\delta - k\alpha_{up}^* = \Big(Pr_{up}(\hat{\alpha}_p, \alpha_{up}^*) - Pr_p(\hat{\alpha}_p)\Big)p\delta < 0$, we have $\alpha_p^* = \alpha_0$.

Plugging in $\alpha_p^* = \alpha_0$, the platform's problem becomes

$$\begin{aligned} \max_{\alpha_0 \in (0,1)} M(\alpha_0) = \max_{\alpha_0 \in (0,1)} \Big(& \int_{-s}^{s} \min\{D(\epsilon), n\theta(1-\alpha_0) + n(1-\theta)(1-\alpha_{up}^*)\} dF(\epsilon) \\ & + n\theta\alpha_0 + n(1-\theta)\alpha_{up}^* \Big), \\ s.t.\ U_p^*(\alpha_0) \ge U_{up}^*(\alpha_{up}^*), & \text{with the constraint binding if } \theta < 1. \end{aligned}$$

In the following, we first prove that given the service target α_0, all drivers participate the qualification scheme, i.e., $\theta^*(\alpha_0) = 1$. Then, we substitute $\theta^*(\alpha_0) = 1$ and characterize the optimal decision on the service target α_0.

1). *Proof of* $\theta^*(\alpha_0) = 1$

For simplicity, we will use θ^* to represent $\theta^*(\alpha_0)$ hereafter. Suppose (α_0^*, θ^*) is the optimal outcome, where $\theta^* \in (0,1)$ and $\alpha_0^* \in (\alpha_{up}^*, 1)$. Then, the total matching volume is shown in Equation (A.7), and subject to $U_p^* = U_{up}^*$, i.e., $Pr_p^* p\delta(1-\alpha_0^*) + p\delta\alpha_0^* - \frac{1}{2}k{\alpha_0^*}^2 = (1-Pr_{up}^*)^2\frac{p^2\delta^2}{2k} + p\delta Pr_{up}^*$, where $\alpha_0^* > \alpha_{up}^* = \frac{p\delta}{k}(1-Pr_{up}^*)$, $Pr_p^* = \int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n\theta^*(1-\alpha_0^*)}\}dF(\epsilon)$ and $Pr_{up}^* = \int_{n\theta^*(1-\alpha_0^*)-L}^{s}\min\{1, \frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)}\}dF(\epsilon)$.

$$M^* = \begin{cases} n, & \text{if } n\theta^*(1-\alpha_0^*) + n(1-\theta^*)(1-\alpha_{up}^*) - L < -s, \\ n - \int_{-s}^{n(1-\alpha_0^*)+n(1-\theta^*)(\alpha_0^*-\alpha_{up}^*)-L} F(\epsilon)d\epsilon, & \text{if } -s < n\theta^*(1-\alpha_0^*) + n(1-\theta^*)(1-\alpha_{up}^*) - L < s, \\ L + n\theta^*\alpha_0^* + n(1-\theta^*)\alpha_{up}^*, & \text{if } n\theta^*(1-\alpha_0^*) + n(1-\theta^*)(1-\alpha_{up}^*) - L > s, \end{cases} \tag{A.7}$$

Let $\alpha_0' = \alpha_0^* - (1-\theta^*)(\alpha_0^* - \alpha_{up}^*)$. We first prove $\theta' = \theta|_{\alpha_0' = \alpha_0^* - (1-\theta^*)(\alpha_0^* - \alpha_{up}^*)} = 1$. That is, all drivers participate in the scheme given the peak target α_0'. In the following, we prove this by showing $U_p' > U_{up}'$ in two steps.

- First, we prove $U_{up}^* > U_{up}'$. Note that $\frac{D(\epsilon)}{n(1-\alpha_0')} > \frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)}$ as long as $D(\epsilon) < n(1-\alpha_0')$, thus $Pr_p' = \int_{-s}^{n(1-\alpha_0')-L}\frac{D(\epsilon)}{n(1-\alpha_0')}dF(\epsilon) + \int_{n(1-\alpha_0')-L}^{s}dF(\epsilon) > \int_{n\theta^*(1-\alpha_0^*)-L}^{n(1-\alpha_0')-L}\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)}dF(\epsilon) + \int_{n(1-\alpha_0')-L}^{s}dF(\epsilon) = Pr_{up}^*$. We also know that $Pr_{up}^* = \int_{n\theta^*(1-\alpha_0^*)-L}^{n(1-\alpha_0')-L}\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)}dF(\epsilon) + \int_{n(1-\alpha_0')-L}^{s}dF(\epsilon) > \int_{n(1-\alpha_0')-L}^{s}dF(\epsilon) = Pr_{up}'$. Therefore, $Pr_p' > Pr_{up}^* > Pr_{up}'$. An unprioritized driver obtains utility $U_{up} = (1-Pr_{up})^2\frac{p^2\delta^2}{2k} + p\delta Pr_{up}$ by working $\alpha_{up} = \frac{p\delta}{k}(1-Pr_{up})$ during peak period, which increases in Pr_{up}, so we have $U_{up}^* > U_{up}'$. Thus, we have proved $U_{up}^* > U_{up}'$.
- Next, we prove $U_p' > U_{up}^*$. To prove $U_p' > U_{up}^*$ (i.e., $Pr'p\delta(1-\alpha_0') + p\delta\alpha_0' - \frac{1}{2}k{\alpha_0'}^2 > Pr_{up}^* p\delta(1-\alpha_{up}^*) + p\delta\alpha_{up}^* - \frac{1}{2}k{\alpha_{up}^*}^2$), it is equivalent to prove $(\alpha_0' - \alpha_{up}^*)^2 < 2\frac{p\delta}{k}(Pr' - Pr_{up}^*)(1-\alpha_0')$. For the left-hand side of the inequality, $(\alpha_0' - \alpha_{up}^*)^2 = (\theta^*)^2(\alpha_0^* - \alpha_{up}^*)^2 = (\theta^*)^2(2\frac{p\delta}{k}(Pr_p^* - Pr_{up}^*)(1-\alpha_0^*))$, where the second equality is derived from $U_p^* = U_{up}^*$ (i.e., $Pr_p^* p\delta(1-\alpha_0^*) + p\delta\alpha_0^* - \frac{1}{2}k{\alpha_0^*}^2 = Pr_{up}^* p\delta(1-\alpha_{up}^*) + p\delta\alpha_{up}^* - \frac{1}{2}k{\alpha_{up}^*}^2$). Therefore, proving $U_p' > U_{up}^*$ is equivalent to prove $(Pr_p^* - Pr_{up}^*)(1-\alpha_0^*)\theta^{*2} < (Pr' - Pr_{up}^*)(1-\alpha_0')$. Note that $(Pr' - Pr_{up}^*)(1-\alpha_0') = (1-\alpha_0')\frac{\int_{n\theta^*(1-\alpha_0^*)-L}^{n(1-\alpha_0')-L}F(\epsilon)d\epsilon}{n(1-\theta^*)(1-\alpha_{up}^*)} - \frac{\int_{-s}^{n(1-\alpha_0')-L}F(\epsilon)d\epsilon}{n} = (\theta^*(1-\alpha_0^*) + (1-\theta^*)(1-\alpha_{up}^*))\frac{\int_{n\theta^*(1-\alpha_0^*)-L}^{n(1-\alpha_0')-L}F(\epsilon)d\epsilon}{n(1-\theta^*)(1-\alpha_{up}^*)} - \frac{\int_{-s}^{n(1-\alpha_0')-L}F(\epsilon)d\epsilon}{n} = \theta^*(1-\alpha_0^*)\frac{\int_{n\theta^*(1-\alpha_0^*)-L}^{n(1-\alpha_0')-L}F(\epsilon)d\epsilon}{n(1-\theta^*)(1-\alpha_{up}^*)} - \frac{\int_{-s}^{n\theta^*(1-\alpha_0^*)-L}F(\epsilon)d\epsilon}{n}$ and $(Pr_p^* - Pr_{up}^*)(1-\alpha_0^*)\theta^{*2} = \theta^*\left(\theta^*(1-\alpha_0^*)\frac{\int_{n\theta^*(1-\alpha_0^*)-L}^{n(1-\alpha_0')-L}F(\epsilon)d\epsilon}{n(1-\theta^*)(1-\alpha_{up}^*)} - \frac{\int_{-s}^{n\theta^*(1-\alpha_0^*)-L}F(\epsilon)d\epsilon}{n}\right) < (Pr' - Pr_{up}^*)(1-\alpha_0')$. Thus, we have proved $U_p' > U_{up}^*$.

Therefore, $U_p' > U_{up}'$ due to $U_{up}^* > U_{up}'$ and $U_p' > U_{up}^*$. Since the utility of participating in the scheme is strictly greater than the utility of not participating, given the peak target α_0', all drivers will choose to participate in the scheme. That is, $\theta' = 1$.

Given the peak target $\alpha_0' = \alpha_0^* - (1-\theta^*)(\alpha_0^* - \alpha_{up}^*)$ and $\theta' = 1$, the total matching volume of M' is shown in Equation (A.8), and subject to $U_p' \geq U_{up}'$, i.e., $Pr_p' p\delta(1-\alpha_0') + p\delta\alpha_0' - \frac{1}{2}k{\alpha_0'}^2 \geq (1-Pr_{up}')^2\frac{p^2\delta^2}{2k} + p\delta Pr_{up}'$, where $\alpha_0' > \frac{p\delta}{k}(1-Pr_{up}')$, $Pr_p' = \int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n(1-\alpha_0')}\}dF(\epsilon)$ and $Pr_{up}' = \int_{n(1-\alpha_0')-L}^{s}dF(\epsilon)$. We can easily show $M' = M^*$.

$$M' = \begin{cases} n, & \text{if } n(1-\alpha_0') - L < -s, \\ n - \int_{-s}^{n(1-\alpha_0')-L} F(\epsilon)d\epsilon, & \text{if } -s < n(1-\alpha_0') - L < s, \\ L + n\alpha_0', & \text{if } n(1-\alpha_0') - L > s, \end{cases} \tag{A.8}$$

Given $\alpha_0' = \alpha_0^* - (1-\theta^*)(\alpha_0^* - \alpha_{up}^*)$, we have shown $\theta' = 1$ and $M' = M^*$. Since $\frac{dM'}{d\alpha_0'} > 0$, the platform can increase the matching volume by choosing a peak target greater than α_0', in which case the total

matching volume exceeds M^*. This contradicts with that (α_0^*, θ^*) is the optimal outcome. Therefore, $\theta^* = 1$ in equilibrium.

2). *Characterization of* α_0^*

Next, we plug in $\theta^*(\alpha_0) = 1$ and characterize the optimal α_0 by considering the following three cases, which are determined by the relative sizes of demand $(L+\epsilon)$ and supply $(n\theta(1-\alpha_0))$ during the off-peak period.

- *Case 1:* $n(1-\alpha_0) \leq \min\{D(\epsilon)\} = L - s$.

  In this case, $Pr_{up} = \int_{n(1-\alpha_0)-L}^{s} dF(\epsilon) = 1$, so $U_{up}^* = p\delta \geq U_p(\alpha_0)$ for any α_0 that satisfies $n(1-\alpha_0) \leq L - s$. Therefore, *Case 1* will not occur in equilibrium.

- *Case 2:* $L - s = \min\{D(\epsilon)\} < n(1-\alpha_0) \leq \max\{D(\epsilon)\} = L + s$.

  In this case, an unprioritized driver will have a probability of $Pr_{up} = \int_{n(1-\alpha_0)-L}^{s} dF(\epsilon) = 1 - F[n(1-\alpha_0) - L]$ to receive service requests during the off-peak period and optimally obtain $U_{up}^* = p\delta(1 - F[n(1-\alpha_0) - L]) + (F[n(1-\alpha_0) - L])^2 \frac{p^2\delta^2}{2k}$ by working $\alpha_{up}^* = \frac{p\delta}{k} F[n(1-\alpha_0) - L]$ during the off-peak period. Note that $U_p(\alpha_0) = (1 - \frac{1}{n}\int_{-s}^{n(1-\alpha_0)-L} F(\epsilon)d\epsilon)p\delta - \frac{1}{2}k\alpha_0^2$, and $\frac{dU_p(\alpha_0)}{d\alpha_0} = p\delta F[n(1-\alpha_0) - L] - k\alpha_0 < p\delta F[n(1-\alpha_0) - L] - k\alpha_{up}^* = 0$, while $\frac{dU_{up}^*}{d\alpha_0} = np\delta(1 - \frac{p\delta}{k} F[n(1-\alpha_0) - L])F'[n(1-\alpha_0) - L] > 0$, thus $U_p(\alpha_0) - U_{up}^* = 0$ decreases in α_0 and has at most one unique solution. Solving $U_p(\alpha_0) - U_{up}^* \geq 0$ leads to $\alpha_0 \leq \alpha_0^p$, where α_0^p satisfies $(1 - \frac{1}{n}\int_{-s}^{n(1-\alpha_0^p)-L} F(\epsilon)d\epsilon)p\delta - \frac{1}{2}k\alpha_0^{p2} = p\delta(1 - F[n(1-\alpha_0^p) - L]) + (F[n(1-\alpha_0^p) - L])^2 \frac{p^2\delta^2}{2k}$. The equilibrium occurs in this case only if the following two conditions are satisfied: (i) $U_p(\alpha_0) > U_{up}^*$ when $\alpha_0 = 1 - \frac{L+s}{n}$ and (ii) $U_p(\alpha_0) < U_{up}^*$ when $\alpha_0 = 1 - \frac{L-s}{n}$, which leads to $k \leq k_3$. So if $k \leq k_3$, $\alpha_0^* = \alpha_0^p$.

- *Case 3:* $n(1-\alpha_0) > \max\{D(\epsilon)\} = L + s$.

  In this case, there is no residual demand. Unprioritized drivers will not receive service requests during the off-peak period. Therefore, this case is equivalent to that under qualification scheme. Therefore, the analysis is the same as the *Case 3* in the proof of Lemma 3.

We substitute α_0^*, denoted by α_C^P, into $U^P(\alpha_0)$ and $M^P(\alpha_0)$, and obtain U_C^P and M_C^P, respectively. The results are provided in Lemma 4. ■

Proof of Proposition 1.

By comparing k_1 and k_3, we obtain $k_1 < k_3$. Next, we compare the performances under different scenarios.

1). *Qualification scheme v.s. Benchmark*

The following comparison is based on Lemma 1 and Lemma 3.

Peak supply: since $\alpha_C^B < \frac{p\delta}{k}$ and $\alpha_0^Q > \frac{p\delta}{k}$, we have $\alpha_0^Q > \alpha_C^B$.

Driver utility: we show $U_C^B > U_C^Q$ below.

- If $k \leq k_1$, $U_C^B = (1 - \frac{\int_{-s}^{n(1-\alpha^b)-L} F(\epsilon)d\epsilon}{n})p\delta - \frac{1}{2}k\alpha^{b2} = U_C^B(\alpha^b) > U_C^B(1 - \frac{L-s}{n}) \geq U_C^Q$, where $\alpha^b \in (1 - \frac{L+s}{n}, 1 - \frac{L-s}{n})$.

- If $k_1 < k \le k_2$, $U_C^B = \frac{-k^2+2\frac{kL}{n}p\delta+2\delta kp+\delta^2p^2+(k-p\delta)\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{4k} > U_C^B(\alpha^b) > U_C^B(1-\frac{L-s}{n}) \ge U_C^Q$.
- If $k_1 < k \le k_2$, $U_C^B = \frac{-k^2+2\frac{kL}{n}p\delta+2\delta kp+\delta^2p^2+(k-p\delta)\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{4k} > U_C^B(\alpha^b) > U_C^B(1-\frac{L-s}{n}) \ge U_C^Q$.
- If $k > k_2$, $U_C^B = \frac{-k^2+2\frac{kL}{n}p\delta+2\delta kp+\delta^2p^2+(k-p\delta)\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{4k} > \frac{p^2\delta^2}{2k} = U_C^Q$.

Total matching volume: we show $M_C^B < M_C^Q$ below.

- If $k \le k_1$, $M_C^B = n - \int_{-s}^{n(1-\alpha^b)-L} F(\epsilon)d\epsilon < n - \int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon \le M_C^Q$ due to $\alpha_0^q > \alpha^b$.
- If $k_1 < k \le k_2$, $M_C^B = L + n\frac{k+p\delta-\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{2k} < n = M_C^Q$.
- If $k_2 < k \le k_3$, $M_C^B = L + n\frac{k+p\delta-\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{2k} < n - s < n - \int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon = M_C^Q$.
- If $k > k_3$, $M_C^B = L + n\frac{k+p\delta-\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{2k} < L + n\frac{p\delta}{k} + \sqrt{2nL\frac{p\delta}{k}} = M_C^Q$.

2). *Prioritization scheme v.s. Benchmark*

The following comparison is based on Lemma 1 and Lemma 4.

Peak supply: we show $\alpha_0^p > \alpha_C^B$ below.

- If $k \le k_1$, note $\alpha_0^p > \frac{p\delta}{k}F[n(1-\alpha_0^p)-L]$ and $\alpha^b = \frac{p\delta}{k}\frac{\int_{-s}^{n(1-\alpha^b)-L}F(\epsilon)d\epsilon}{n(1-\alpha^b)} < \frac{p\delta}{k}F[n(1-\alpha^b)-L]$, so we have $\alpha_0^p > \alpha^b$.
- If $k_1 < k \le k_3$, $\alpha_C^B = \frac{k+p\delta-\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{2k} < 1 - \frac{L+s}{n} < \alpha^b < \alpha_0^p$.
- If $k > k_3$, $\alpha_0^p > \frac{p\delta}{k} > \alpha_C^B$.

Driver utility: we show $U_C^B > U_C^P$ below.

- If $k \le k_1$, note $\alpha^b = \frac{p\delta}{k}\frac{\int_{-s}^{n(1-\alpha^b)-L}F(\epsilon)d\epsilon}{n(1-\alpha^b)}$, so $(1-\frac{1}{n}\int_{-s}^{n(1-\alpha)-L}F(\epsilon)d\epsilon)p\delta - \frac{1}{2}k\alpha^2 = (1-\frac{k}{p\delta}\alpha^b(1-\alpha^b))p\delta - \frac{1}{2}k\alpha^2 = p\delta - k\alpha^b + \frac{1}{2}k{\alpha^b}^2$; note $U_C^P = \frac{p\delta(2kPr+(1-Pr)^2p\delta)}{2k} = \frac{p\delta(2k(1-F)+F^2p\delta)}{2k} = p\delta - p\delta F + \frac{p^2\delta^2}{2k}F^2$, where $F = [n(1-\alpha_0^p)-L]$. Because $\frac{p\delta F}{k} > a^b$, $p\delta - k\alpha^b + \frac{1}{2}k{\alpha^b}^2 - (p\delta - p\delta F + \frac{p^2\delta^2}{2k}F^2) = p\delta F - k\alpha^b - \frac{1}{2k}((p\delta F)^2 - k{\alpha^b}^2) = (p\delta F - k\alpha^b)(1-\frac{p\delta F+k\alpha^b}{2k}) > 0$, so $U_C^B > U_C^P$.
- If $k > k_3$, $U_C^B = \frac{-k^2+2\delta k(L+1)p+(k-p\delta)\sqrt{(k-p\delta)^2+4kLp\delta}+\delta^2p^2}{4k} > \frac{p^2\delta^2}{2k} = U_C^P$.

Total matching volume: we show $M_C^B < M_C^P$ below.

- If $k \le k_1$, $M_C^B < M_C^P$ due to $\alpha^b < \alpha_0^p$.
- If $k_1 < k \le k_3$, $M_C^B = L + n\frac{k+p\delta-\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{2k} < n - s < n - \int_{-s}^{n(1-\alpha_0^p)-L} F(\epsilon)d\epsilon = M_C^P$.
- If $k > k_3$, $M_C^B = L + n\frac{k+p\delta-\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta}}{2k} < L + n\frac{p\delta}{k} + \sqrt{2nL\frac{p\delta}{k}} = M_C^P$.

3). *Prioritization scheme v.s. Qualification scheme*

The following comparison is based on Lemma 3 and Lemma 4.

Peak supply: since α_0^p satisfies $(1-\frac{1}{n}\int_{-s}^{n(1-\alpha_0)-L}F(\epsilon)d\epsilon)p\delta - \frac{1}{2}k\alpha_0^2 = U^*$, where $U^* = \frac{p\delta(2k\int_{n(1-\alpha_0)-L}^{s}dF(\epsilon)+(1-\int_{n(1-\alpha_0)-L}^{s}dF(\epsilon))^2p\delta)}{2k} \ge \frac{p^2\delta^2}{2k}$, we have $\alpha_0^q > \alpha_0^p$. Thus, $\alpha_0^Q > \alpha_0^p$.

Driver utility: we show $U_C^Q \le U_C^P$ below.

- If $k \le k_2$, $U_C^Q = U_p(1-\frac{L-s}{n}) < U_p(\alpha_0^p) = U_C^P$.
- If $k_2 < k \le k_3$, $U_C^Q = \frac{p^2\delta^2}{2k} < U_C^P$.
- If $k > k_3$, $U_C^Q = U_C^P$.

Total matching volume: since $\alpha_0^q > \alpha_0^p$, $M_C^Q \ge M_C^P$ always holds. ■

Proof of Lemma 5.

Note that $U_O^B(\alpha) = p\delta\alpha - \frac{1}{2}k\alpha^2$. Using FOC to solve for the optimal α, we have $\alpha_O^B = \frac{p\delta}{k}$, $U_O^B = \frac{p^2\delta^2}{2k}$, and $M_O^B = \int_{-s}^{s} D(\epsilon)dF(\epsilon) + n\alpha_O^B = L + n\frac{p\delta}{k}$. ■

Proof of Proposition 2.

The following comparison is based on Lemma 1 and Lemma 5.

Peak supply in an open system is greater than that in a closed system, since $\alpha_O^B = \frac{p\delta}{k} > \alpha_C^B$.

Total matching volume is higher under an open system compared to a closed system, as shown below. The total matching volume in an open system is given by $M_O^B = L + n\frac{p\delta}{k}$, while in a closed system it is

$$M_C^B = \begin{cases} n - \int_{-s}^{n(1-\alpha^b)-L} F(\epsilon)d\epsilon, & \text{if } k \le k_1 \\ L + \frac{n(k+p\delta-\sqrt{(k-p\delta)^2+4\frac{kL}{n}p\delta})}{2k}, & \text{if } k > k_1 \end{cases}.$$

If $k \le k_1$, $M_O^B - M_C^B = \int_{-s}^{s} D(\epsilon)dF(\epsilon) + n\frac{p\delta}{k} - \int_{-s}^{s} \min\{D(\epsilon), n(1-\alpha^b)\}dF(\epsilon) - n\alpha^b > n(\frac{p\delta}{k} - \alpha^b) > 0$. If $k > k_1$, $M_O^B = L + n\frac{p\delta}{k} > L + \frac{n(k+p\delta-\sqrt{(k-p\delta)^2+4kLp\delta})}{2k} = M^B$. Thus, we have proved $M_O^B > M_C^B$.

Driver utility is higher in a closed system, since $U_C^B > \frac{p^2\delta^2}{2k} > U_O^B$. ■

Proof of Lemma 6.

In an open system, unprioritized drivers face the same problem as unqualified drivers because they will not receive any service request during the off-peak period due to the infinite number of part-time drivers in the off-peak period. The problem of the prioritized driver is also the same as the qualified driver in a closed system (see Lemma 3). Therefore, when full-time drivers' utilization is less than 1, i.e., $k > k_2$, the prioritization scheme in an open system is equivalent to the qualification scheme in a closed system. However, by Lemma 3, when $k \le k_2$, the optimal peak target is binding to $1 - \frac{L-s}{n}$ in closed system (i.e., $\alpha_0^Q = 1 - \frac{L-s}{n}$) because raising peak target cannot improve the total matching volume anymore, while in an open system, the total matching volume is $M_O^P(\alpha_0) = L + n\alpha_0$, which will further increase as the peak target α_0 increases. Therefore, when $k \le k_2$, in an open system with the prioritization scheme, the platform can further increase the peak target until the prioritized driver obtains the same utility as the unprioritized driver, i.e., $p\delta - \frac{1}{2}k\alpha_0^2 = \frac{p^2\delta^2}{2k}$. On solving this equation, we have $\alpha_0^* = \sqrt{\frac{p\delta}{k}(2 - \frac{p\delta}{k})}$. The equilibrium peak supply, denoted by ${\alpha_0}_O^P$, is ${\alpha_0}_O^P = \begin{cases} \sqrt{\frac{p\delta}{k}(2-\frac{p\delta}{k})}, & \text{if } k \le k_2 \\ \alpha_0^q, & \text{if } k_2 < k \le k_3 \\ \frac{p\delta}{k} + \sqrt{2\frac{L}{n}\frac{p\delta}{k}}, & \text{if } k > k_3 \end{cases}$, $U_C^P = \frac{p^2\delta^2}{2k}$, and $M_O^P = L + n{\alpha_0}_O^P$. ■

Proof of Proposition 3.

1). *Qualification scheme v.s. Benchmark*

The following comparison is based on Lemma 3 and Lemma 5.

To compare the *total matching volume*, we formulate the difference between the matching volume in an open system and a closed system as shown below:

$$\Delta M(k) = M_O^Q - M_O^B = \begin{cases} n(1-\frac{p\delta}{k}) - L, & \text{if } k \le k_2, \\ n(1-\frac{p\delta}{k}) - L - \int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon, & \text{if } k_2 < k \le k_3, \\ \sqrt{2nL\frac{p\delta}{k}}, & \text{if } k > k_3. \end{cases}$$

It can be shown that $k_3 > \frac{p\delta}{1-\frac{L+s}{n}}$. We next analyze $\Delta M(k)$ based on the following two cases.

Case 1: $\frac{p\delta}{k} < 1 - \frac{L+s}{n}$. This case is equivalent to $k > \frac{p\delta}{1-\frac{L+s}{n}}$. In this case, we can show $\Delta M(k) > 0$ as follows.

- If $\frac{p\delta}{1-\frac{L+s}{n}} < k \le k_2$, $\Delta M(k) > n(1-(1-\frac{L+s}{n})) - L = s > 0$;
- If $k_2 < k \le k_3$, $\Delta M(k) = n(1-\frac{p\delta}{k}) - L - \int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon > n(1-\alpha_0^q) - L - \int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon > 0$;
- If $k > k_3$, $\Delta M(k) > 0$. $M_O^Q > M_O^B$ always holds.

Case 2: $\frac{p\delta}{k} \ge 1 - \frac{L+s}{n}$

In this case, $\Delta M(k) = \begin{cases} n(1-\frac{p\delta}{k}) - L, & \text{if } k \le k_2 \\ n - \int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon - L - n\frac{p\delta}{k}, & \text{if } k_2 < k \le \frac{p\delta}{1-\frac{L+s}{n}} \end{cases}$.

- If $k \le k_2$, solving $\Delta M(k) \ge (<) 0$ yields $k \ge (<) \frac{p\delta}{1-L/n}$.
- If $k_2 < k \le \frac{p\delta}{1-\frac{L+s}{n}}$, the optimal peak target α_0^q solves $(1 - \frac{1}{n}\int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon)p\delta - \frac{1}{2}k\alpha_0^{q2} = \frac{p^2\delta^2}{2k}$. By taking the derivative of this equation with respect to k and using the implicit differentiation method, we obtain $\frac{d\alpha_0^q}{dk} = \frac{1}{2}\frac{\alpha_0^{q2} - \frac{p^2\delta^2}{k^2}}{p\delta F[n(1-\alpha_0^q)-L] - k\alpha_0^q}$. Based on this, we have $\frac{d\Delta M(k)}{dk} = \frac{\partial \Delta M(k)}{\partial k} + \frac{\partial \Delta M(k)}{\partial \alpha_0^q}\frac{d\alpha_0^q}{dk} = n\frac{2\alpha_0^q\frac{p\delta}{k} - (\alpha_0^{q2} + \frac{p^2\delta^2}{k^2})F[n(1-\alpha_0^q)-L]}{2k(\alpha_0^q - \frac{p\delta}{k}F[n(1-\alpha_0^q)-L])}$. First consider $k = k_2$. In this case, we have $\alpha_0^q = 1 - \frac{L-s}{n}$ and $F[n(1-\alpha_0^q)-L] = 0$, $\frac{d\Delta M(k)}{dk} = \frac{np\delta}{k^2} > 0$, Next consider $k = \frac{p\delta}{1-\frac{L+s}{n}}$. In this case, we have $\alpha_0^q \in (1-\frac{L+s}{n}, 1-\frac{L-s}{n})$ and $F[n(1-\alpha_0^q)-L] \in (0,1)$, so $\frac{d\Delta M(k)}{dk}$ may be negative. Thus, $\Delta M(k)$ either always increases or first increases then decreases in $k \in (k_2, \frac{p\delta}{1-\frac{L+s}{n}}]$.

  Note that $\Delta M(\frac{p\delta}{1-\frac{L+s}{n}}) = n - \int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon - L - \frac{np\delta}{\frac{p\delta}{1-\frac{L+s}{n}}} = s - \int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon)p\delta > 0$, $\Delta M(k_2) = n(1-\frac{p\delta}{k_2}) - L \ge (<) 0$ if $s \le (>)\sqrt{n^2-L^2} + L - n$. Thus, if $s \le \sqrt{n^2-L^2} + L - n$, $\Delta M \ge 0$. If $s > \sqrt{n^2-L^2} + L - n$, $\Delta M = 0$ has a unique solution, denoted by $\tilde{k}$, which satisfies $n - \int_{-s}^{n(1-\tilde{\alpha}_0^q)-L} F(\epsilon)d\epsilon - L - n\frac{p\delta}{\tilde{k}} = 0$, where $\tilde{\alpha}_0^q > \frac{p\delta}{\tilde{k}}$ and $\tilde{\alpha}_0^q$ solves $(1 - \frac{1}{n}\int_{-s}^{n(1-\tilde{\alpha}_0^q)-L} F(\epsilon)d\epsilon)p\delta - \frac{1}{2}\tilde{k}\tilde{\alpha}_0^{q2} = \frac{p^2\delta^2}{2\tilde{k}}$. Therefore, $M_O^Q \ge M_O^B$ if $k_4 < k < \frac{p\delta}{1-\frac{L+s}{n}}$, where $k_4 = \begin{cases} \frac{p\delta}{1-L/n}, & \text{if } s \le \sqrt{n^2-L^2}+L-n \\ \tilde{k}, & \text{if } s > \sqrt{n^2-L^2}+L-n \end{cases}$.

Combining the results from the above two cases, we have proved $M_O^Q > M_O^B$ if $k > k_4$ and $M_O^Q \le M_O^B$ if $k \le k_4$.

For the *driver utility*, we compare $U_O^B = \frac{p^2\delta^2}{2k}$ and $U_C^Q = \begin{cases} p\delta - \frac{1}{2}k(1-\frac{L-s}{n})^2, & \text{if } k \le k_2, \\ \frac{p^2\delta^2}{2k}, & \text{if } k > k_2. \end{cases}$. If $k \le k_2$, we can easily prove $U_O^B = \frac{p^2\delta^2}{2k} \le U_C^Q$. If $k > k_2$, we have $U_O^B = U_C^Q$.

2). *Prioritization scheme v.s. Benchmark*

The following comparison is based on Lemma 5 and Lemma 6.

To compare the *total matching volume*, we formulate the difference between the prioritization scheme and the benchmark scenario in an open system as shown below, which is always positive, so $M_O^P > M_O^B$

always holds.

$$M_O^P - M_O^B = \begin{cases} n(\sqrt{\frac{p\delta}{k}(2-\frac{p\delta}{k})} - \frac{p\delta}{k}), & \text{if } k \le k_2 \\ n(\alpha_0^q - \frac{p\delta}{k}), & \text{if } k_2 < k \le k_3 \\ n\sqrt{2\frac{L}{n}\frac{p\delta}{k}}, & \text{if } k > k_3 \end{cases} \cdot$$

The *driver utility* under the prioritization scheme and the benchmark scenario is equal, since $U_O^P = U_O^B = \frac{p^2\delta^2}{2k}$.

3). *Qualification scheme v.s. Prioritization scheme*

The following comparison is based on Lemma 3 and Lemma 6.

Total matching volume: we prove that $M_O^P > M_O^Q$ if $k < k_3$, and $M_O^P = M_O^Q$ if $\ge k_3$.

- If $k \le k_2$, we can show that $M_O^P - M_O^Q = L + n\sqrt{\frac{p\delta}{k}(2-\frac{p\delta}{k})} - n > 0$;
- If $k_2 < k < k_3$, $M_O^P - M_O^Q = \int_{-s}^{s} D(\epsilon)d\epsilon + n\alpha_0^q - (\int_{-s}^{s} \min\{D(\epsilon), n(1-\alpha_0^q)\}d\epsilon + n\alpha_0^q) > 0$;
- If $k \ge k_3$, $M_O^P - M_O^Q = 0$.

Driver utility: we prove that $U_O^P < U_C^Q$ if $k < k_2$ and $U_O^P = U_C^Q$ if $k \ge k_2$.

- If $k < k_2$, $U_O^P - U_C^Q = \frac{p^2\delta^2}{2k} - (p\delta - \frac{1}{2}k(1-\frac{L-s}{n})^2) < 0$;
- If $k \ge k_2$, $U_O^P - U_C^Q = 0$. ■

Appendix B. Incentive Scheme vs. Surge Pricing

Surge pricing in a closed system. Under surge pricing, the ride-hailing platform can adjust the price multiplier σ to shift the supply from the off-peak period to the peak period. If the price multiplier is not very high (i.e., $D_p - \sigma p \ge n\hat{\alpha}$), then each driver can receive ride request with certainty during the peak period. If the price multiplier is sufficiently high (i.e., $D_p - \sigma p < n\hat{\alpha}$), then each driver can receive service requests with a probability of $\frac{D_p - \sigma p}{n\hat{\alpha}}$ during the peak period. Therefore, a driver's utility from working α during the peak period is $\min\{1, \frac{D_p-\sigma p}{n\hat{\alpha}}\}\sigma p\delta\alpha - \frac{1}{2}k\alpha^2$.

Under surge pricing, given the peak price multiplier σ, an individual driver's problem is $\max_{\alpha\in[0,1]} U_C^{SP}(\alpha)$, where $U_C^{SP}(\alpha) = \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n(1-\hat{\alpha})}\}dF(\epsilon) \cdot p\delta(1-\alpha) + \min\{1, \frac{D_p-\sigma p}{n\hat{\alpha}}\} \cdot \sigma p\delta\alpha - \frac{1}{2}k\alpha^2$. The total matching volume is $M_C^{SP}(\alpha) = \int_{-s}^{s} \min\{D(\epsilon), n(1-\hat{\alpha})\}dF(\epsilon) + \min\{D_p - \sigma p, n\hat{\alpha}\}$. The platform profit is $\pi_C^{SP}(\sigma) = \int_{-s}^{s} \min\{D(\epsilon), n(1-\hat{\alpha})\}dF(\epsilon) \cdot p(1-\delta) + \min\{D_p - \sigma p, n\hat{\alpha}\} \cdot \sigma p(1-\delta)$. Maximizing $\pi_C^{SP}(\sigma)$ yields the equilibrium results in Lemma A.1. The proof of Lemma A.1 can be found in SA.A of the Supplementary Analysis.

LEMMA A.1. *In a closed system, when the platform implements surge pricing, in equilibrium,*

- *peak working time:*

(1) when $D_p \ge p\sigma + n$,

$$\alpha_C^{SP*} = \begin{cases} \frac{k+\sigma p\delta - \sqrt{(k-\sigma p\delta)^2 + \frac{4kL}{n}p\delta}}{2k} \equiv \alpha_{C1}, & \text{if } k \ge \frac{np\delta}{D_p - p\sigma}(\sigma - \frac{L}{n-(D_p-p\sigma)}) \equiv k_2^s, \\ \alpha_{C2}, & \text{if } \frac{np\delta(\sigma-1)}{n-L+s} \equiv k_4^s \le k < k_2^s, \\ \frac{p\delta(\sigma-1)}{k} \equiv \alpha_{C3}, & \text{if } p\delta(\sigma-1) \le k < k_4^s, \\ 1, & \text{if } k < p\delta(\sigma-1); \end{cases}$$

(2) when $p\sigma+n-L+s\le D_p<p\sigma+n$,

$$\alpha_C^{SP*}=\begin{cases}\alpha_{C1}, & \text{if } k\ge k_2^s,\\ \alpha_{C2}, & \text{if } k_4^s\le k<k_2^s,\\ \alpha_{C3}, & \text{if } \frac{np\delta(\sigma-1)}{D_p-p\sigma}\equiv k_5^s\le k<k_4^s,\\ \alpha_{C6}, & \text{if either } (p\sigma+\frac{n}{\sigma}<p\sigma+n-L+s,\ p\sigma+n-L+s\le D_p<p\sigma+n\\ & \quad \text{and } p\delta(\frac{\sigma(D_p-p\sigma)}{n}-1)\equiv k_8^s\le k<k_5^s)\\ & \text{or } (p\sigma+\frac{n}{\sigma}\ge p\sigma+n-L+s,\ p\sigma+n-L+s\le D_p<p\sigma+n \text{ and } k<k_5^s)\\ 1, & \text{if either } (p\sigma+\frac{n}{\sigma}<p\sigma+n-L+s,\ p\sigma+n-L+s\le D_p<p\sigma+n \text{ and } k<k_8^s)\\ & \text{or } (p\sigma+\frac{n}{\sigma}\ge p\sigma+n-L+s,\ p\sigma+\frac{n}{\sigma}\le D_p<p\sigma+n \text{ and } k<k_8^s)\end{cases}$$

(3) when $p\sigma+\frac{L}{\sigma}(\frac{n}{L+s}-1)\le D_p<p\sigma+n-L+s$,

(3.1) when $p\sigma+\frac{L}{\sigma}(\frac{n}{L+s}-1)\le D_p<p\sigma+n-L-s$,

$$\alpha_C^{SP*}=\begin{cases}\alpha_{C1}, & \text{if } k\ge\frac{np\delta}{D_p-p\sigma}(\sigma-\frac{L}{n-(D_p-p\sigma)})\equiv k_1^s,\\ \alpha_{C4}, & \text{if } \frac{np\delta}{n-L-s}(\frac{\sigma(D_p-p\sigma)}{n-L-s}-\frac{L}{L+s})\equiv k_6^s\le k<k_1^s,\end{cases}$$

(3.2) when $p\sigma+n-L-s\le D_p<p\sigma+n-L+s$,

$$\alpha_C^{SP*}=\begin{cases}\alpha_{C1}, & \text{if } k\ge k_2^s,\\ \alpha_{C2}, & \text{if } \frac{np\delta}{D_p-p\sigma}(\sigma-1+\frac{\int_{-s}^{n-D_p+p\sigma-L}F(\epsilon)d\epsilon}{n-D_p+p\sigma})\equiv k_3^s\le k<k_2^s,\end{cases}$$

(3.3) when $p\sigma+\frac{n-L+s}{\sigma}<p\sigma+n-L-s$, *if either* $(p\sigma+\frac{L}{\sigma}(\frac{n}{L+s}-1)\le D_p<p\sigma+\frac{n-L+s}{\sigma}$ *and* $k<k_6^s)$, *or* $(p\sigma+\frac{n-L+s}{\sigma}\le D_p<p\sigma+n-L-s$ *and* $\frac{np\delta}{n-L+s}(\frac{\sigma(D_p-p\sigma)}{n-L+s}-1)\equiv k_7^s\le k<k_6^s)$, *or* $(p\sigma+n-L-s\le D_p<p\sigma+n-L+s$ *and* $k_7^s\le k<k_3^s)$, *then* $\alpha_C^{SP*}=\alpha_{C5}$;

(3.4) when $p\sigma+\frac{n-L+s}{\sigma}\ge p\sigma+n-L-s$, *if either* $(p\sigma+\frac{L}{\sigma}(\frac{n}{L+s}-1)\le D_p<p\sigma+n-L-s$ *and* $k<k_6^s)$,*or* $(p\sigma+n-L-s\le D_p<p\sigma+\frac{n-L+s}{\sigma}$ *and* $k<k_3^s)$, *or* $(p\sigma+\frac{n-L+s}{\sigma}\le D_p<p\sigma+n-L+s$ *and* $k_7^s\le k<k_3^s)$, *then* $\alpha_C^{SP*}=\alpha_{C5}$;

(3.5) when $p\sigma+\frac{n}{\sigma}<p\sigma+n-L+s$,

$$\alpha_C^{SP*}=\begin{cases}\alpha_{C6}, & \text{if either } (p\sigma+\frac{n-L+s}{\sigma}\le D_p<p\sigma+\frac{n}{\sigma} \text{ and } k<k_7^s),\\ & \quad \text{or } (p\sigma+\frac{n}{\sigma}\le D_p<p\sigma+n-L+s \text{ and } k_8^s\le k<k_7^s);\\ 1, & \text{if } (p\sigma+\frac{n}{\sigma}\le D_p<p\sigma+n-L+s \text{ and } k<k_8^s);\end{cases}$$

(3.6) when $p\sigma+\frac{n}{\sigma}\ge p\sigma+n-L+s$, *then* $\alpha_C^{SP*}=\alpha_{C6}$ *when* $k<k_7^s$.

(4) when $p\sigma\le D_p<p\sigma+\frac{L}{\sigma}(\frac{n}{L+s}-1)$,

$$\alpha_C^{SP*}=\begin{cases}\alpha_{C1}, & \text{if } k\ge k_1^s,\\ \alpha_{C4}, & \text{if } k<k_1^s;\end{cases}$$

- *driver utility:*

$$U_C^{SP*}=\begin{cases}\frac{k-p\delta\sigma}{4k}\sqrt{\frac{4kLp\delta}{n}+(k-p\delta\sigma)^2}+\frac{p\delta\sigma}{2}+\frac{(p\delta\sigma)^2}{4k}+\frac{Lp\delta}{2n}-\frac{k}{4}, & if\ \alpha_C^{SP*}=\alpha_{C1},\\ \sigma p\delta\alpha_{C2}-\frac{1}{2}k\alpha_{C2}^2+\left(1-\frac{\int_{-s}^{n(1-\alpha_{C2})-L}F(\epsilon)d\epsilon}{n(1-\alpha_{C2})}\right)p\delta(1-\alpha_{C2}), & if\ \alpha_C^{SP*}=\alpha_{C2},\\ p\delta(\frac{p\delta(\sigma-1)^2}{2k}+1), & if\ \alpha_C^{SP*}=\alpha_{C3},\\ \frac{D_p-\sigma p}{n}\sigma p\delta-\frac{1}{2}k\alpha_{C4}^2+\frac{L}{n}p\delta, & if\ \alpha_C^{SP*}=\alpha_{C4},\\ \frac{D_p-\sigma p}{n}\sigma p\delta-\frac{1}{2}k\alpha_{C5}^2+\left(1-\frac{\int_{-s}^{n(1-\alpha_{C5})-L}F(\epsilon)d\epsilon}{n(1-\alpha_{C5})}\right)p\delta(1-\alpha_{C5}), & if\ \alpha_C^{SP*}=\alpha_{C5},\\ (\frac{D_p-\sigma p}{n}\sigma p\delta-\frac{1}{2}k\alpha_{C6}^2)+p\delta(1-\alpha_{C6}), & if\ \alpha_C^{SP*}=\alpha_{C6},\\ p\delta\sigma\min\{1,\frac{D_p-p\sigma}{n}\}-\frac{k}{2}, & if\ \alpha_C^{SP*}=1;\end{cases}$$

- *total matching volume:*

$$M_C^{SP*}=\begin{cases}L+\frac{n}{2k}(k+p\delta\sigma-\sqrt{\frac{4kLp\delta}{n}+(k-p\delta\sigma)^2}), & if\ \alpha_C^{SP*}=\alpha_{C1},\\ \int_{-s}^{s}\min\{L+\epsilon,n(1-\alpha_{C2})\}dF(\epsilon)+n\alpha_{C2}, & if\ \alpha_C^{SP*}=\alpha_{C2},\\ n, & if\ \alpha_C^{SP*}=\alpha_{C3},\\ L+D_p-\sigma p, & if\ \alpha_C^{SP*}=\alpha_{C4},\\ \int_{-s}^{s}\min\{L+\epsilon,n(1-\alpha_{C5})\}dF(\epsilon)+D_p-\sigma p, & if\ \alpha_C^{SP*}=\alpha_{C5},\\ n(1+\frac{p\delta}{2k}-\frac{\sqrt{np\delta(np\delta+4D_pk\sigma-4kp\sigma^2)}}{2kn})+D_p-\sigma p, & if\ \alpha_C^{SP*}=\alpha_{C6},\\ min\{n,D_p-\sigma p\}, & if\ \alpha_C^{SP*}=1;\end{cases}$$

- *platform profit:*

$$\pi_C^{SP*}=\begin{cases}\left(L+\frac{n\sigma}{2k}(k+p\delta\sigma-\sqrt{\frac{4kLp\delta}{n}+(k-p\delta\sigma)^2})\right)p(1-\delta), & if\ \alpha_C^{SP*}=\alpha_{C1},\\ \left(\int_{-s}^{s}\min\{L+\epsilon,n(1-\alpha_{C2})\}dF(\epsilon)+n\alpha_{C2}\sigma\right)p(1-\delta), & if\ \alpha_C^{SP*}=\alpha_{C2},\\ \frac{n(k+p\delta(\sigma-1)^2)}{k}p(1-\delta), & if\ \alpha_C^{SP*}=\alpha_{C3},\\ (L+(D_p-\sigma p)\sigma)\,p(1-\delta), & if\ \alpha_C^{SP*}=\alpha_{C4},\\ \left(\int_{-s}^{s}\min\{L+\epsilon,n(1-\alpha_{C5})\}dF(\epsilon)+(D_p-\sigma p)\sigma\right)p(1-\delta), & if\ \alpha_C^{SP*}=\alpha_{C5},\\ \left(n(1+\frac{p\delta}{2k}-\frac{\sqrt{np\delta(np\delta+4D_pk\sigma-4kp\sigma^2)}}{2kn})+(D_p-\sigma p)\sigma\right)p(1-\delta), & if\ \alpha_C^{SP*}=\alpha_{C6},\\ \min\{n,D_p-\sigma p\}\sigma p(1-\delta), & if\ \alpha_C^{SP*}=1;\end{cases}$$

where $\alpha_{C1}=\frac{k+\sigma p\delta-\sqrt{(k-\sigma p\delta)^2+\frac{4kL}{n}p\delta}}{2k}$; α_{C2} satisfies $\alpha_{C2}=\frac{p\delta}{k}(\sigma-1+\frac{\int_{-s}^{n(1-\alpha_{C2})-L}F(\epsilon)d\epsilon}{n(1-\alpha_{C2})})$; $\alpha_{C3}=\frac{p\delta(\sigma-1)}{k}$; α_{C4} satisfies $\alpha_{C4}=\frac{p\delta}{kn}(\frac{\sigma(D_p-p\sigma)}{\alpha_{C4}}-\frac{L}{1-\alpha_{C4}})$; α_{C5} satisfies $\alpha_{C5}=\frac{p\delta\sigma(D_p-p\sigma)}{kn\alpha_{C5}}-\frac{p\delta}{k}(1-\frac{\int_{-s}^{n(1-\alpha_{C5})-L}F(\epsilon)d\epsilon}{n(1-\alpha_{C5})})$; $\alpha_{C6}=-\frac{p\delta}{2k}+\frac{\sqrt{np\delta(np\delta+4D_pk\sigma-4kp\sigma^2)}}{2kn}$.

Surge pricing in an open system. Under an open system, a full-time driver's problem is $\underset{\alpha\in[0,1]}{max}\ U_O^{SP}(\alpha)$, where $U_O^{SP}(\alpha)=\min\{1,\frac{D_p-\sigma p}{n\hat{\alpha}}\}\cdot\sigma p\delta\alpha-\frac{1}{2}k\alpha^2$. The total matching volume is $M_O^{SP}(\alpha)=L+\min\{n\hat{\alpha},D_p-\sigma p\}$. The platform profit is $\pi_O^{SP}=L\cdot p(1-\delta)+\min\{n\hat{\alpha},D_p-\sigma p\}\cdot\sigma p(1-\delta)$. Maximizing π_O^{SP} yields the equilibrium results in Lemma A.2. The proof of Lemma A.2 can be found in SA.A of the Supplementary Analysis.

LEMMA A.2. *In an open system, when the platform implements surge pricing, in equilibrium,*

- *peak working time of a full time driver:*

$$\alpha_O^{SP*}=\begin{cases}\frac{p\delta\sigma}{k}, & \text{if either } (p\sigma<D_p\le n+p\sigma \text{ and } k\ge p\delta\sigma\frac{n}{D_p-p\sigma})\\ & \text{or } (D_p>n+p\sigma \text{ and } k\ge p\delta\sigma),\\ \sqrt{\frac{p\delta\sigma(D_p-p\sigma)}{kn}}, & \text{if } p\sigma<D_p\le n+p\sigma \text{ and } p\delta\sigma\frac{D_p-p\sigma}{n}\le k\le p\delta\sigma\frac{n}{D_p-p\sigma},\\ 1, & \text{if either } (p\sigma<D_p\le n+p\sigma \text{ and } k\le p\delta\sigma\frac{D_p-p\sigma}{n})\\ & \text{or } (D_p\ge n+p\sigma \text{ and } k<p\delta\sigma);\end{cases}$$

- *the utility of a full time driver:*

$$U_O^{SP*}=\begin{cases}\frac{(p\delta\sigma)^2}{2k}, & \text{if either } (p\sigma<D_p\le n+p\sigma \text{ and } k\ge p\delta\sigma\frac{n}{D_p-p\sigma})\\ & \text{or } (D_p>n+p\sigma \text{ and } k\ge p\delta\sigma),\\ \frac{p\delta\sigma}{2}\frac{D_p-p\sigma}{n}, & \text{if } p\sigma<D_p\le n+p\sigma \text{ and } p\delta\sigma\frac{D_p-p\sigma}{n}\le k\le p\delta\sigma\frac{n}{D_p-p\sigma},\\ p\delta\sigma-\frac{k}{2}, & \text{if } D_p\ge n+p\sigma \text{ and } k<p\delta\sigma,\\ p\delta\sigma\frac{D_p-p\sigma}{n}-\frac{k}{2} & \text{if } p\sigma<D_p\le n+p\sigma \text{ and } k\le p\delta\sigma\frac{D_p-p\sigma}{n},\end{cases}$$

- *total matching volume:*

$$M_O^{SP*}=\begin{cases}L+\frac{np\delta\sigma}{k}, & \text{if either } (p\sigma<D_p\le n+p\sigma \text{ and } k\ge p\delta\sigma\frac{n}{D_p-p\sigma})\\ & \text{or } (D_p>n+p\sigma \text{ and } k\ge p\delta\sigma)\\ L+n, & \text{if } D_p>n+p\sigma \text{ and } k<p\delta\sigma\\ L+D_p-\sigma p, & \text{if } p\sigma<D_p\le n+p\sigma \text{ and } k\le p\delta\sigma\frac{n}{D_p-p\sigma}\end{cases}$$

- *platform profit:*

$$\pi_O^{SP*}=\begin{cases}p(1-\delta)(L+\frac{np\delta\sigma^2}{k}), & \text{if either } (p\sigma<D_p\le n+p\sigma \text{ and } k\ge p\delta\sigma\frac{n}{D_p-p\sigma})\\ & \text{or } (D_p>n+p\sigma \text{ and } k\ge p\delta\sigma)\\ p(1-\delta)(L+n\sigma), & \text{if } D_p>n+p\sigma \text{ and } k<p\delta\sigma\\ p(1-\delta)(L+(D_p-\sigma p)\sigma), & \text{if } p\sigma<D_p\le n+p\sigma \text{ and } k\le p\delta\sigma\frac{n}{D_p-p\sigma}\end{cases}$$

Appendix C. Heterogeneous Congestion Costs

In this section, we relax the assumption that all drivers have the same congestion costs. Specifically, we assume that a fraction $\phi \in (0,1)$ of full-time drivers have high congestion costs with a cost coefficient k_h and refer to them as high-type drivers. The remaining $1-\phi$ full-time drivers have low congestion costs with a cost coefficient k_l (where $k_l < k_h$) and are referred to as low-type drivers. With this modification, we examine the robustness of our main model predictions. We find that the model prediction of an open system, that the prioritization scheme leads to weakly higher matching volume, continues to hold given drivers' heterogeneity in congestion costs. The prediction of a closed system, that the qualification scheme leads to weakly higher matching volume, requires certain boundary conditions. Therefore, we focus on presenting the analysis and discussing the boundary conditions of a closed system.

Given a closed system, we first show that for both schemes, if the platform optimally chooses a peak work target that will motivate both types of drivers to participate in the scheme, our results continue to hold as long as k_l is not too small.

Lemma A.3. *In a closed system and given that the price per unit of service time is fixed, if the platform's optimal work target α_0 involves both types of drivers under both the qualification and prioritization schemes, then the equilibrium characterizations in Lemma 3 and 4 continue to hold with k replaced by k_h as long as k_l is not too small, i.e., $k_l > \frac{p\delta(1-\int_{-s}^{s}\min\{1,\frac{D(\epsilon)}{n(1-\alpha_0^{Pb})}\}F(\epsilon)d\epsilon)}{\alpha_0^{Pb}}$, where α_0^{Pb} satisfies $\int_{-s}^{s}\min\{1,\frac{D(\epsilon)}{n(1-\alpha_0^{Pb})}\}dF(\epsilon)p\delta(1-\alpha_0^{Pb}) + p\delta\alpha_0^{Pb} - \frac{1}{2}k_h{\alpha_0^{Pb}}^2 = p(1-F[n(1-\alpha_0^{Pb})-L]) + (F[n(1-\alpha_0^{Pb})-L])^2\frac{p^2\delta^2}{2k_h}$. Furthermore, the qualification scheme leads to a higher matching volume than the prioritization scheme.*

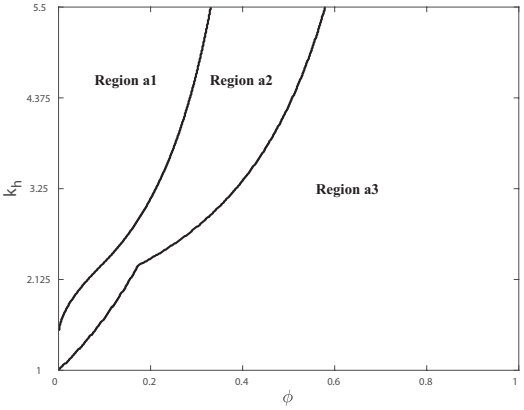
The proof of Lemma A.3 can be found in SA.B.2 of the Supplementary Analysis. We then numerically explore the optimal driver composition for each incentive scheme[1] and visualize the result in Figure A1(a). Two observations are made. First, for both incentive schemes, the platform engages only low-type drivers if and only if ϕ is low and k_h is large, and otherwise engages both types of drivers. The intuition is as follows: with small ϕ and large k_h, the platform needs to significantly lower the peak target to engage both types of drivers, which is not profitable compared to setting a high peak target to engage only low-type drivers, especially given that the proportion of high-type drivers is small. Second, the parameter region in which the platform engages both types of drivers is smaller under a prioritization scheme than under a qualification scheme. This reflects a qualification scheme's higher effectiveness in engaging drivers, consistent with the insight from our main model that the platform can set a higher peak target under a qualification scheme.

Figure A1(b) further visualizes the region where the qualification scheme leads to a higher total matching volume (i.e., Region b2), which aligns with the main model prediction. It is seen that all three driver compositions shown in Figure A1(a) are possible when the qualification scheme outperforms the prioritization scheme. First, when both schemes involve all drivers (i.e., Region a3 in Figure A1(a)), the

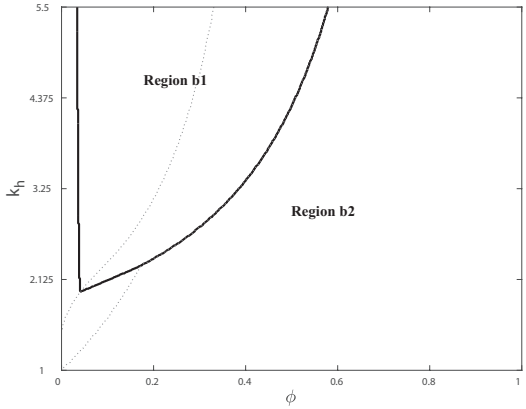
[1] We have also tested ten different sets of parameters for robustness checks and found very consistent patterns. The detailed results can be found in SA.D.5 of the Supplementary Analysis.

Figure A1 Numerical Analysis with Heterogeneous Drivers. ($D_0 = 1, p = 0.5, \delta = 0.8, n = 1.3, k_l = 1, \epsilon \sim U[-0.1, 0.1]$)

(a) The optimal composition of drivers



(b) The optimal implementation of incentive schemes



(a) **Region a1**: The platform induces only low-type drivers to participate under both schemes; **Region a2**: The platform induces both types of drivers to participate under a qualification scheme, but only low-type drivers under a prioritization scheme; **Region a3**: The platform induces both types of drivers to participate under both schemes. (b) **Region b1**: The prioritization scheme leads to the highest matching volume; **Region b2**: The qualification scheme leads to the highest matching volume.

qualification scheme outperforms the prioritization scheme. The reason is the same as in the main model, that the qualification scheme can set a higher service target and increase the overall matching volume to a higher level. Second, when both schemes induce only low-type drivers to participate (i.e., Region a1 in Figure A1(a)), if the fraction of high-type drivers is low, the qualification scheme outperforms the prioritization scheme. That is, if the majority of drivers have the same congestion cost, our result will not be sensitive to the addition of a small number of drivers who have higher congestion costs. Third, when platform engages both types of drivers under a qualification scheme, but only low-type drivers under a prioritization scheme (i.e., Region a2 in Figure A1(a)), if the congestion cost of high-type drivers is not significantly higher than that of low-type drivers, the qualification scheme outperforms the prioritization scheme. This is because a qualification scheme not only attracts more drivers to participate, but also enables a higher work target when the congestion cost of high-type drivers is not significantly large. Lastly, if the high-type drivers have large congestion costs (compared to low-type drivers) and the proportion of high and low-type drivers is relatively balanced (i.e., Region b1), we find that the prioritization scheme leads to a higher matching volume. In this case, the prioritization scheme uses a high work target to engage only low-type drivers, but can still utilize high-type drivers to provide service during the off-peak period. In contrast, the qualification scheme either engages both types of drivers, which requires lowering the work target due to the sheer number of high-type drivers, or engages only low-type drivers, which excludes high-type drivers from providing service during the off-peak period. In both situations, the qualification scheme results in a lower matching volume.

In summary, the analysis suggests that the boundary condition for our main model result of a closed system (i.e., the qualification scheme generates a weakly higher matching volume) is that the heterogeneity in congestion costs among drivers is not significant, or that the majority of drivers have either high or low congestion costs.

Supplementary Analysis for "Balancing Supply with Demand on Ride-hailing Platforms in Markets with Price Regulations: An Operational Approach"

In this file, we first provide the proofs of two lemmas related to surge pricing in Section SA.A. We then relax certain model assumptions and examine the robustness and boundary conditions of our main model results in Section SA.B. Specifically, in Section SA.B.1, we relax the assumption of an infinite number of part-time drivers and provide the proof of Proposition 4 from Section 6.2, which examines the scenario with a finite number of part-time drivers. In Section SA.B.2, we provide the proof of Lemma A.3, which considers full-time drivers with heterogeneous congestion costs. In Section SA.B.3, we explore the scenario where drivers who are constrained to work during peak periods are included, as well as the case where a finite subset of drivers have the flexibility to choose in which period to work, but work fewer hours than full-time drivers. In Section SA.B.4, we relax the assumption that the peak period is always under-supplied, even when full-time drivers devote all their time to the peak period (i.e., $(D_p > n)$), and investigate the scenario with volatile peak demand. In Section SA.B.5, we relax the assumption that the daily peak period is long enough for the platform to always set the target as the global optimal solution, and consider how a capped peak period affects the platform's optimal decisions. Next, we provide justifications for the model setting. Specifically, in Section SA.C.1, we prove that a prioritization scheme is superior to a randomized prioritization scheme. In Section SA.C.2, we show that a full-time driver will optimally use all of their working time each day, even if they are allowed to choose a shorter total working time. Finally, in Section SA.D, we provide numerical analysis and results to compare the incentive schemes and surge pricing, and conduct robustness checks in the case of heterogeneous congestion costs.

In the following, the equilibrium results are indicated with superscript $*$. We use subscripts C, O to represent *Closed system* and *Open system* respectively, subscripts $q/uq, p/up$ to represent *qualified/unqualified drivers* and *prioritized/unprioritized drivers* respectively, and use superscripts B, Q, P to represent *Benchmark, Qualification scheme*, and *Prioritization scheme*, respectively.

SA.A Proofs of Lemmas on Surge Pricing

Proof of Lemma A.1

Depending on the demand-supply condition during the peak and off-peak periods, there are in total 6 cases as summarized in Table SA.1.[1] We will analyze each case and obtain driver's optimal decision (α_C^{SP*}).

[1] For Case 3 and Case 6, the optimal value of α equals 1 under certain conditions, which we introduce within the respective cases.

Table SA.1 Driver utility under surge pricing in a closed system

Off-Peak \ Peak	under-supplied $\hat{\alpha} \le \frac{D_p - p\sigma}{n}$	over-supplied $\hat{\alpha} > \frac{D_p - p\sigma}{n}$
over-supplied $\hat{\alpha} \le 1 - \frac{L+s}{n}$	**Case 1**: $(\sigma p\delta\alpha - \frac{1}{2}k\alpha^2) + \frac{L}{n(1-\hat{\alpha})} p\delta(1-\alpha)$	**Case 4**: $(\frac{D_p - \sigma p}{n\hat{\alpha}}\sigma p\delta\alpha - \frac{1}{2}k\alpha^2) + \frac{L}{n(1-\hat{\alpha})} p\delta(1-\alpha)$
over- or under-supplied $1 - \frac{L+s}{n} < \hat{\alpha} \le 1 - \frac{L-s}{n}$	**Case 2**: $(\sigma p\delta\alpha - \frac{1}{2}k\alpha^2) + \left(1 - \frac{\int_{-s}^{n(1-\hat{\alpha})-L} F(\epsilon)d\epsilon}{n(1-\hat{\alpha})}\right) p\delta(1-\alpha)$	**Case 5**: $(\frac{D_p - \sigma p}{n\hat{\alpha}}\sigma p\delta\alpha - \frac{1}{2}k\alpha^2) + \left(1 - \frac{\int_{-s}^{n(1-\hat{\alpha})-L} F(\epsilon)d\epsilon}{n(1-\hat{\alpha})}\right) p\delta(1-\alpha)$
under-supplied $\hat{\alpha} > 1 - \frac{L-s}{n}$	**Case 3**: $(\sigma p\delta\alpha - \frac{1}{2}k\alpha^2) + p\delta(1-\alpha)$	**Case 6**: $(\frac{D_p - \sigma p}{n\hat{\alpha}}\sigma p\delta\alpha - \frac{1}{2}k\alpha^2) + p\delta(1-\alpha)$

Case 1: $\hat{\alpha} \le \frac{D_p - p\sigma}{n}$ and $\hat{\alpha} \le 1 - \frac{L+s}{n}$.

In this case, the peak market is under-supplied and the off-peak market is over-supplied. A driver's utility is $U_C^{SP}(\alpha) = (\sigma p\delta\alpha - \frac{1}{2}k\alpha^2) + \frac{L}{n(1-\hat{\alpha})} p\delta(1-\alpha)$. Since $U_C^{SP}(\alpha)$ is concave in α, by using FOC, we obtain $\alpha(\hat{\alpha}) = \frac{p\delta}{k}(\sigma - \frac{L}{n(1-\hat{\alpha})})$. Equating $\alpha(\hat{\alpha}) = \hat{\alpha}$ yields the optimal α, which is denoted by $\alpha_{C1}^{\dagger}$ as shown below (the other solution exceeds 1 and thus is infeasible),

$$\alpha_C^{\dagger} = \frac{k + \sigma p\delta - \sqrt{(k - \sigma p\delta)^2 + \frac{4kL}{n} p\delta}}{2k} \equiv \alpha_{C1}.$$

The following conditions should be satisfied: $\alpha_C^{\dagger} \le \frac{D_p - p\sigma}{n}$, $\alpha_C^{\dagger} \le 1 - \frac{L+s}{n}$ and $0 \le \alpha_C^{\dagger} \le 1$. That is, either $(p\sigma < D_p \le p\sigma + n - L - s$ and $k \ge \frac{np\delta}{D_p - p\sigma}(\sigma - \frac{L}{n-(D_p - p\sigma)}) \equiv k_1^s)$ or $(D_p > p\sigma + n - L - s$ and $k \ge \frac{np\delta}{L+s}(\frac{(\sigma-1)L + \sigma s}{n-L-s}) \equiv k_2^s)$.

In this case, a driver's utility is $U_C^{SP*} = \frac{k - p\delta\sigma}{4k}\sqrt{\frac{4kLp\delta}{n} + (k - p\delta\sigma)^2} + \frac{p\delta\sigma}{2} + \frac{(p\delta\sigma)^2}{4k} + \frac{Lp\delta}{2n} - \frac{k}{4}$, the total matching volume is $M_C^{SP*} = L + \frac{n}{2k}\left(k + p\delta\sigma - \sqrt{\frac{4kLp\delta}{n} + (k - p\delta\sigma)^2}\right)$, and the platform profit is $\pi_C^{SP*} = \left(L + \frac{n\sigma}{2k}(k + p\delta\sigma - \sqrt{\frac{4kLp\delta}{n} + (k - p\delta\sigma)^2})\right) p(1-\delta)$.

Case 2: $\hat{\alpha} \le \frac{D_p - p\sigma}{n}$ and $1 - \frac{L+s}{n} < \hat{\alpha} \le 1 - \frac{L-s}{n}$.

In this case, the peak market is under-supplied and the off-peak market could be either over-supplied or under-supplied depending on the realized ϵ. A driver's utility is $U_C^{SP}(\alpha) = (\sigma p\delta\alpha - \frac{1}{2}k\alpha^2) + \left(1 - \frac{\int_{-s}^{n(1-\hat{\alpha})-L} F(\epsilon)d\epsilon}{n(1-\hat{\alpha})}\right) p\delta(1-\alpha)$, which is concave in α. Solving FOC, we have $\alpha(\hat{\alpha}) = \frac{p\delta}{k}A(\hat{\alpha})$, where $A(\hat{\alpha}) = \sigma - 1 + \frac{\int_{-s}^{n(1-\hat{\alpha})-L} F(\epsilon)d\epsilon}{n(1-\hat{\alpha})}$. The equilibrium condition requires $\alpha(\hat{\alpha}) = \hat{\alpha}$, whose solution is denoted as $\alpha^{\dagger}$. We can show that $\frac{dA(\hat{\alpha})}{d\hat{\alpha}} < 0$. Therefore, for $\alpha(\hat{\alpha}) = \frac{p\delta}{k}A(\hat{\alpha})$, LHS (RHS) increases (decreases) in $\hat{\alpha}$, and there is at most one solution.

To guarantee the existence of an equilibrium, the following conditions should hold. (a) $\frac{D_p - p\sigma}{n} \ge 1 - \frac{L+s}{n}$ to guarantee the existence of this case. (b) On the left boundary of $\hat{\alpha}$ (i.e. $\hat{\alpha} = 1 - \frac{L+s}{n}$), LHS is less than or equal to RHS. That is, $1 - \frac{L+s}{n} < \frac{p\delta}{k}\left(\sigma - 1 + \frac{s}{L+s}\right)$. (c) On the right boundary of $\hat{\alpha}$, LHS is greater than or equal to RHS. That is, if the right boundary is $1 - \frac{L-s}{n}$ (i.e., $1 - \frac{L-s}{n} \le \frac{D_p - p\sigma}{n}$), then $1 - \frac{L-s}{n} \ge \frac{p\delta}{k}(\sigma - 1)$;

if the right boundary is $\frac{D_p-p\sigma}{n}$ (i.e. $\frac{D_p-p\sigma}{n} \le 1-\frac{L-s}{n}$), then $\frac{D_p-p\sigma}{n} \ge \frac{p\delta}{k}(\sigma-1+\frac{\int_{-s}^{n-D_p+p\sigma-L} F(\epsilon)d\epsilon}{n-D_p+p\sigma})$. If the above conditions are satisfied, then $\alpha^\dagger$ is the optimal solution, where $\alpha^\dagger$ satisfies

$$\alpha^\dagger = \frac{p\delta}{k}(\sigma-1+\frac{\int_{-s}^{n(1-\alpha^\dagger)-L} F(\epsilon)d\epsilon}{n(1-\alpha^\dagger)}) \equiv \alpha_{C2}.$$

And the following conditions should be satisfied: either ($p\sigma+n-L-s \le D_p \le p\sigma+n-L+s$ and $k_3^s \equiv \frac{np\delta}{D_p-p\sigma}(\sigma-1+\frac{\int_{-s}^{n-D_p+p\sigma-L} F(\epsilon)d\epsilon}{n-D_p+p\sigma}) \le k \le k_2^s$) or ($D_p \ge p\sigma+n-L+s$ and $k_4^s \equiv \frac{np\delta(\sigma-1)}{n-L+s} \le k \le k_2^s$).

In this case, a driver's utility is $U_C^{SP*} = \sigma p\delta\alpha_{C2} - \frac{1}{2}k\alpha_{C2}^2 + \left(1-\frac{\int_{-s}^{n(1-\alpha_{C2})-L} F(\epsilon)d\epsilon}{n(1-\alpha_{C2})}\right)p\delta(1-\alpha_{C2})$, the total matching volume is $M_C^{SP*} = \int_{-s}^{s}\min\{L+\epsilon, n(1-\alpha_{C2})\}dF(\epsilon) + n\alpha_{C2}$, and the platform profit is $\pi_C^{SP*} = \left(\int_{-s}^{s}\min\{L+\epsilon, n(1-\alpha_{C2})\}dF(\epsilon) + n\alpha_{C2}\sigma\right)p(1-\delta)$.

Case 3: $\hat{\alpha} \le \frac{D_p-p\sigma}{n}$ and $\hat{\alpha} > 1-\frac{L-s}{n}$.

In this case, both the peak and off-peak markets are under-supplied. A driver's utility is $U_C^{SP}(\alpha) = p\delta\,(1+(\sigma-1)\alpha) - \frac{1}{2}k\alpha^2$. By employing FOC, we have

$$\alpha^\dagger = \frac{p\delta(\sigma-1)}{k} \equiv \alpha_{C3}.$$

The equilibrium condition requires $\alpha(\hat{\alpha}) = \alpha^\dagger = \hat{\alpha}$, and the following conditions should be satisfied: $\alpha^\dagger \le \frac{D_p-p\sigma}{n}$, $\alpha^\dagger > 1-\frac{L-s}{n}$ and $0 \le \alpha^\dagger \le 1$. This is equivalent to ($p\sigma+n-L+s < D_p \le p\sigma+n$ and $k_5^s \equiv \frac{np\delta(\sigma-1)}{D_p-p\sigma} \le k < k_4^s$) or ($D_p > p\sigma+n$ and $p\delta(\sigma-1) \le k < k_4^s$}). In this case, a driver's utility is $U_C^{SP*} = p\delta(\frac{p\delta(\sigma-1)^2}{2k}+1)$, the total matching volume is $M_C^{SP*} = n$, and the platform profit is $\pi_C^{SP*} = \frac{np(1-\delta)(k+p\delta(\sigma-1)^2)}{k}$.

Besides, we can also show that the optimal α equals 1 if $D_p > p\sigma+n$ and $k < p\delta(\sigma-1)$. Under such condition, the driver utility is $U_C^{SP*} = p\delta\sigma - \frac{k}{2}$, the matching volumne is $M_C^{SP*} = n$, and the platform profit is $\pi_C^{SP*} = n\sigma p(1-\delta)$.

Case 4: $\hat{\alpha} > \frac{D_p-p\sigma}{n}$ and $\hat{\alpha} \le 1-\frac{L+s}{n}$.

In this case, both the peak and off-peak periods are over-supplied. A driver's utility is $U_C^{SP}(\alpha) = (\frac{D_p-\sigma p}{n\hat{\alpha}}\sigma p\delta\alpha - \frac{1}{2}k\alpha^2) + \frac{L}{n(1-\hat{\alpha})}p\delta(1-\alpha)$, which is concave in α. By the FOC, we have $\alpha(\hat{\alpha}) = \frac{p\delta}{kn}A(\hat{\alpha})$, where $A(\hat{\alpha}) = \frac{\sigma(D_p-p\sigma)}{\hat{\alpha}} - \frac{L}{1-\hat{\alpha}}$. We can show that $\frac{dA(\hat{\alpha})}{d\hat{\alpha}} < 0$. Therefore, for $\alpha(\hat{\alpha}) = \frac{p\delta}{kn}A(\hat{\alpha})$, LHS (RHS) increases (decreases) in $\hat{\alpha}$ and there is at most one solution. Equating $\alpha(\hat{\alpha}) = \hat{\alpha}$ yields the optimal α, which is denoted by $\alpha^\dagger$ and

$$\alpha^\dagger = \frac{p\delta}{kn}(\frac{\sigma(D_p-p\sigma)}{\alpha^\dagger} - \frac{L}{1-\alpha^\dagger}) \equiv \alpha_{C4}.$$

To guarantee the existence of an equilibrium, the following conditions should hold. (a) $\frac{D_p-p\sigma}{n} \le 1-\frac{L+s}{n}$ to guarantee the existence of this case. (b) On the left boundary of $\hat{\alpha}$ (i.e. $\hat{\alpha} = \frac{D_p-p\sigma}{n}$), LHS is less than or equal to RHS. That is, $\frac{D_p-p\sigma}{n} \le \frac{p\delta}{kn}(n\sigma - \frac{L}{1-\frac{D_p-p\sigma}{n}})$. (c) On the right boundary of $\hat{\alpha}$ (i.e., $\hat{\alpha} = 1-\frac{L+s}{n}$), LHS is greater than or equal to RHS. That is, $1-\frac{L+s}{n} \ge \frac{p\delta}{kn}(\frac{\sigma(D_p-p\sigma)}{1-\frac{L+s}{n}} - \frac{Ln}{L+s})$. Hence, the following

conditions should be satisfied: either ($p\sigma < D_p \le p\sigma + \frac{L}{\sigma}(\frac{n}{L+s} - 1)$ and $k < k_1^s$) or ($p\sigma + \frac{L}{\sigma}(\frac{n}{L+s} - 1) < D_p < p\sigma + n - L - s$ and $k_6^s \equiv \frac{np\delta}{n-L-s}(\frac{\sigma(D_p - p\sigma)}{n-L-s} - \frac{L}{L+s}) \le k \le k_1^s$).

In this case, a driver's utility is $U_C^{SP*} = \frac{D_p - \sigma p}{n}\sigma p\delta - \frac{1}{2}k\alpha_{C4}^2 + \frac{Lp\delta}{n}$, the total matching volume is $M_C^{SP*} = L + D_p - \sigma p$, and the platform profit is $\pi_C^{SP*} = (L + (D_p - \sigma p)\sigma)\, p(1-\delta)$.

Case 5: $\hat{\alpha} > \frac{D_p - p\sigma}{n}$ and $1 - \frac{L+s}{n} < \hat{\alpha} \le 1 - \frac{L-s}{n}$.

In this case, the peak period is always over-supplied, and the off-peak period can be either over or under supplied depending on the realization of ϵ. A driver's utility is $U_C^{SP}(\alpha) = (\frac{D_p - \sigma p}{n\hat{\alpha}}\sigma p\delta\alpha - \frac{1}{2}k\alpha^2) + \left(1 - \frac{\int_{-s}^{n(1-\hat{\alpha})-L} F(\epsilon)d\epsilon}{n(1-\hat{\alpha})}\right) p\delta(1-\alpha)$, which is concave in α. By the FOC, we have $\alpha(\hat{\alpha}) = \frac{p\delta}{k}A(\hat{\alpha})$, where $A(\hat{\alpha}) = \frac{\sigma(D_p - p\sigma)}{n\hat{\alpha}} - 1 + \frac{\int_{-s}^{n(1-\hat{\alpha})-L} F(\epsilon)d\epsilon}{n(1-\hat{\alpha})}$. The equilibrium outcome requires $\alpha(\hat{\alpha}) = \hat{\alpha}$, whose solution is denoted as $\alpha^\dagger$. We can show that $\frac{dA(\hat{\alpha})}{d\hat{\alpha}} < 0$. Therefore, for $\hat{\alpha} = \frac{p\delta}{k}A(\hat{\alpha})$, LHS (RHS) increases (decreases) in $\hat{\alpha}$, and there is at most one solution.

To guarantee the existence of an equilibrium, the following conditions should hold. (a) $\frac{D_p - p\sigma}{n} \le 1 - \frac{L-s}{n}$ to guarantee the existence of this case. (b) On the left boundary of $\hat{\alpha}$, LHS is less than or equal RHS. That is, if the left boundary is $1 - \frac{L+s}{n}$ (i.e. $1 - \frac{L+s}{n} \ge \frac{D_p - p\sigma}{n}$), then $1 - \frac{L+s}{n} \le \frac{p\delta}{k}(\frac{\sigma(D_p - p\sigma)}{n-L-s} - 1 + \frac{s}{L+s})$; if the left boundary is $\frac{D_p - p\sigma}{n}$ (i.e. $1 - \frac{L+s}{n} \le \frac{D_p - p\sigma}{n}$), then $\frac{D_p - p\sigma}{n} \le \frac{p\delta}{k}(\sigma - 1 + \frac{\int_{-s}^{n-D_p+p\sigma-L} F(\epsilon)d\epsilon}{n-D_p+p\sigma})$. (c) On the right boundary of $\hat{\alpha}$ (i.e. $\hat{\alpha} = 1 - \frac{L-s}{n}$), LHS is greater than or equal RHS. That is, $1 - \frac{L-s}{n} \ge \frac{p\delta}{k}(\frac{\sigma(D_p - p\sigma)}{n-L+s} - 1)$. If the above conditions are satisfied, then $\alpha^\dagger$ is the optimal solution, where $\alpha^\dagger$ satisfies

$$\alpha^\dagger = \frac{p\delta\sigma(D_p - p\sigma)}{kn\alpha^\dagger} - \frac{p\delta}{k}\left(1 - \frac{\int_{-s}^{n(1-\alpha^\dagger)-L} F(\epsilon)d\epsilon}{n(1-\alpha^\dagger)}\right) \equiv \alpha_{C5}.$$

And the following conditions should be satisfied: (1) if $p\sigma + \frac{n-L+s}{\sigma} \le p\sigma + n - L - s$, then either ($p\sigma + \frac{L}{\sigma}(\frac{n}{L+s} - 1) < D_p \le p\sigma + \frac{n-L+s}{\sigma}$ and $k \le k_6^s$) or ($p\sigma + \frac{n-L+s}{\sigma} < D_p \le p\sigma + n - L - s$ and $k_7^s \equiv \frac{np\delta}{n-L+s}(\frac{\sigma(D_p - p\sigma)}{n-L+s} - 1) \le k \le k_6^s$) or ($p\sigma + n - L - s < D_p < p\sigma + n - L + s$ and $k_7^s < k < k_3^s$); (2) if $p\sigma + \frac{n-L+s}{\sigma} > p\sigma + n - L - s$, then either ($p\sigma + \frac{L}{\sigma}(\frac{n}{L+s} - 1) < D_p \le p\sigma + n - L - s$ and $k < k_6^s$) or ($p\sigma + n - L - s \le D_p \le p\sigma + \frac{n-L+s}{\sigma}$ and $k \le k_3^s$) or ($p\sigma + \frac{n-L+s}{\sigma} < D_p \le p\sigma + n - L + s$ and $k_7^s \le k \le k_3^s$).

In this case, a driver's utility is $U_C^{SP*} = \frac{D_p - \sigma p}{n}\sigma p\delta - \frac{1}{2}k\alpha_{C5}^2 + \left(1 - \frac{\int_{-s}^{n(1-\alpha_{C5})-L} F(\epsilon)d\epsilon}{n(1-\alpha_{C5})}\right) p\delta(1-\alpha_{C5})$, the total matching volume is $M_C^{SP*} = \int_{-s}^{s} \min\{L + \epsilon, n(1-\alpha_{C5})\}dF(\epsilon) + D_p - \sigma p$ and the platform profit is $\pi_C^{SP*} = \left(\int_{-s}^{s} \min\{L + \epsilon, n(1-\alpha_{C5})\}dF(\epsilon) + (D_p - \sigma p)\sigma\right) p(1-\delta)$.

Case 6: $\hat{\alpha} > \frac{D_p - p\sigma}{n}$ and $\hat{\alpha} > 1 - \frac{L-s}{n}$.

In this case, the peak period is over-supplied and the off-peak period is under-supplied. A driver's utility is $U_C^{SP}(\alpha) = (\frac{D_p - \sigma p}{n\hat{\alpha}}\sigma p\delta\alpha - \frac{1}{2}k\alpha^2) + p\delta(1-\alpha)$, which is concave in α. By the FOC, we have $\alpha(\hat{\alpha}) = \frac{p\delta}{k}(\sigma\frac{D_p - p\sigma}{\hat{\alpha}n} - 1)$. Equating $\alpha(\hat{\alpha}) = \hat{\alpha}$ yields the optimal α, which is denoted by $\alpha^\dagger$ and

$$\alpha^\dagger = -\frac{p\delta}{2k} + \frac{\sqrt{np\delta(np\delta + 4D_pk\sigma - 4kp\sigma^2)}}{2kn} \equiv \alpha_{c6}.$$

The following conditions should be satisfied: $\alpha^\dagger > \frac{D_p - p\sigma}{n}$, $\alpha^\dagger > 1 - \frac{L-s}{n}$ and $0 \le \alpha^\dagger \le 1$, which is equivalent to the conditions below: (1) If $p\sigma + \frac{n}{\sigma} < p\sigma + n - L + s$, then either ($p\sigma + \frac{n-L+s}{\sigma} < D_p < p\sigma + \frac{n}{\sigma}$ and $k < k_7^s$)

or ($p\sigma + \frac{n}{\sigma} < D_p < p\sigma + n - L + s$ and $k_8^s \equiv p\delta(\frac{\sigma(D_p - p\sigma)}{n} - 1) \le k < k_7^s$) or ($p\sigma + n - L + s < D_p \le n + p\sigma$ and $k_8^s \le k < k_5^s$); (2) If $p\sigma + \frac{n}{\sigma} > p\sigma + n - L + s$, then either ($p\sigma + \frac{n-L+s}{\sigma} < D_p < p\sigma + n - L + s$ and $k < k_7^s$) or ($p\sigma + n - L + s < D_p < p\sigma + \frac{n}{\sigma}$ and $k < k_5^s$) or ($p\sigma + \frac{n}{\sigma} < D_p \le n + p\sigma$ and $k_8^s \le k < k_5^s$). Under the above conditions, a driver's utility is $U_C^{SP*} = (\frac{D_p - \sigma p}{n}\sigma p\delta - \frac{1}{2}k\alpha_{C6}^2) + p\delta(1 - \alpha_{C6})$, the total matching volume is $M_C^{SP*} = n(1 + \frac{p\delta}{2k} - \frac{\sqrt{np\delta(np\delta + 4D_pk\sigma - 4kp\sigma^2)}}{2kn}) + D_p - \sigma p$, and the platform profit is $\pi_C^{SP*} = \left(n(1 + \frac{p\delta}{2k} - \frac{\sqrt{np\delta(np\delta + 4D_pk\sigma - 4kp\sigma^2)}}{2kn}) + (D_p - \sigma p)\sigma\right) p(1 - \delta)$.

In addition, we can also show that the optimal α equals 1 if either ($p\sigma + \frac{n}{\sigma} < p\sigma + n - L + s$, $p\sigma + \frac{n}{\sigma} < D_p < p\sigma + n - L + s$ and $k < k_8^s$) or ($p\sigma + \frac{n}{\sigma} < p\sigma + n - L + s$, $p\sigma + n - L + s < D_p \le p\sigma + n$ and $k < k_8^s$) or ($p\sigma + \frac{n}{\sigma} > p\sigma + n - L + s$, $p\sigma + \frac{n}{\sigma} < D_p < p\sigma + n$ and $k < k_8^s$). Under the above conditions, a driver's utility is $U_C^{SP*} = \frac{p\delta\sigma(D_p - p\sigma)}{n} - \frac{k}{2}$, the total matching volume is $M_C^{SP*} = D_p - \sigma p$ and the platform profit is $\pi_C^{SP*} = (D_p - \sigma p)\sigma p(1 - \delta)$. ■

Proof of Lemma A.2

Depending on the demand-supply condition in the peak period, there are 2 cases as shown below.

Case 1: $\hat{\alpha} \le \frac{D_p - p\sigma}{n}$.

In this case, the peak market is under-supplied. A driver's utility is $U_O^{SP}(\alpha) = \sigma p\delta\alpha - \frac{1}{2}k\alpha^2$. Using FOC, we obtain $\alpha_O^\dagger = \frac{p\delta\sigma}{k}$. The equilibrium condition requires $\alpha(\hat{\alpha}) = \hat{\alpha} = \alpha_O^\dagger$, so $\alpha_O^\dagger \le \frac{D_p - p\sigma}{n}$ and $\alpha_O^\dagger \ge 0$ should hold. That is, either ($p\sigma < D_p \le n + p\sigma$ and $k \ge p\delta\sigma\frac{n}{D_p - p\sigma}$) or ($D_p > n + p\sigma$ and $k \ge p\delta\sigma$). In this case, a full-time driver's utility is $U_O^{SP*} = \frac{(p\delta\sigma)^2}{2k}$, the total matching volume is $M_O^{SP*} = L + \frac{np\delta\sigma}{k}$, and the platform profit is $\pi_O^{SP*} = p(1 - \delta)(L + \frac{np\delta\sigma^2}{k})$.

Besides, we can also show that the optimal α equals 1 if $D_p > n + p\sigma$ and $k < p\delta\sigma$. In this case, a driver's utility is $U_O^{SP*} = \sigma p\delta - \frac{1}{2}k$, the total matching volume is $M_O^{SP*} = L + n$, and the platform profit is $\pi_O^{SP*} = p(1 - \delta)(L + n\sigma)$.

Case 2: $\hat{\alpha} > \frac{D_p - p\sigma}{n}$.

In this case, the peak market is over-supplied. A driver's utility is $U_O^{SP}(\alpha) = (\frac{D_p - \sigma p}{n\hat{\alpha}})\sigma p\delta\alpha - \frac{1}{2}k\alpha^2$. Using FOC, we obtain $\alpha_O^\dagger = \frac{p\delta\sigma(D_p - p\sigma)}{\hat{\alpha}kn}$. The equilibrium condition requires $\alpha(\hat{\alpha}) = \hat{\alpha} = \alpha_O^\dagger$, so we obtain $\alpha_O^\dagger = \sqrt{\frac{p\delta\sigma(D_p - p\sigma)}{kn}}$. To guarantee the solution falls within the range of this case, conditions $\alpha_O^\dagger > \frac{D_p - p\sigma}{n}$ and $\alpha_O^\dagger \le 1$ should hold. That is, $p\sigma < D_p \le n + p\sigma$ and $p\delta\sigma\frac{D_p - p\sigma}{n} \le k \le p\delta\sigma\frac{n}{D_p - p\sigma}$. In this case, a driver's utility is $U_O^{SP*} = \frac{p\delta\sigma}{2}\frac{D_p - p\sigma}{n}$, the total matching volume is $M_O^{SP*} = L + D_p - \sigma p$, and the platform profit is $\pi_O^{SP*} = p(1 - \delta)(L + (D_p - \sigma p)\sigma)$.

Besides, we can further show that the optimal α equals 1 if $p\sigma < D_p \le n + p\sigma$ and $k < p\delta\sigma\frac{D_p - p\sigma}{n}$. In this case, a driver's utility is $U_O^{SP*} = p\delta\sigma\frac{D_p - p\sigma}{n} - \frac{k}{2}$, the total matching volume is $M_O^{SP*} = L + D_p - \sigma p$, and the platform profit is $\pi_O^{SP*} = p(1 - \delta)(L + (D_p - \sigma p)\sigma)$. ■

SA.B Supplementary Proofs and Analyses on Model Extensions

SA.B.1 Finite part-time drivers

Proof of Proposition 4.

To explicitly incorporate a finite number of part-time drivers, we denote the ratio between part-time drivers and full-time drivers in an open system as ρ (>0). The total number of part-time drivers is then ρn. We denote the fraction of time per day that a part-time driver works as γ (note that we normalize the daily working hours of a full-time driver to 1, so $\gamma<1$). Therefore, each part-time driver provides a supply of γ per day, and the total daily supply from part-time drivers is $\rho n\gamma$.

Observing α_0, full-time drivers decide whether to participate the scheme, and we denote the proportions of prioritized and unprioritized drivers as $\theta\in(0,1]$ and $1-\theta$, respectively. $\hat{\alpha}_p$ and $\hat{\alpha}_{up}$ represent the average peak working time of prioriritized and unprioriritzed drivers, respectively. The optimization problems of a prioritized driver and an unprioritized driver are as follows.

1). For a prioritized driver, given $\alpha_0,\theta,\hat{\alpha}_p$, the problem is $\underset{\alpha_p\geq\alpha_0}{max}\ U_p(\alpha_p)=\underset{\alpha_p\geq\alpha_0}{max}\Big(Pr_p(\hat{\alpha}_p)p\delta(1-\alpha_p)+p\delta\alpha_p-\frac{1}{2}k\alpha_p^2\Big)$, where $Pr_p(\hat{\alpha}_p)=\int_{-s}^{n\theta(1-\hat{\alpha}_p)-L}\min\{1,\frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)}\}dF(\epsilon)$.

2). For an unprioritized driver, given $\alpha_0,\theta,\hat{\alpha}_p,\hat{\alpha}_{up}$, the problem is $\underset{\alpha_{up}<\alpha_0}{max}\ U_{up}(\alpha_{up})=Pr_{up}(\hat{\alpha}_p,\hat{\alpha}_{up})p\delta(1-\alpha_{up})+p\delta\alpha_{up}-\frac{1}{2}k\alpha_{up}^2$, where $Pr_{up}(\hat{\alpha}_p,\hat{\alpha}_{up})=\int_{n\theta(1-\hat{\alpha}_p)-L}^{s}\min\{1,\frac{D(\epsilon)-n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})+n\rho\gamma}\}dF(\epsilon)$.

We start by deriving α_{up}^* and proving $\alpha_p^*=\alpha_0$ as shown below. For the unprioritized driver's problem, solving FOC yields $\alpha_{up}(\hat{\alpha}_p,\hat{\alpha}_{up})=\frac{p\delta}{k}(1-Pr_{up}(\hat{\alpha}_p,\hat{\alpha}_{up}))$. Equating $\alpha_{up}(\hat{\alpha}_p,\hat{\alpha}_{up})=\hat{\alpha}_{up}$ yields $\alpha_{up}^\dagger$, which satisfies $\alpha_{up}^\dagger=\frac{p\delta}{k}(1-Pr_{up}(\hat{\alpha}_p,\alpha_{up}^\dagger))$. If $\alpha_0\leq\alpha_{up}^\dagger$, the peak target is so small that all drivers will be equally prioritized, and the prioritization scheme becomes ineffective. Thus, the service target $\alpha_0>\alpha_{up}^\dagger$ should be satisfied and the optimal working amount of an unprioritized driver in the peak period is $\alpha_{up}^*=\alpha_{up}^\dagger$. Then, we show that $\alpha_p^*=\alpha_0$. Note that $Pr_p(\hat{\alpha}_p)=\int_{-s}^{n\theta(1-\hat{\alpha}_p)-L}\frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)}dF(\epsilon)+\int_{n\theta(1-\hat{\alpha}_p)-L}^{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})+n\rho\gamma-L}dF(\epsilon)+\int_{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})+n\rho\gamma-L}^{s}dF(\epsilon)>\int_{n\theta(1-\hat{\alpha}_p)-L}^{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})+n\rho\gamma-L}\frac{D(\epsilon)-n\theta(1-\hat{\alpha}_0)}{n(1-\theta)(1-\hat{\alpha}_{up})+n\rho\gamma}dF(\epsilon)+\int_{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})+n\rho\gamma-L}^{s}dF(\epsilon)=Pr_{up}(\hat{\alpha}_p,\hat{\alpha}_{up})$. For the prioritized driver's problem, since $\frac{dU_p(\alpha_p)}{d\alpha_p}=\Big(1-Pr_p(\hat{\alpha}_p)\Big)p\delta-k\alpha_p\leq\Big(1-Pr_p(\hat{\alpha}_p)\Big)p\delta-k\alpha_0<\Big(1-Pr_p(\hat{\alpha}_p)\Big)p\delta-k\alpha_{up}^*=\Big(Pr_{up}(\hat{\alpha}_p,\alpha_{up}^*)-Pr_p(\hat{\alpha}_p)\Big)p\delta<0$, we have $\alpha_p^*=\alpha_0$.

Plugging in $\alpha_p^*=\alpha_0$, the platform's optimization problem becomes

$$\underset{\alpha_0\in(0,1)}{max}\ M(\alpha_0)=\underset{\alpha_0\in(0,1)}{max}\Big(\int_{-s}^{s}\min\{D(\epsilon),n\theta(1-\alpha_0)+n(1-\theta)(1-\alpha_{up}^*)+n\rho\gamma\}dF(\epsilon)+n\theta\alpha_0+n(1-\theta)\alpha_{up}^*\Big),$$

$$s.t.\ U_p^*(\alpha_0)\geq U_{up}^*(\alpha_{up}^*),\text{with the constraint binding if }\theta<1.$$

In the following, we first prove that, given the service target α_0, all drivers participate the qualification scheme, i.e., $\theta^*(\alpha_0)=1$. Then, we substitute $\theta^*(\alpha_0)=1$ and characterize the optimal decision on the service target α_0.

1). *Proof of* $\theta^*(\alpha_0)=1$

For simplicity, we will use θ^* to represent $\theta^*(\alpha_0)$ hereafter. Suppose (α_0^*,θ^*) is the optimal outcome, where $\theta^*\in(0,1)$ and $\alpha_0^*\in(\alpha_{up}^*,1)$. Then, the total matching volume is shown in Equation (SA.1), and subject to $U_p^*=U_{up}^*$, i.e., $Pr_p^*p\delta(1-\alpha_0^*)+p\delta\alpha_0^*-\frac{1}{2}k{\alpha_0^*}^2=(1-Pr_{up}^*)^2\frac{p^2\delta^2}{2k}+p\delta Pr_{up}^*$, where $\alpha_0^*>\alpha_{up}^*=\frac{p\delta}{k}(1-Pr_{up}^*)$, $Pr_p^*=\int_{-s}^{s}\min\{1,\frac{D(\epsilon)}{n\theta^*(1-\alpha_0^*)}\}dF(\epsilon)$ and $Pr_{up}^*=\int_{n\theta^*(1-\alpha_0^*)-L}^{s}\min\{1,\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma}\}dF(\epsilon)$.

$$M^*=\begin{cases}n+n\rho\gamma, & \text{if } n\theta^*(1-\alpha_0^*)+n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma-L<-s,\\ n+n\rho\gamma-\int_{-s}^{n\theta^*(1-\alpha_0^*)+n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma-L}F(\epsilon)d\epsilon, & \text{if } -s<n\theta^*(1-\alpha_0^*)+n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma-L<s,\\ L+n\theta^*\alpha_0^*+n(1-\theta^*)\alpha_{up}^*, & \text{if } n\theta^*(1-\alpha_0^*)+n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma-L>s,\end{cases}\quad\text{(SA.1)}$$

Let $\alpha_0'=\alpha_0^*-(1-\theta^*)(\alpha_0^*-\alpha_{up}^*)$. We first prove $\theta'=\theta|_{\alpha_0'=\alpha_0^*-(1-\theta^*)(\alpha_0^*-\alpha_{up}^*)}=1$. That is, all drivers participate in the scheme given the peak target α_0'. In the following, we prove this by showing $U_p'>U_{up}'$ in two steps.

- First, we prove $U_{up}^*>U_{up}'$. Note that $\frac{D(\epsilon)}{n(1-\alpha_0')}>\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma}$ as long as $D(\epsilon)<n(1-\alpha_0')$, thus $Pr_{up}^*=\int_{n\theta^*(1-\alpha_0^*)-L}^{n(1-\alpha_0')+n\rho\gamma-L}\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma}dF(\epsilon)+\int_{n(1-\alpha_0')+n\rho\gamma-L}^{s}dF(\epsilon)=\left(\int_{n\theta^*(1-\alpha_0^*)-L}^{n(1-\alpha_0')-L}\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma}dF(\epsilon)+\int_{n(1-\alpha_0')-L}^{n(1-\alpha_0')+n\rho\gamma-L}\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma}dF(\epsilon)\right)+\left(\int_{n(1-\alpha_0')-L}^{s}dF(\epsilon)-\int_{n(1-\alpha_0')-L}^{n(1-\alpha_0')+n\rho\gamma-L}dF(\epsilon)\right)=\int_{n\theta^*(1-\alpha_0^*)-L}^{n(1-\alpha_0')-L}\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma}dF(\epsilon)+\int_{n(1-\alpha_0')-L}^{s}dF(\epsilon)-\left(\int_{n(1-\alpha_0')-L}^{n(1-\alpha_0')+n\rho\gamma-L}dF(\epsilon)-\int_{n(1-\alpha_0')-L}^{n(1-\alpha_0')+n\rho\gamma-L}\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma}dF(\epsilon)\right)<\int_{n\theta^*(1-\alpha_0^*)-L}^{n(1-\alpha_0')-L}\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma}dF(\epsilon)+\int_{n(1-\alpha_0')-L}^{s}dF(\epsilon)<\int_{-s}^{n(1-\alpha_0')-L}\frac{D(\epsilon)}{n(1-\alpha_0')}dF(\epsilon)+\int_{n(1-\alpha_0')-L}^{s}dF(\epsilon)=Pr_p'$. Note that $\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma}>\frac{D(\epsilon)-n(1-\alpha_0')}{n\rho\gamma}$ as long as $D(\epsilon)<n(1-\alpha_0')+n\rho\gamma$, thus $Pr_{up}^*=\int_{n\theta^*(1-\alpha_0^*)-L}^{n(1-\alpha_0')+n\rho\gamma-L}\frac{D(\epsilon)-n\theta^*(1-\alpha_0^*)}{n(1-\theta^*)(1-\alpha_{up}^*)+n\rho\gamma}dF(\epsilon)+\int_{n(1-\alpha_0')+n\rho\gamma-L}^{s}dF(\epsilon)>\int_{n(1-\alpha_0')-L}^{n(1-\alpha_0')+n\rho\gamma-L}\frac{D(\epsilon)-n(1-\alpha_0')}{n\rho\gamma}dF(\epsilon)+\int_{n(1-\alpha_0')+n\rho\gamma-L}^{s}dF(\epsilon)=Pr_{up}'$. Therefore, $Pr_p'>Pr_{up}^*>Pr_{up}'$. Because the optimal utility for an unprioritized driver is $U_{up}=(1-Pr_{up})^2\frac{p^2\delta^2}{2k}+p\delta Pr_{up}$, which increases in Pr_{up}, then we have $U_{up}^*>U_{up}'$.
- Next, we prove $U_p'>U_{up}^*$. To prove $U_p'>U_{up}^*$ (i.e., $Pr'p\delta(1-\alpha_0')+p\delta\alpha_0'-\frac{1}{2}k{\alpha_0'}^2>Pr_{up}^*p\delta(1-\alpha_{up}^*)+p\delta\alpha_{up}^*-\frac{1}{2}k{\alpha_{up}^*}^2$), it is equivalent to prove $(\alpha_0'-\alpha_{up}^*)^2<2\frac{p\delta}{k}(Pr'-Pr_{up}^*)(1-\alpha_0')$. For the left-hand side of the inequality, $(\alpha_0'-\alpha_{up}^*)^2=(\theta^*)^2(\alpha_0^*-\alpha_{up}^*)^2=(\theta^*)^2(2\frac{p\delta}{k}(Pr_p^*-Pr_{up}^*)(1-\alpha_0^*))$, where the second equality is derived from $U_p^*=U_{up}^*$ (i.e., $Pr_p^*p\delta(1-\alpha_0^*)+p\delta\alpha_0^*-\frac{1}{2}k{\alpha_0^*}^2=Pr_{up}^*p\delta(1-\alpha_{up}^*)+p\delta\alpha_{up}^*-\frac{1}{2}k{\alpha_{up}^*}^2$). Therefore, proving $U_p'>U_{up}^*$ is equivalent to proving $(\alpha_0^*-\alpha_{up}^*)^2{\theta^*}^2<2\frac{p\delta}{k}(Pr'-Pr_{up}^*)(1-\alpha_0')$, i.e, $(Pr_p^*-Pr_{up}^*)(1-\alpha_0^*){\theta^*}^2<(Pr'-Pr_{up}^*)(1-\alpha_0')$.

  Let $\theta^*(1-\alpha_0^*)\equiv x,(1-\theta^*)(1-\alpha_{up}^*)\equiv y$, and $\rho\gamma\equiv z$, $x,y,z>0$. Through similar procedure, we have $Pr'-Pr_{up}^*=\frac{\int_{nx-L}^{nx+ny+nz-L}F(\epsilon)d\epsilon}{ny+nz}-\frac{\int_{-s}^{nx+ny-L}F(\epsilon)d\epsilon}{nx+ny}$ and $Pr_p^*-Pr_{up}^*=\frac{\int_{nx-L}^{nx+ny+nz-L}F(\epsilon)d\epsilon}{ny+nz}-\frac{\int_{-s}^{nx-L}F(\epsilon)d\epsilon}{x}$. Because $(Pr'-Pr_{up}^*)(1-\alpha_0')-(Pr_p^*-Pr_{up}^*)(1-\alpha_0^*){\theta^*}^2>(Pr'-Pr_{up}^*)(1-\alpha_0')-(Pr_p^*-Pr_{up}^*)(1-\alpha_0^*)\theta^*=(x+y)\frac{\int_{nx-L}^{nx+ny+nz-L}F(\epsilon)d\epsilon}{ny+nz}-\frac{\int_{-s}^{nx+ny-L}F(\epsilon)d\epsilon}{n}-\left(x\frac{\int_{x-L}^{nx+ny+nz-L}F(\epsilon)d\epsilon}{ny+nz}-\frac{\int_{-s}^{nx-L}F(\epsilon)d\epsilon}{n}\right)=y\frac{\int_{nx-L}^{nx+ny+nz-L}F(\epsilon)d\epsilon}{ny+nz}-\frac{\int_{nx-L}^{nx+ny-L}F(\epsilon)d\epsilon}{n}=\left(y\int_{nx-L}^{nx+ny+nz-L}F(\epsilon)d\epsilon-(y+z)\int_{nx-L}^{nx+ny-L}F(\epsilon)d\epsilon\right)/(ny+nz)=\left(y\int_{nx+ny-L}^{nx+ny+nz-L}F(\epsilon)d\epsilon-z\int_{nx-L}^{nx+ny-L}F(\epsilon)d\epsilon\right)/(ny+nz)$. Let $H[z]=y\int_{nx+ny-L}^{nx+ny+nz-L}F(\epsilon)d\epsilon-z\int_{nx-L}^{nx+ny-L}F(\epsilon)d\epsilon$ $(z>0)$, $\frac{dH[z]}{dz}=nyF[nx+ny+nz-L]-\int_{nx-L}^{nx+ny-L}F(\epsilon)d\epsilon>nyF[nx+ny-L]-\int_{nx-L}^{nx+ny-L}F(\epsilon)d\epsilon>0$, so $H[z]>H[0]=0$ and thus $(Pr'-Pr_{up}^*)(1-\alpha_0')>(Pr_p^*-Pr_{up}^*)(1-\alpha_0^*){\theta^*}^2$. Thus, we have proved $U_p'>U_{up}^*$.

Therefore, we have proved $U'_p > U'_{up}$ due to $U^*_{up} > U'_{up}$ and $U'_p > U^*_{up}$. Since the utility of participating in the scheme is strictly greater than the utility of not participating, given the peak target α'_0, all drivers will choose to participate in the scheme. That is, $\theta' = 1$.

Given the peak target $\alpha'_0 = \alpha^*_0 - (1-\theta^*)(\alpha^*_0 - \alpha^*_{up})$ and $\theta' = 1$, the total matching volume of M' is shown in Equation (SA.2), and subject to $U'_p \geq U'_{up}$, i.e., $Pr'_p p\delta(1-\alpha'_0) + p\delta\alpha'_0 - \frac{1}{2}k{\alpha'_0}^2 \geq (1-Pr'_{up})^2\frac{p^2\delta^2}{2k} + p\delta Pr'_{up}$, where $\alpha'_0 > \frac{p\delta}{k}(1-Pr'_{up})$, $Pr'_p = \int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n(1-\alpha'_0)}\}dF(\epsilon)$ and $Pr'_{up} = \int_{n(1-\alpha'_0)-L}^{s}\min\{1, \frac{D(\epsilon)-n(1-\alpha'_0)}{n\rho\gamma}\}dF(\epsilon)$. We can easily show $M' = M^*$.

$$M' = \begin{cases} n + n\rho\gamma, & \text{if } n(1-\alpha'_0) + n\rho\gamma - L < -s, \\ n + n\rho\gamma - \int_{-s}^{n(1-\alpha'_0)-L} F(\epsilon)d\epsilon, & \text{if } -s < n(1-\alpha'_0) + n\rho\gamma - L < s, \\ L + n\alpha'_0, & \text{if } n(1-\alpha'_0) + n\rho\gamma - L > s, \end{cases} \tag{SA.2}$$

Given the peak target $\alpha'_0 = \alpha^*_0 - (1-\theta^*)(\alpha^*_0 - \alpha^*_{up})$, we have shown $\theta' = 1$ and $M' = M^*$. Since $\frac{dM'}{d\alpha'_0} > 0$, the platform can enhance the matching volume by choosing a peak target greater than α'_0, in which case the total matching volume exceeds M^*. This contradicts with that (α^*_0, θ^*) is the optimal outcome. Therefore, $\theta^* = 1$ in equilibrium.

2). *Characterization of optimal α_0, denoted by ${\alpha_0}_O^{P\dagger}$*

Next, we plug in $\theta^*(\alpha_0) = 1$ and characterize the optimal α_0. Note that given α_0, being an unprioritized driver will optimally obtain $U^*_{up} = p\delta Pr_{up} + (1-Pr_{up})^2\frac{p^2\delta^2}{2k}$ by working $\alpha^*_{up}(\alpha_0) = \frac{p\delta(1-Pr_{up})}{k}$ during peak, where $Pr_{up} = \int_{n(1-\alpha_0)-L}^{s}\min\{1, \frac{D(\epsilon)-n(1-\alpha_0)}{n\rho\gamma}\}dF(\epsilon)$. A prioritized driver obtains $U_p(\alpha_0) = Pr_p p\delta(1-\alpha_0) + p\delta\alpha_0 - \frac{1}{2}k\alpha_0^2$, where $Pr_p = \int_{-s}^{n(1-\alpha_0)-L}\min\{1, \frac{D(\epsilon)}{n(1-\alpha_0)}\}dF(\epsilon)$. Because $\frac{dU_p(\alpha_0)}{d\alpha_0} < 0$ and $\frac{dU^*_{up}}{d\alpha_0} > 0$, there should be at most one unique solution of $U_p(\alpha_0) = U^*_{up}$, and the optimal peak target, denoted by ${\alpha_0}_O^{P\dagger}$, solves $U_p({\alpha_0}_O^{P\dagger}) = U^*_{up}$.

3). *Proof of $\frac{d{\alpha_0}_O^{P\dagger}}{d\rho} \geq 0$*

Then, we show that the optimal peak target ${\alpha_0}_O^{P\dagger}$ increases in the number of part-time drivers ρ by considering the following five cases, which are determined by the relative size of demand $(L+\epsilon)$ and supplies $(n(1-{\alpha_0}_O^{P\dagger}))$ from only full-time drivers and $(n(1-{\alpha_0}_O^{P\dagger}) + n\rho\gamma)$ from both full-time and part-time drivers during the off-peak period. Let $F[n(1-{\alpha_0}_O^{P\dagger}) + n\rho\gamma - L] \equiv F_1$ and $F[n(1-{\alpha_0}_O^{P\dagger}) - L] \equiv F_0$.

- *Case 1: $n(1-{\alpha_0}_O^{P\dagger}) + n\rho\gamma \leq \min\{D(\epsilon)\} = L - s$*

  In this case, $Pr_{up} = \int_{n(1-\alpha_0)-L}^{s} dF(\epsilon) = 1$, so $U^*_{up} = p\delta > U_p(\alpha_0)$. Therefore, ${\alpha_0}_O^{P\dagger}$ does not exist and thus this case will not occur in equilibrium.

- *Case 2: $n(1-{\alpha_0}_O^{P\dagger}) \leq \min\{D(\epsilon)\} = L - s$ and $L - s = \min\{D(\epsilon)\} < n(1-{\alpha_0}_O^{P\dagger}) + n\rho\gamma \leq \max\{D(\epsilon)\}$*

  In this case, $U_p({\alpha_0}_O^{P\dagger}) = U^*_{up}({\alpha_0}_O^{P\dagger})$ is equivalent to $p\delta(1-{\alpha_0}_O^{P\dagger}) + p\delta{\alpha_0}_O^{P\dagger} - \frac{1}{2}k{{\alpha_0}_O^{P\dagger}}^2 = p\delta Pr_{up} + (1-Pr_{up})^2\frac{p^2\delta^2}{2k}$, where $Pr_{up} = 1 - \frac{\int_{-s}^{n(1-{\alpha_0}_O^{P\dagger})+n\rho\gamma-L} F(\epsilon)d\epsilon}{n\rho\gamma}$. Differentiating ${\alpha_0}_O^{P\dagger}$ with respect to ρ and using the implicit differentiation method yield $\frac{d{\alpha_0}_O^{P\dagger}}{d\rho} = \frac{p\delta\left(n\rho\gamma F_1 - \int_{-s}^{n(1-{\alpha_0}_O^{P\dagger})+n\gamma\rho-L} F(\epsilon)\,d\epsilon\right)\left(kn\gamma\rho - p\delta\int_{-s}^{n(1-{\alpha_0}_O^{P\dagger})+n\gamma\rho-L} F(\epsilon)\,d\epsilon\right)}{n\rho(n{\alpha_0}_O^{P\dagger}k^2\rho^2\gamma^2 + p\delta F_1(kn\gamma\rho - p\delta\int_{-s}^{n(1-{\alpha_0}_O^{P\dagger})+n\gamma\rho-L} F(\epsilon)d\epsilon))} > 0$ due to $F_1 > \frac{\int_{-s}^{n(1-{\alpha_0}_O^{P\dagger})+n\gamma\rho-L} F(\epsilon)d\epsilon}{n\rho\gamma}$.

- *Case 3:* $n(1-\alpha_{0_O}^{P\dagger}) \le \min\{D(\epsilon)\} = L - s$ *and* $n(1-\alpha_{0_O}^{P\dagger}) + n\rho\gamma > \max\{D(\epsilon)\} = L + s$

  In this case, $U_p(\alpha_{0_O}^{P\dagger}) = U_{up}^*(\alpha_{0_O}^{P\dagger})$ is equivalent to $p\delta(1-\alpha_{0_O}^{P\dagger}) + p\delta\alpha_{0_O}^{P\dagger} - \frac{1}{2}k{\alpha_{0_O}^{P\dagger}}^2 = p\delta Pr_{up} + (1 - Pr_{up})^2\frac{p^2\delta^2}{2k}$, where $Pr_{up} = \frac{L-n(1-\alpha_{0_O}^{P\dagger})}{n\rho\gamma}$. Differentiating $\alpha_{0_O}^{P\dagger}$ with respect to ρ and using the implicit differentiation method yield $\frac{d\alpha_{0_O}^{P\dagger}}{d\rho} = \frac{p\delta(L-n(1-\alpha_{0_O}^{P\dagger}))(np\delta(\alpha_{0_O}^{P\dagger}-\gamma\rho-1)+\gamma kn\rho+Lp\delta)}{n\rho\left(n\left(\alpha_{0_O}^{P\dagger}\gamma^2k^2\rho^2+p^2\delta^2(\alpha_{0_O}^{P\dagger}-\gamma\rho-1)+\gamma kp\delta\rho\right)+Lp^2\delta^2\right)} > 0$ due to $L > n(1-\alpha_{0_O}^{P\dagger})$.

- *Case 4:* $\min\{D(\epsilon)\} < n(1-\alpha_{0_O}^{P\dagger}) \le \max\{D(\epsilon)\}$ *and* $\min\{D(\epsilon)\} < n(1-\alpha_{0_O}^{P\dagger}) + n\rho\gamma \le \max\{D(\epsilon)\}$

  In this case, $U_p(\alpha_{0_O}^{P\dagger}) = U_{up}^*(\alpha_{0_O}^{P\dagger})$ is equivalent to $(1 - \frac{\int_{-s}^{n(1-\alpha_0)-L}F(\epsilon)d\epsilon}{n(1-\alpha_0)})p\delta(1-\alpha_{0_O}^{P\dagger}) + p\delta\alpha_{0_O}^{P\dagger} - \frac{1}{2}k{\alpha_{0_O}^{P\dagger}}^2 = p\delta Pr_{up} + (1-Pr_{up})^2\frac{p^2\delta^2}{2k}$, where $Pr_{up} = 1 - \frac{\int_{n(1-\alpha_{0_O}^{P\dagger})-L}^{n(1-\alpha_{0_O}^{P\dagger})+n\rho\gamma-L}F(\epsilon)d\epsilon}{n\rho\gamma}$. Differentiating $\alpha_{0_O}^{P\dagger}$ with respect to ρ and using the implicit differentiation method yield $\frac{d\alpha_{0_O}^{P\dagger}}{d\rho} = \frac{p\delta(n\rho\gamma F_1 - \int_{n(1-\alpha_{0_O}^{P\dagger})-L}^{n(1-\alpha_{0_O}^{P\dagger})+n\rho\gamma-L}F(\epsilon)d\epsilon)(kn\rho\gamma - p\delta\int_{n(1-\alpha_{0_O}^{P\dagger})-L}^{n(1-\alpha_{0_O}^{P\dagger})+n\rho\gamma-L}F(\epsilon)d\epsilon)}{n\rho(p\delta(F_1-F_0)(kn\rho\gamma - p\delta\int_{n(1-\alpha_{0_O}^{P\dagger})-L}^{n(1-\alpha_{0_O}^{P\dagger})+n\rho\gamma-L}F(\epsilon)d\epsilon + kn\rho^2\gamma^2(k\alpha_{0_O}^{P\dagger} - p\delta F_0))} > 0$ due to $F_1 > \frac{\int_{-s}^{n(1-\alpha_{0_O}^{P\dagger})+n\gamma\rho-L}F(\epsilon)d\epsilon}{n\rho\gamma}$ and $\alpha_{0_O}^{P\dagger} > \frac{p\delta}{k}(1-Pr_{up}) = \frac{p\delta}{k}\frac{\int_{n(1-\alpha_{0_O}^{P\dagger})-L}^{n(1-\alpha_{0_O}^{P\dagger})+n\rho\gamma-L}F(\epsilon)d\epsilon}{n\rho\gamma} > \frac{p\delta}{k}F_0$.

- *Case 5:* $L - s = \min\{D(\epsilon)\} < n(1-\alpha_{0_O}^{P\dagger}) \le \max\{D(\epsilon)\} = L + s$ *and* $n(1-\alpha_{0_O}^{P\dagger}) + n\rho\gamma > \max\{D(\epsilon)\}$

  In this case, $U_p(\alpha_{0_O}^{P\dagger}) = U_{up}^*(\alpha_{0_O}^{P\dagger})$ is equivalent to $(1 - \frac{\int_{-s}^{n(1-\alpha_0)-L}F(\epsilon)d\epsilon}{n(1-\alpha_0)})p\delta(1-\alpha_{0_O}^{P\dagger}) + p\delta\alpha_{0_O}^{P\dagger} - \frac{1}{2}k{\alpha_{0_O}^{P\dagger}}^2 = p\delta Pr_{up} + (1-Pr_{up})^2\frac{p^2\delta^2}{2k}$, where $Pr_{up} = \frac{L-n(1-\alpha_{0_O}^{P\dagger}\alpha_{0_O}^{P\dagger})+\int_{-s}^{n(1-\alpha_{0_O}^{P\dagger})-L}F(\epsilon)d\epsilon}{n\rho\gamma}$. Differentiating $\alpha_{0_O}^{P\dagger}$ with respect to ρ and using the implicit differentiation method yield $\frac{d\alpha_{0_O}^{P\dagger}}{d\rho} = \frac{p\delta\left(\int_{-s}^{n\left(1-\alpha_{0_O}^{P\dagger}\right)-L}F(\epsilon)\,d\epsilon + L - \left(1-\alpha_{0_O}^{P\dagger}\right)n\right)\left(p\delta\left(\int_{-s}^{n\left(1-\alpha_{0_O}^{P\dagger}\right)-L}F(\epsilon)\,d\epsilon + L - n\left(1-\alpha_{0_O}^{P\dagger}\right)\right)+\gamma n\rho(k-p\delta)\right)}{n\rho\left((1-F_0)p^2\delta^2\left(\int_{-s}^{n(1-\alpha_{0_O}^{P\dagger})-L}F(\epsilon)d\epsilon + L - n\left(1-\alpha_{0_O}^{P\dagger}\right)\right)+\gamma^2kn\rho^2\left(\alpha_{0_O}^{P\dagger}k - F_0p\delta\right)+\gamma(1-F_0)np\delta\rho(k-p\delta)\right)} > 0.$

- *Case 6:* $n(1-\alpha_{0_O}^{P\dagger}) > \max\{D(\epsilon)\} = L + s$

  In this case, $U_p(\alpha_{0_O}^{P\dagger}) = U_{up}^*(\alpha_{0_O}^{P\dagger})$ is equivalent to $\frac{L}{n(1-\alpha_{0_O}^{P\dagger})}p\delta(1-\alpha_{0_O}^{P\dagger}) + p\delta\alpha_{0_O}^{P\dagger} - \frac{1}{2}k{\alpha_{0_O}^{P\dagger}}^2 = \frac{p^2\delta^2}{2k}$. Differentiating $\alpha_{0_O}^{P\dagger}$ with respect to ρ and using the implicit differentiation method yield $\frac{d\alpha_{0_O}^{P\dagger}}{d\rho} = 0$.

Then, for $M_O^{P\dagger} = M(\alpha_{0_O}^{P\dagger}) = \int_{-s}^{s}\min\{D(\epsilon), n(1-\alpha_{0_O}^{P\dagger}) + n\rho\gamma\}dF(\epsilon) + n\alpha_{0_O}^{P\dagger}$, we have $\frac{\partial M_O^{P\dagger}}{\partial\rho} > 0$ and $\frac{\partial M_O^{P\dagger}}{\partial\alpha_{0_O}^{P\dagger}} > 0$, so $\frac{dM_O^{P\dagger}}{d\rho} = \frac{\partial M_O^{P\dagger}}{\partial\rho} + \frac{\partial M_O^{P\dagger}}{\partial\alpha_{0_O}^{P\dagger}}\frac{d\alpha_{0_O}^{P\dagger}}{d\rho} > 0$. Note $M_O^{P\dagger}|_{\rho=0} = M_C^P < M_C^Q$ and $M_O^{P\dagger}|_{\rho\to\infty} = M_O^P > M_O^Q = M_C^Q$, so there exists ρ^* that satisfies $M_O^{P\dagger}|_{\rho=\rho^*} = M_C^Q$ so that $M_O^{P\dagger} > M_C^Q$ if $\rho > \rho^*$, where ρ^* satisfies $\int_{-s}^{s}\min\{D(\epsilon), n(1-\alpha_{0_O}^{P\dagger}(\rho^*)) + n\rho^*\gamma\}dF(\epsilon) + n\alpha_{0_O}^{P\dagger}(\rho^*) = M_C^Q$ and M_C^Q can be found in Lemma 3. ■

SA.B.2 Heterogeneous congestion cost

Proof of Lemma A.3.

1). *Qualification scheme*

For an individual unqualified driver, the optimal working time during the peak period is given by: $\alpha_{iuq}^* = \frac{p\delta}{k_i}$, and the corresponding utility is: $U_{iuq}^* = \frac{p^2\delta^2}{2k_i}$, $i \in \{h, l\}$. To involve both types of drivers, the peak target α_0 must satisfy $\alpha_0 \in (\frac{p\delta}{k_h}, 1)$. Upon observing α_0, a proportion $\theta_i \in (0, 1]$ of drivers in group $i \in \{h, l\}$ choose to become qualified, while the remaining proportion $1-\theta_i$ remain unqualified.

We first show that a high-type qualified driver optimally works binding to α_0, i.e., $\alpha^*_{hq} = \alpha_0$. For a high-type qualified driver, the optimization problem is given by: $\underset{\alpha_{hq}\geq\alpha_0}{max}\ U_{hq}(\alpha_{hq}) = \underset{\alpha_{hq}\geq\alpha_0}{max}\ Pr^Q p\delta(1-\alpha_{hq}) + p\delta\alpha_{hq} - \frac{1}{2}k_h\alpha_{hq}^2$, where $Pr^Q = \int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n\phi\theta_h(1-\hat{\alpha}_{hq})+n(1-\phi)\theta_l(1-\hat{\alpha}_{lq})}\}dF(\epsilon)$ and $\hat{\alpha}_{iq}$ denote the average working time during the peak period of the driver in group i. Note that $\frac{dU_{hq}(\alpha_{hq})}{d\alpha_{hq}} = (1-Pr^Q)p\delta - k_h\alpha_{hq} < (1-Pr^Q)p\delta - k_h\frac{p\delta}{k_h} < 0$, so $\alpha^*_{hq} = \alpha_0$.

We next show that all low-type drivers choose to be qualified, i.e., $\theta^*_l = 1$. If $\alpha_0 < \frac{p\delta}{k_l}$, low-type drivers are naturally qualified, so $\theta^*_l = 1$. If $\alpha_0 \geq \frac{p\delta}{k_l}$, we also find that a low-type qualified driver optimally works binding to α_0 using the same technique as in the proof of Lemma 3. Then, the utility for a qualified driver of type i is given by: $U_{iq} = Pr^Q p\delta(1-\alpha_0) + p\delta\alpha_0 - \frac{1}{2}k_i\alpha_0^2$, where $Pr^Q = \int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n\phi\theta_h(1-\alpha_0)+n(1-\phi)\theta_l(1-\alpha_0)}\}dF(\epsilon)$. To involve both types of drivers, there must be $U_{iq} - U^*_{iuq} \geq 0$. We have $\frac{d(U_{iq}-U^*_{iuq})}{dk_i} = -\frac{1}{2}(\alpha_0^2 - \frac{p^2\delta^2}{k_i^2}) < 0$, so $U_{lq} - U^*_{luq} > U_{hq} - U^*_{huq} \geq 0$ and thus $\theta^*_l = 1$. Note that a low-type driver spontaneously works up to $\frac{p\delta(1-Pr^Q)}{k_l}$ if $\alpha_0 < \frac{p\delta(1-Pr^Q)}{k_l}$, which is equivalent to $k_l < \frac{p\delta(1-Pr^Q)}{\alpha_0}$. To stay focused, we consider the case where the congestion cost for low-type drivers is not too small, i.e., $k_l \geq \frac{p\delta(1-Pr^Q)}{\alpha_0}$, such that they spontaneously choose to work exactly at α_0.

Through the same procedure as that with homogeneous cost in Lemma 3, we have $\theta^*_h = 1$ in equilibrium. Therefore, the optimal peak target, denoted by α_0^{Qb}, should satisfy $U_{hq}(\alpha_0^{Qb}) = U^*_{huq}$, i.e., $\int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n(1-\alpha_0^{Qb})}\}dF(\epsilon)p\delta(1-\alpha_0^{Qb}) + p\delta\alpha_0^{Qb} - \frac{1}{2}k_h{\alpha_0^{Qb}}^2 = \frac{p^2\delta^2}{2k_h}$, in which case low-type drivers will obtain a higher utility than being the unqualified. We observe that if replacing k with k_h, α_0^Q (see Lemma 3) and α_0^{Qb} are determined by the same equation. Then, the total matching volume is

$$M_C^{Qb} = \int_{-s}^{s}\min\{D(\epsilon), n(1-\alpha_0^{Qb})\}dF(\epsilon) + n\alpha_0^{Qb}.$$

2). <u>*Prioritization scheme*</u>

Denote by $\hat{\alpha}_{hp}, \hat{\alpha}_{lp}$ and $\hat{\alpha}_{hup}, \hat{\alpha}_{lup}$ the average peak work time of prioritized high-type and low-type drivers, as well as unprioritized high-type and low-type drivers, respectively. For a prioritized driver, the optimization problem is given by: $\underset{\alpha_{ip}\geq\alpha_0}{max}\ U_{ip}(\alpha_{ip}) = \underset{\alpha_{ip}\geq\alpha_0}{max}\ \left(Pr^P p\delta(1-\alpha_{ip}) + p\delta\alpha_{ip} - \frac{1}{2}k_i\alpha_{ip}^2\right)$, $i \in \{l,h\}$, where $Pr^P = \int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n(1-\phi)\theta_l(1-\hat{\alpha}_{lp})+n\phi\theta_h(1-\hat{\alpha}_{hp})}\}dF(\epsilon)$. For an unprioritized driver, the optimization problem is given by: $\underset{\alpha_{iup}<\alpha_0}{max}\ U_{iup}(\alpha_{iup}) = \underset{\alpha_{iup}<\alpha_0}{max}\ \left(Pr_{up}p\delta(1-\alpha_{iup}) + p\delta\alpha_{iup} - \frac{1}{2}k_i\alpha_{iup}^2\right)$, where $Pr_{up} = \int_{n(1-\phi)\theta_l(1-\hat{\alpha}_{lp})+n\phi\theta_h(1-\hat{\alpha}_{hp})-L}^{s}\min\{1, \frac{D(\epsilon)-n(1-\phi)\theta_l(1-\hat{\alpha}_{lp})-n\phi\theta_h(1-\hat{\alpha}_{hp})}{n(1-\phi)(1-\theta_l)(1-\hat{\alpha}_{lup})+n\phi(1-\theta_h)(1-\hat{\alpha}_{hup})}\}dF(\epsilon)$. Similar to the proof of Lemma 4, we can show $Pr^P > \int_{n(1-\phi)\theta_l(1-\hat{\alpha}_{lp})+n\phi\theta_h(1-\hat{\alpha}_{hp})-L}^{s}dF(\epsilon) > Pr_{up}$. Then, because $U_{iup}(\alpha_{iup})$ is concave in α_{iup}, solving FOC yields $\alpha^*_{iup} = \frac{p\delta(1-Pr_{up})}{k_i}$. We focus on the scenario that α_0 is not too small (i.e., $\alpha_0 > \frac{p\delta(1-Pr_{up})}{k_i}$) and thus can differentiate two prioritized and unprioritized drivers for each type of drivers. Plugging $\alpha^*_{iup} = \frac{p\delta(1-Pr_{up})}{k_i}$ in $U_{ip}(\alpha_{ip})$ yields $U^*_{ip} = p\delta Pr_{up} + (1-Pr_{up})^2\frac{p^2\delta^2}{2k_i}$. We next show that a prioritized driver optimally works α_0 during peak periods. Because $U_{ip}(\alpha_{ip})$ is concave in α_{ip} and $\frac{dU_{ip}(\alpha_{ip})}{d\alpha_{ip}} = (1-Pr^P)p\delta - k_i\alpha_{ip} \leq (1-Pr^P)p\delta - k_i\alpha_0 < (1-Pr^P)p\delta - k_i\frac{p\delta(1-Pr_{up})}{k_i} =$

$(1-Pr^P)p\delta - k_i\frac{p\delta(1-Pr_{up})}{k_i} = (Pr_{up}-Pr^P)p\delta < 0$, so $\alpha_{ip}^* = \alpha_0$. To involve both types, there must be $U_{ip} - U_{iup}^* \geq 0$, i.e., $Pr^P p\delta(1-\alpha_0) + p\delta\alpha_0 - \frac{1}{2}k_i\alpha_0^2 - (p\delta Pr_{up} + (1-Pr_{up})^2\frac{p^2\delta^2}{2k_i}) \geq 0$. Note the LHS of the inequality decreases in k_i, so if $U_{hp} - U_{hup}^* = 0$, $U_{lp} - U_{lup}^* > 0$ always holds. Therefore, if the platform involves high-type drivers to the prioritization scheme, all low-type drivers will also participate the scheme, i.e., $\theta_l^* = 1$. Similarly, we consider the case where low-type drivers' congestion cost is not too small and they spontaneously work binding to α_0, i.e., $k_l \geq \frac{p\delta(1-Pr^P)}{\alpha_0}$. Then, using the same technique as that with homogeneous cost in Lemma 4, we also have $\theta_h^* = 1$.

Therefore, the optimal peak target, denoted by α_0^{Pb}, should satisfy $U_{hp}(\alpha_0^{Pb}) = U_{hup}^*$, i.e., $\int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n(1-\alpha_0^{Pb})}\}dF(\epsilon)p\delta(1-\alpha_0^{Pb}) + p\delta\alpha_0^{Pb} - \frac{1}{2}k_h{\alpha_0^{Pb}}^2 = p\delta(1-F[n(1-\alpha_0^{Pb})-L])) + (F[n(1-\alpha_0^{Pb})-L])^2\frac{p^2\delta^2}{2k_h}$, in which case low-type drivers will obtain a higher utility than being the unprioritized. We observe that if replacing k with k_h, α_0^P (see Lemma 4) and α_0^{Pb} are determined by the same equation. Then, the total matching volume is

$$M_C^{Pb} = \int_{-s}^{s}\min\{D(\epsilon), n(1-\alpha_0^{Pb})\}dF(\epsilon) + n\alpha_0^{Pb}.$$

3). *Comparison*

Note that α_0^{Qb} is determined by $\int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n(1-\alpha_0^{Qb})}\}dF(\epsilon)p\delta(1-\alpha_0^{Qb}) + p\delta\alpha_0^{Qb} - \frac{1}{2}k_h{\alpha_0^{Qb}}^2 = \frac{p^2\delta^2}{2k_h}$, and α_0^{Pb} is determined by $\int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n(1-\alpha_0^{Pb})}\}dF(\epsilon)p\delta(1-\alpha_0^{Pb}) + p\delta\alpha_0^{Pb} - \frac{1}{2}k_h{\alpha_0^{Pb}}^2 = p(1-F[n(1-\alpha_0^{Pb})-L]) + (F[n(1-\alpha_0^{Pb})-L])^2\frac{p^2\delta^2}{2k_h}$. Similar to the proof of Proposition 1, we can show that $\alpha_0^{Qb} > \alpha_0^{Pb}$. Note that the thresholds of k_l in the incentive scheme $j \in \{Q, P\}$ is $\frac{p\delta(1-\int_{-s}^{s}\min\{1, D(\epsilon)/n(1-\alpha_0^{jb})\}F(\epsilon)d\epsilon)}{\alpha_0^{jb}}$ decreases in α_0, so $\frac{p\delta(1-\int_{-s}^{s}\min\{1, D(\epsilon)/n(1-\alpha_0^{Pb})\}F(\epsilon)d\epsilon)}{\alpha_0^{Pb}} > \frac{p\delta(1-\int_{-s}^{s}\min\{1, D(\epsilon)/n(1-\alpha_0^{Qb})\}F(\epsilon)d\epsilon)}{\alpha_0^{Qb}}$ and thus the low-type drivers always work up to α_0 as long as $k_l > \frac{p\delta(1-\int_{-s}^{s}\min\{1, D(\epsilon)/n(1-\alpha_0^{Pb})\}F(\epsilon)d\epsilon)}{\alpha_0^{Pb}}$. In addition, because $\frac{dM_C^{jb}}{d\alpha_0^{jb}} > 0$, we have $M_C^{Qb} > M_C^{Pb}$. ■

SA.B.3 Discussion on alternative driver composition

In this section, we first consider drivers who are constrained to work during peak periods. Next, we consider a finite number of drivers who have the flexibility to choose in which period to work, but not both, and work fewer hours than full-time drivers, which we refer to as flexible part-time drivers in this section. We examine the robustness of our main results by exploring the above scenarios.

(1) Consider drivers who are constrained to work during peak periods

Suppose the size of drivers who can only work during peak periods is m (> 0). Note that in our main model, the peak demand is assumed to be infinite, resulting in a consistent under-supply during peak periods. Therefore, including these drivers simply increases supply during peak periods without affecting the decisions of full-time drivers or part-time drivers who are constrained to work during off-peak periods. As a result, the full-time drivers' working time decisions during peak periods in the benchmark scenario, the platform's decision on peak target α_0 when implementing incentive schemes, and the full-time drivers'

utility across all scenarios remain unchanged. However, the platform's total matching volume increases from M^i to $M^i + m$, where $M^i, i \in \{B, Q, P\}$, represents the equilibrium total matching volume under the benchmark, the qualification scheme, and the prioritization scheme, respectively.

(2) Consider a finite subset of flexible part-time drivers

We assume that a finite subset of part-time drivers are flexible and have the option to choose in which period to work, but not both, due to their limited working hours. We denote the size of these flexible part-time drivers as βn, where $\beta > 0$ represents the ratio of the size of this subset of part-time drivers to that of the full-time drivers, whose size is denoted as n. Recall that the total daily working time of a full-time driver is normalized to 1. We denote the daily working time of a part-time driver as γ and assume that $\gamma \in (0, 1)$. The remaining part-time drivers, without the luxury of choosing the working period, can only work during off-peak periods.

Since there are infinite part-time drivers who are restricted to working during the off-peak period, flexible part-time drivers will choose to work during the peak period, because the expected profit is zero if serving during off-peak times.

Note that the key issue this study trying to address is the supply-demand imbalance, especially the supply shortage during the peak peirod. Therefore, as long as the service supply is smaller than the demand even when these flexible part-time drivers provide service during the peak period (i.e., $D_p - p > n + \beta n\gamma$), a full-time driver's time allocation between peak and off-peak periods remains unchanged and our main results remain robust. However, if with the participation of flexible part-time drivers, some "peak hours" become oversupplied, that is, the supply during these "peak hours" is enough to meet the demand even without any platform intervention (incentive scheme, surge pricing, etc.), then these so-called "peak hours" are not the focus of our study and should not be defined as peak hours when implementing incentive schemes.

SA.B.4 Volatile peak demand

In the main model, we assume the peak period is always under-supplied even when full-time drivers devote all their time to the peak period (i.e. $D_p > n$). In this section, we relax this assumption by considering the scenario with volatile peak demand. Specifically, we suppose the peak demand is $D_l \in (n - L + s, n)$ with probability $\beta \in (0, 1)$, and $D_h \geq n$ with probability $1 - \beta$.

We first show that, in a closed system, the platform sets a (weakly) higher peak period work target when employing a qualification scheme, as opposed to a prioritization scheme. Additionally, implementing a qualification scheme increases the total matching volume compared to the prioritization scheme. These results are consistent with the findings in Proposition 1. Next, we show that, in an open system, implementing a prioritization scheme (weakly) increases the platform's total matching volume compared to the qualification scheme, which is consistent with the conclusion in Proposition 3.

1). *Closed system*

Suppose upon observing α_0, a proportion $\theta \in (0,1)$ and $1-\theta$ of full-time drivers choose to participate and not participate the scheme respectively.

- *Qualification scheme*

  Given the average work time during the peak period $\hat{\alpha}_{uq}$ and $\hat{\alpha}_q$ of unqualified drivers and qualified drivers respectively, for an unqualified driver, the optimization problem is given by: $\underset{\alpha_{uq}<\alpha_0}{max}\ U_{uq}(\alpha_{uq}) = \underset{\alpha_{uq}<\alpha_0}{max}\ Pr_{peak}(\hat{\alpha}_q,\hat{\alpha}_{uq})\cdot p\delta\alpha_{uq} - \frac{1}{2}k\alpha_{uq}^2$, where $Pr_{peak}(\hat{\alpha}_q,\hat{\alpha}_{uq}) = \beta\min\{\frac{D_l}{n(1-\theta)\hat{\alpha}_{uq}+n\theta\hat{\alpha}_q},1\}+1-\beta$. Because $U_{uq}(\alpha_{uq})$ is concave in α_{uq}, we solve FOC and obtain $\alpha_{uq}^{\dagger}(\hat{\alpha}_{uq}) = \frac{p\delta}{k}Pr_{peak}(\hat{\alpha}_q,\hat{\alpha}_{uq})$. Similarly, we here focus on the case where the service target $\alpha_0 > \alpha_{uq}^{\dagger}(\hat{\alpha}_{uq})$. The equilibrium condition requires $\alpha_{uq}^{\dagger}(\hat{\alpha}_{uq}) = \hat{\alpha}_{uq}$, which yields α_{uq}^* that solves $\alpha_{uq}^* = \frac{p\delta}{k}Pr_{peak}(\hat{\alpha}_q,\alpha_{uq}^*)$ and thus $U_{uq}^* = \frac{p^2\delta^2}{2k}(Pr_{peak}(\hat{\alpha}_q,\alpha_{uq}^*))^2$. For a qualified driver, the optimization problem is given by: $\underset{\alpha_q\geq\alpha_0}{max}\ U_q(\alpha_q) = \underset{\alpha_q\geq\alpha_0}{max}\ Pr^Q(\hat{\alpha}_q)p\delta(1-\alpha_q)+Pr_{peak}(\hat{\alpha}_q,\alpha_{uq}^*)\cdot p\delta\alpha_q - \frac{1}{2}k\alpha_q^2$, where $Pr^Q(\hat{\alpha}_q) = \int_{-s}^{s}\min\{1,\frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_q)}\}dF(\epsilon)$. Note that $\frac{dU_q(\alpha_q)}{d\alpha_q} = (Pr_{peak}(\hat{\alpha}_q,\alpha_{uq}^*)-Pr^Q(\hat{\alpha}_q))p\delta - k\alpha_q \leq (Pr_{peak}(\hat{\alpha}_q,\alpha_{uq}^*)-Pr^Q(\hat{\alpha}_q))p\delta - k\alpha_0 < (Pr_{peak}(\hat{\alpha}_q,\alpha_{uq}^*)-Pr^Q(\hat{\alpha}_q))p\delta - k\frac{p\delta}{k}Pr_{peak}(\hat{\alpha}_q,\alpha_{uq}^*) < 0$, so we have $\alpha_q^* = \hat{\alpha}_q = \alpha_0$. Then, the total matching volume can be written as

$$\begin{aligned} M_C^Q(\theta(\alpha_0)) = &\int_{-s}^{s}\min\{D(\epsilon),n\theta(1-\alpha_0)\}dF(\epsilon) + (1-\beta)(n\theta\alpha_0 + n(1-\theta)\alpha_{uq}^*) \\ &+ \beta\min\{D_l, n\theta\alpha_0 + n(1-\theta)\alpha_{uq}^*\}. \end{aligned}$$

  Through the similar procedure as in the proof of Lemma 3, we can show $\theta^*(\alpha_0) = 1$. The total matching volume becomes $M_C^Q(\alpha_0) = \int_{-s}^{s}\min\{D(\epsilon),n(1-\alpha_0)\}dF(\epsilon) + (1-\beta)n\alpha_0 + \beta\min\{D_l, n\alpha_0\}$, the platform then maximizes $M_C^Q(\alpha_0)$ by setting α_0 so that $U_q(\alpha_0) \geq U_{uq}^*$. The process of deriving optimal α_0^Q is similar to the proof of Lemma 3.

- *Prioritization scheme*

  Denote by $\hat{\alpha}_{up}$ and $\hat{\alpha}_p$ the average working time of unprioritized drivers and prioritized drivers during the peak period respectively. Given $\hat{\alpha}_{up}$ and $\hat{\alpha}_p$, the optimization problem of an unprioritized driver is given by: $\underset{\alpha_{up}<\alpha_0}{max}\ U_{up}(\alpha_{up}) = \underset{\alpha_{up}<\alpha_0}{max}\ Pr_{up}(\hat{\alpha}_p,\hat{\alpha}_{up})\cdot p\delta(1-\alpha_{up}) + Pr_{peak}(\hat{\alpha}_p,\hat{\alpha}_{up})\cdot p\delta\alpha_{up} - \frac{1}{2}k\alpha_{up}^2$, where $Pr_{up}(\hat{\alpha}_p,\hat{\alpha}_{up}) = \int_{n\theta(1-\hat{\alpha}_p)-L}^{s}\min\{1,\frac{D(\epsilon)-n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}\}dF(\epsilon)$ and $Pr_{peak}(\hat{\alpha}_p,\hat{\alpha}_{up}) = \beta\min\{\frac{D_l}{n(1-\theta)\hat{\alpha}_{up}+n\theta\hat{\alpha}_p},1\}+1-\beta$. We focus on a case when β is not too large so that $Pr_{peak}(\hat{\alpha}_p,\hat{\alpha}_{up}) > Pr_{up}(\hat{\alpha}_p,\hat{\alpha}_{up})$. Because U_{up} is concave in α_{up}, we solve FOC and obtain $\alpha_{up}^{\dagger}(\hat{\alpha}_{up}) = \frac{p\delta}{k}(Pr_{peak}(\hat{\alpha}_p,\hat{\alpha}_{up}) - Pr_{up}(\hat{\alpha}_p,\hat{\alpha}_{up}))$. Similarly, we here focus on the case where the service target $\alpha_0 > \alpha_{up}^{\dagger}(\hat{\alpha}_{up})$. The equilibrium condition requires $\alpha_{up}^{\dagger}(\hat{\alpha}_{up}) = \hat{\alpha}_{up}$, which yields α_{up}^* that solves $\alpha_{up}^* = \frac{p\delta}{k}(Pr_{peak}(\hat{\alpha}_p,\alpha_{up}^*) - Pr_{up}(\hat{\alpha}_p,\alpha_{up}^*))$ and thus $U_{up}^* = p\delta Pr_{up}(\hat{\alpha}_p,\alpha_{up}^*) + (Pr_{peak}(\hat{\alpha}_p,\alpha_{up}^*) - Pr_{up}(\hat{\alpha}_p,\alpha_{up}^*))^2\frac{p^2\delta^2}{2k}$. For a prioritized driver, the optimization problem is given by: $\underset{\alpha_p\geq\alpha_0}{max}\ U_p(\alpha_p) = \underset{\alpha_p\geq\alpha_0}{max}\ Pr^P(\hat{\alpha}_p)p\delta(1-\alpha_p) + Pr_{peak}(\hat{\alpha}_p,\alpha_{up}^*)p\delta\alpha_p - \frac{1}{2}k\alpha_p^2$, where $Pr^P(\hat{\alpha}_p) = \int_{-s}^{s}\min\{1,\frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)}\}dF(\epsilon)$. Note that $\frac{dU_p(\alpha_p)}{d\alpha_p} = (Pr_{peak}(\hat{\alpha}_p,\alpha_{up}^*) - Pr^P(\hat{\alpha}_p))p\delta - k\alpha_p \leq (Pr_{peak}(\hat{\alpha}_p,\alpha_{up}^*) - Pr^P(\hat{\alpha}_p))p\delta - k\alpha_0 < (Pr_{peak}(\hat{\alpha}_p,\alpha_{up}^*) - Pr^P(\hat{\alpha}_p))p\delta - k\frac{p\delta}{k}(Pr_{peak}(\hat{\alpha}_p,\alpha_{up}^*) - Pr_{up}(\hat{\alpha}_p,\alpha_{up}^*)) = (Pr_{up}(\hat{\alpha}_p,\alpha_{up}^*) - Pr^P(\hat{\alpha}_p))p\delta$.

Because $Pr^P(\hat{\alpha}_p) > \int_{n\theta(1-\hat{\alpha}_p)-L}^{s} dF(\epsilon) \geq Pr_{up}(\hat{\alpha}_p, \alpha^*_{up})$ and thus $\frac{dU_p(\alpha_p)}{d\alpha_p} < 0$, we have $\alpha^*_p = \hat{\alpha}_p = \alpha_0$. Then, the total matching volume can be written as

$$M_C^P(\theta(\alpha_0)) = \int_{-s}^{s} \min\{D(\epsilon), n\theta(1-\alpha_0) + n(1-\theta)(1-\alpha^*_{up})\}dF(\epsilon) + (1-\beta)(n\theta\alpha_0 + n(1-\theta)\alpha^*_{up}) + \beta\min\{D_l, n\theta\alpha_0 + n(1-\theta)\alpha^*_{up}\}.$$

Through the similar procedure as in the proof of Lemma 4, we can show $\theta^*(\alpha_0) = 1$. The total matching volume becomes $M_C^P(\alpha_0) = \int_{-s}^{s} \min\{D(\epsilon), n(1-\alpha_0)\}dF(\epsilon) + (1-\beta)n\alpha_0 + \beta\min\{D_l, n\alpha_0\}$, the platform then maximizes $M_C^P(\alpha_0)$ by setting α_0 so that $U_p(\alpha_0) \geq U^*_{up}$. The process of deriving optimal α_0^P is similar to the proof of Lemma 4.

- *Comparison* Through a similar analysis as in the proof of Proposition 1, we can show $\alpha_0^Q > \alpha_0^P$, and thus $M_C^Q > M_C^P$.

2). *Open system*

- *Qualification scheme*

  The analysis is the same as that in a closed system, since part-time drivers will not be assigned any service requests.

- *Prioritization scheme*

  Denote by $\hat{\alpha}_{up}$ and $\hat{\alpha}_p$ the average work time of unprioritized drivers and prioritized drivers during the peak period respectively. Given $\hat{\alpha}_{up}$ and $\hat{\alpha}_p$, for a prioritized driver, the optimization problem is given by: $\underset{\alpha_p \geq \alpha_0}{max}\ U_p(\alpha_p) = Pr^P(\hat{\alpha}_p)p\delta(1-\alpha_p) + Pr_{peak}(\hat{\alpha}_p, \hat{\alpha}_{up})p\delta\alpha_p - \frac{1}{2}k\alpha_p^2$, where $Pr_{peak}(\hat{\alpha}_p, \hat{\alpha}_{up}) = (\beta\min\{\frac{D_l}{n(1-\theta)\hat{\alpha}_{up} + n\theta\hat{\alpha}_p}, 1\} + 1 - \beta)$ and $Pr^P(\hat{\alpha}_p) = \int_{-s}^{s}\min\{1, \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)}\}dF(\epsilon)$. For an unprioritized driver, the optimization problem is given by: $\underset{\alpha_{up}<\alpha_0}{max}\ U_{up}(\alpha_{up}) = \underset{\alpha_{up}<\alpha_0}{max}\ Pr_{peak}(\hat{\alpha}_p, \hat{\alpha}_{up})p\delta\alpha_{up} - \frac{1}{2}k\alpha_{up}^2$. Because U_{up} is concave in α_{up}, we solve FOC and obtain $\alpha^\dagger_{up}(\hat{\alpha}_{up}) = \frac{p\delta}{k}Pr_{peak}(\hat{\alpha}_p, \hat{\alpha}_{up})$. Similarly, we here focus on the case where the service target $\alpha_0 > \alpha^\dagger_{up}(\hat{\alpha}_{up})$. The equilibrium condition requires $\alpha^\dagger_{up}(\hat{\alpha}_{up}) = \hat{\alpha}_{up}$, which yields $\alpha^*_{up} = \frac{p\delta}{k}Pr_{peak}(\hat{\alpha}_p, \alpha^*_{up})$ and thus $U^*_{up} = \frac{p^2\delta^2}{2k}(Pr_{peak}(\hat{\alpha}_p, \alpha^*_{up}))^2$. Note that $\frac{dU_p(\alpha_p)}{d\alpha_p} = (Pr_{peak}(\hat{\alpha}_p, \alpha^*_{up}) - Pr^P(\hat{\alpha}_p))p\delta - k\alpha_p \leq (Pr_{peak}(\hat{\alpha}_p, \alpha^*_{up}) - Pr^P(\hat{\alpha}_p))p\delta - k\alpha_0 < (Pr_{peak}(\hat{\alpha}_p, \alpha^*_{up}) - Pr^P(\hat{\alpha}_p))p\delta - k\frac{p\delta}{k}Pr_{peak}(\hat{\alpha}_p, \alpha^*_{up}) < 0$, so we have $\alpha^*_p = \hat{\alpha}_p = \alpha_0$. Then, the total matching volume becomes

$$M_O^P(\theta(\alpha_0)) = \int_{-s}^{s} D(\epsilon)dF(\epsilon) + (1-\beta)(n\theta\alpha_0 + n(1-\theta)\alpha^*_{up}) + \beta\min\{D_l, n\theta\alpha_0 + n(1-\theta)\alpha^*_{up}\}.$$

Through similar procedures as in the proof of Lemma 4, we can show $\theta^*(\alpha_0) = 1$. The total matching volume becomes $M_O^P(\alpha_0) = \int_{-s}^{s} D(\epsilon)dF(\epsilon) + (1-\beta)n\alpha_0 + \beta\min\{D_l, n\alpha_0\}$, the platform then maximizes $M_O^P(\alpha_0)$ by setting α_0 so that $U_p(\alpha_0) \geq U^*_{up}$. The process of deriving optimal α_{0O}^P is similar to the proof of Lemma 6.

- *Comparison* Through a similar analysis as in the proof of Proposition 3, we can show $\alpha_0^Q \leq \alpha_{0O}^P$, and thus $M_C^Q \leq M_C^P$. ■

SA.B.5 Length of peak period is capped and less than 1

Note that the peak target is attainable only if it does not exceed the length of peak period, otherwise no drivers will be able to meet the qualification/prioritization criteria. In the main model, we have an underlying assumption that the daily peak period is considerably long so that the platform can always set the target as the global optimal solution. In reality, the peak period in some cities may only last for a short period of time, thus accounts for a small fraction of a day. In this part, we relax the assumption and consider how a capped peak period within a day alters the platform's adoption preference over the two schemes and drivers' problems. Let $\bar{\alpha} \in (\frac{p\delta}{k}, 1)$ denote the fraction of peak period within a day. The reason for adding such a constraint on $\bar{\alpha}$ is, in either closed system or open system without any incentive scheme, drivers' supply during the peak period should not exceed the length of peak period $\bar{\alpha}$, otherwise it is meaningless to shift supply to peak hours. In the following, we capture how $\bar{\alpha}$ impact equilibrium outcomes and the platform's choice over the two schemes.

We start from the closed system. Recall that the optimal target work time for peak hours are α_0^Q and α_0^P ($\leq \alpha_0^Q$) under the qualification scheme and the prioritization scheme respectively, the length of the peak period $\bar{\alpha}$ may have three distribution intervals, $\bar{\alpha} \geq 1 - \frac{L-s}{n}$ (*long peak*), $1 - \frac{L+s}{n} \leq \bar{\alpha} < 1 - \frac{L-s}{n}$ (*intermediate peak*), and $\bar{\alpha} < 1 - \frac{L+s}{n}$ (*short peak*). With *short peak*, the peak target α_0 should satisfy $\alpha_0 \leq \bar{\alpha} < 1 - \frac{L+s}{n}$, in which case the equilibrium total matching volume are $M_C^Q = M_C^P = L + n \cdot min\{\frac{p\delta}{k} + \sqrt{2\frac{L}{n}\frac{p\delta}{k}}, \bar{\alpha}\}$, thus it is the same to use either scheme. With *intermediate peak*, $M^Q = \begin{cases} n - \int_{-s}^{n(1-min\{\alpha_0^q, \bar{\alpha}\})-L} F(\epsilon)d\epsilon, & \text{if } k \leq k_3 \\ L + n\frac{p\delta}{k} + \sqrt{2nL\frac{p\delta}{k}}, & \text{if } k > k_3 \end{cases}$ and $M_C^P = \begin{cases} n - \int_{-s}^{n(1-min\{\alpha_0^p, \bar{\alpha}\})-L} F(\epsilon)d\epsilon, & \text{if } k \leq k_3 \\ L + n\frac{p\delta}{k} + \sqrt{2nL\frac{p\delta}{k}}, & \text{if } k > k_3 \end{cases}$, using qualification scheme is not worse than using prioritization scheme. With *long peak*, as we discussed in base model, the platform should implement qualification scheme. Hence, qualification scheme weakly dominates prioritization scheme in a closed system when the length of peak period is capped.

We now look at the open system. It is worth noting that qualification scheme plays the same role in both closed and open systems. Furthermore, the two incentive schemes are equivalent in the open system when the peak target α_0 is constrained to be smaller than $1 - \frac{L-s}{n}$ according to the proof of Lemma 3 and Lemma 6. Thus, both schemes are equivalent if $\bar{\alpha} \leq 1 - \frac{L-s}{n}$. With *long peak* (i.e., $\bar{\alpha} > 1 - \frac{L-s}{n}$), $M_O^P = \begin{cases} L + n \cdot min\{\sqrt{\frac{p\delta}{k}(2 - \frac{p\delta}{k})}, \bar{\alpha}\}, & \text{if } k \leq k_2 \\ L + n\alpha_0^q, & \text{if } k_2 < k \leq k_3 \\ L + n\frac{p\delta}{k} + \sqrt{2nL\frac{p\delta}{k}}, & \text{if } k \geq k_3 \end{cases}$ and $M_O^Q = M_C^Q$ as presented in Lemma 3, and we can show that $M_O^P \geq M_O^Q$. Hence, prioritization scheme weakly dominates qualification scheme in an open system when the length of peak period is capped. ■

SA.C Justification on Model Setting

SA.C.1 Randomized prioritization scheme

In the main model, we consider the prioritization scheme where drivers who meet the peak period service target are guaranteed priority for receiving ride requests during the off-peak period. However, the platform may randomly give such priority to drivers who meet the peak period service target, which we call a randomized prioritization scheme. Specifically, when implementing a randomized prioritization scheme, with probability $\eta \in (0,1)$, drivers who reach the target have priority to get off-peak orders, and with probability $1-\eta$, all drivers have equal chance to get off-peak orders. In this section, we will show that the randomized prioritization scheme is inferior to the prioritization scheme we proposed. Note that $\eta = 1$ ($\eta = 0$) reduces to the prioritization scheme (benchmark scenario) in our base model. We then show that the platform would like to set $\eta = 1$.

We first look at a closed system. Suppose that observing α_0, a proportion of $\theta \in (0,1)$ and $1-\theta$ drivers participate and do not participate the randomized prioritization scheme respectively. Given $\hat{\alpha}_p$ and $\hat{\alpha}_{up}$, the respective optimization problems of an individual unprioritized and prioritized driver are

$$\max_{\alpha_{up}<\alpha_0} U_{up}(\alpha_{up}) = \max_{\alpha_{up}<\alpha_0} \Big(((1-\eta)Pr + \eta Pr_{up})p\delta(1-\alpha_{up}) + p\delta\alpha_{up} - \frac{1}{2}k\alpha_{up}^2\Big), and$$
$$\max_{\alpha_p \geq \alpha_0} U_p(\alpha_p) = \max_{\alpha_p \geq \alpha_0} \Big(((1-\eta)Pr + \eta Pr^P)p\delta(1-\alpha_p) + p\delta\alpha_p - \frac{1}{2}k\alpha_p^2\Big),$$

where $Pr^P(\hat{\alpha}_p) = \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)}\}dF(\epsilon)$, $Pr(\hat{\alpha}_p, \hat{\alpha}_{up}) = \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})}\}dF(\epsilon)$, and $Pr_{up}(\hat{\alpha}_p, \hat{\alpha}_{up}) = \int_{n\theta(1-\hat{\alpha}_p)-L}^{s} \min\{1, \frac{D(\epsilon)-n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}\}dF(\epsilon)$.

First, we prove that $Pr^P(\hat{\alpha}_p) \geq Pr(\hat{\alpha}_p, \hat{\alpha}_{up}) \geq Pr_{up}(\hat{\alpha}_p, \hat{\alpha}_{up})$. Note that $Pr^P(\hat{\alpha}_p) = \int_{-s}^{n\theta(1-\hat{\alpha}_p)-L} \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)}dF(\epsilon) + \int_{n\theta(1-\hat{\alpha}_p)-L}^{s} dF(\epsilon) > \int_{-s}^{n\theta(1-\hat{\alpha}_p)-L} \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})}dF(\epsilon) + \int_{n\theta(1-\hat{\alpha}_p)-L}^{s} \min\{1, \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})}\}dF(\epsilon) = Pr(\hat{\alpha}_p, \hat{\alpha}_{up})$. Moreover, $\frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})} > \frac{D(\epsilon)-n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}$ as long as $D(\epsilon) < n\theta(1-\hat{\alpha}_p) + n(1-\theta)(1-\hat{\alpha}_{up})$, so $\int_{n\theta(1-\hat{\alpha}_p)-L}^{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})-L} \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})}dF(\epsilon) > \int_{n\theta(1-\hat{\alpha}_p)-L}^{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})-L} \frac{D(\epsilon)-n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}dF(\epsilon)$ and thus $Pr(\hat{\alpha}_p, \hat{\alpha}_{up}) > \int_{n\theta(1-\hat{\alpha}_p)-L}^{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})-L} \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})}dF(\epsilon) + \int_{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})-L}^{s} dF(\epsilon) > \int_{n\theta(1-\hat{\alpha}_p)-L}^{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})-L} \frac{D(\epsilon)-n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}dF(\epsilon) + \int_{n\theta(1-\hat{\alpha}_p)+n(1-\theta)(1-\hat{\alpha}_{up})-L}^{s} dF(\epsilon) = Pr_{up}(\hat{\alpha}_p, \hat{\alpha}_{up})$. Therefore, $Pr^P(\hat{\alpha}_p) \geq Pr(\hat{\alpha}_p, \hat{\alpha}_{up}) \geq Pr_{up}(\hat{\alpha}_p, \hat{\alpha}_{up})$.

Next, we prove that prioritized drivers optimally work binding to α_0. For the unprioritized driver, because $U_p(\alpha_p)$ is concave in α_p, we use FOC and obtain $\alpha_{up}^{\dagger}(\hat{\alpha}_{up}) = \frac{p\delta}{k}(1-(1-\eta)Pr - \eta Pr_{up})$. Similarly, we here focus on the case where the service target $\alpha_0 > \alpha_{up}^{\dagger}(\hat{\alpha}_{up})$. The equilibrium condition requires $\alpha_{up}^{\dagger}(\hat{\alpha}_{up}) = \hat{\alpha}_{up}$, which yields $\alpha_{up}^{*} = \frac{p\delta}{k}(1-(1-\eta)Pr - \eta Pr_{up})$ and thus $U_{up}^{*}(\alpha_0) = p\delta((1-\eta)Pr + \eta Pr_{up}) + (1-(1-\eta)Pr - \eta Pr_{up})^2 \frac{p^2\delta}{2k}$. For the prioritized driver, $\frac{dU_p(\alpha_p)}{d\alpha_p} = (1-(1-\eta)Pr - \eta Pr^P)p\delta - k\alpha_p \leq (1-(1-\eta)Pr - \eta Pr^P)p\delta - k\alpha_0 < (1-(1-\eta)Pr - \eta Pr^P)p\delta - k\frac{p\delta}{k}(1-(1-\eta)Pr - \eta Pr_{up}) = p\delta\eta(Pr_{up} - Pr^P) < 0$, so $\alpha_p^{*} = \hat{\alpha}_p = \alpha_0$.

Then, using the same technique as in the proof of Lemma 4, we show that $\theta^*(\alpha_0)=1$. The utility of a prioritized driver becomes $U_p(\alpha_0)=Pr^P p\delta(1-\alpha_0)+p\delta\alpha_0-\frac{1}{2}k\alpha_0^2$ and $U^*_{up}(\alpha_0)=p\delta((1-\eta)Pr+\eta Pr_{up})+(1-(1-\eta)Pr-\eta Pr_{up})^2\frac{p^2\delta}{2k}$, where $Pr^P=Pr=\int_{-s}^{s}\min\{1,\frac{D(\epsilon)}{n(1-\alpha_0)}\}dF(\epsilon)$ and $Pr_{up}=\int_{n(1-\alpha_0)-L}^{s}dF(\epsilon)$. Note that $U_{up}(\alpha_0)^*$ increases in $(1-\eta)Pr+\eta Pr_{up}$ and $(1-\eta)Pr+\eta Pr_{up}$ increases in α_0, so $U^*_{up}(\alpha_0)$ increases in α_0. Similar to the proof of Lemma 4, we have $U_p(\alpha_0)$ decreases in α_0. Hence, $U_p(\alpha_0)=U^*_{up}(\alpha_0)$ has at most one solution and thus the optimal α_0^* solves $U_p(\alpha_0^*)=U^*_{up}(\alpha_0^*)$. Also, note that $U^*_{up}(\alpha_0)$ increases in η given α_0 due to $Pr>Pr_{up}$ but $U_p(\alpha_0)$ does not relate to η, so we can find that the optimal α_0 increases in η, i.e., $\frac{d\alpha_0^*}{d\eta}>0$. Because the total matching volume $M^*=\int_{-s}^{s}\min\{D(\epsilon),n(1-\alpha_0^*)\}dF(\epsilon)+n\alpha_0^*$ increases in α_0^*, we have that the platform would set $\eta^*=1$ in equilibrium.

We then look at an open system. Note that a driver will have near-zero chance to work during off-peak through being unprioritzed due to the sufficient number of part-time drivers during off-peak periods. Because all off-peak will be satisfied in an open system, the platform's objective is to choose a α_0 so that prioritized drivers obtain no less than those unprioritized. Note that the utility of a prioritized driver increases in η but that of an unprioritized driver has nothing to do with η, so the optimal α_0 increases in η and thus $\eta^*=1$. ■

SA.C.2 Fractions of working time a (full-time) driver allocates across two periods sum up to be 1.

In our model, a full time driver's total working time is normalized to 1. In this section, we consider a scenario in which a full time driver can choose to work a shorter total working time $\tau\in[\frac{p\delta}{k},1]$ and show that a full-time driver chooses $\tau=1$ in equilibrium. We assume that the total working time is at least $\frac{p\delta}{k}$ because a driver optimally work $\frac{p\delta}{k}$ during peak periods if he does not have chance to work during off-peak periods. We next show that the driver's optimal utility increases in τ.

1). *Closed system*

- *Benchmark*
  Given the average peak working time $\hat{\alpha}$ of all other drivers, a full-time driver's problem is $U^B(\alpha,\tau)=Pr^B p\delta(\tau-\alpha)+p\delta\alpha-\frac{1}{2}k\alpha^2$, where $Pr^B=\int_{-s}^{s}\min\{1,\frac{D(\epsilon)}{n(\tau-\hat{\alpha})}\}dF(\epsilon)$. Using the same technique as the proof of Lemma 1, we can rewrite $U^B(\alpha)$ as

$$U^B(\alpha,\tau)=\begin{cases}p\delta\tau-\frac{1}{2}k\alpha^2, & \text{if } n(\tau-\hat{\alpha})\le L-s,\\ (\tau-(\tau-\alpha)\frac{\int_{-s}^{n(\tau-\hat{\alpha})-L}F(\epsilon)d\epsilon}{n(\tau-\hat{\alpha})})p\delta-\frac{1}{2}k\alpha^2, & \text{if } L-s<n(\tau-\hat{\alpha})\le L+s,\\ (\frac{L}{n(\tau-\hat{\alpha})}(\tau-\alpha)+\alpha)p\delta-\frac{1}{2}k\alpha^2, & \text{if } n(\tau-\hat{\alpha})>L+s.\end{cases} \tag{SA.3}$$

  (1) When $n(\tau-\hat{\alpha})\le L-s$, note that $U^B(\alpha,\tau)$ decreases in α given $\tau\in[\frac{p\delta}{k},1]$ and thus the optimal solution is $\alpha^*(\hat{\alpha})=0$. The equilibrium condition requires $\alpha^*(\hat{\alpha})=\hat{\alpha}$ and thus $\hat{\alpha}=0$, the condition $n(\tau-\hat{\alpha})\le L-s$ becomes $n\tau\le L-s$. If $\tau\le\frac{L-s}{n}$, $\alpha^*(\tau)=0$, and thus $\frac{dU^B(\tau)}{d\tau}=p\delta>0$; if $\tau>\frac{L-s}{n}$, this case will not occur in equilibrium.

  (2) When $L-s<n(\tau-\hat{\alpha})\le L+s$, $U^B(\alpha,\tau)$ is concave in α given $\tau\in[\frac{p\delta}{k},1]$, we solve FOC and obtain $\alpha(\hat{\alpha})=\frac{p\delta}{k}\frac{\int_{-s}^{n(\tau-\hat{\alpha})-L}F(\epsilon)d\epsilon}{n(\tau-\hat{\alpha})}$. Through similar analysis as in the proof of Lemma 1, we know

that there exists at most one solution in this case. The equilibrium occurs in this case only if the following two conditions are satisfied: (i) $\hat{\alpha} < \frac{p\delta}{k}\frac{\int_{-s}^{n(\tau-\hat{\alpha})-L} F(\epsilon)d\epsilon}{n(\tau-\hat{\alpha})}$ when $\hat{\alpha} = \tau - \frac{L+s}{n}$ and (ii) $\hat{\alpha} > \frac{p\delta}{k}\frac{\int_{-s}^{n(\tau-\hat{\alpha})-L} F(\epsilon)d\epsilon}{n(\tau-\hat{\alpha})}$ when $\hat{\alpha} = \tau - \frac{L-s}{n}$, which lead to $k \leq \frac{p\delta s}{(L+s)(\tau-\frac{L+s}{n})} = k_1(\tau)$. By Envelope Theorem, $\frac{dU^B(\tau)}{d\tau} = \frac{\partial U^B(\alpha,\tau)}{\partial \tau} + \frac{\partial U^B(\alpha,\tau)}{\partial \alpha}|_{\alpha=\alpha^*(\tau)} \cdot \frac{\partial \alpha^*(\tau)}{\partial \tau} = (1 - \frac{\int_{-s}^{n(\tau-\alpha^*(\tau))-L} F(\epsilon)d\epsilon}{n(\tau-\alpha^*(\tau))})p\delta > 0$.

(3) When $n(\tau - \hat{\alpha}) > L + s$, $U^B(\alpha, \tau)$ is concave in α given $\tau \in [\frac{p\delta}{k}, 1]$, we solve FOC and obtain $\alpha(\hat{\alpha}) = \frac{p\delta}{k}(1 - \frac{L}{n(\tau-\hat{\alpha})})$. The equilibrium condition requires $\alpha(\hat{\alpha}) = \hat{\alpha}$, which yields the optimal $\alpha(\tau) = \frac{k\tau+p\delta-\sqrt{(k\tau-p\delta)^2+4kLp\delta/n}}{2k} \equiv \alpha_1^\dagger(\tau)$ (the other solution exceeds 1 and thus is infeasible). By solving $n(\tau - \alpha_1^\dagger(\tau)) > L + s$, we obtain $k > k_1(\tau)$. Therefore, when $k > k_1(\tau)$, $\alpha_1^\dagger(\tau)$ is the optimal solution in this case. Then, plugging $\alpha = \alpha_1^\dagger(\tau)$ in $U^B(\alpha, \tau)$ yields $U^B(\tau) = \frac{Lp\delta}{n} + \frac{p\delta(k\tau+p\delta-\sqrt{(k\tau-p\delta)^2+4kLp\delta/n})}{2k} - \frac{(k\tau+p\delta-\sqrt{(k\tau-p\delta)^2+4kLp\delta/n})^2}{8k}$, differentiating which with respect to τ yields $\frac{dU^B(\tau)}{d\tau} = \frac{(p\delta-k\tau+\sqrt{4kLp\delta/n+(p\delta-k\tau)^2})^2}{4\sqrt{4kLp\delta/n+(p\delta-k\tau)^2}} > 0$.

To sum up, $U^B(\tau)$ increases in $\tau \in [\frac{p\delta}{k}, 1]$, so $\tau^* = 1$.

- *Qualification Scheme*

The game sequence is as follows: the platform first determines $\alpha_0 \in (\frac{p\delta}{k}, 1]$, then full-time drivers determine their total working time $\tau \in [\frac{p\delta}{k}, 1]$ and allocate the working time across two periods. Note that the unqualified driver optimally works $\alpha_{uq}^* = \frac{p\delta}{k}$ during peak periods and obtains $U_{uq}^* = \frac{p^2\delta^2}{2k}$, thus α_0 must satisfy $\alpha_0 > \frac{p\delta}{k}$ and qualified drivers will obtain no less than $\frac{p^2\delta^2}{2k}$ to make the qualification scheme valid. If $\tau = \frac{p\delta}{k}$, full-time drivers naturally choose to be unqualified and obtain a not higher utility compared to being qualified. Therefore, for those who choose to be qualified, τ must satisfy $\tau \in (\frac{p\delta}{k}, 1]$. Suppose a proportion of $\theta \in [0, 1]$ and $1 - \theta$ full-time drivers choose to be qualified and unqualified respectively. Given the average working time of qualified driver $\hat{\alpha}_q$, the problem of a qualified driver is $\underset{\alpha \geq \alpha_0}{max}\ U_q(\alpha_q) = Pr^Q \cdot p\delta(\tau - \alpha_q) + p\delta\alpha_q - \frac{1}{2}k\alpha_q^2$, where $Pr^Q = \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n\theta(\tau-\hat{\alpha}_q)}\}dF(\epsilon)$. Using the same technique as in the proofs of Lemma 2 and Lemma 3, we show that qualified drivers optimally work α_0 during peak hours, and given the service $\alpha_0 \in (\frac{p\delta}{k}, 1]$, all drivers participate the qualification scheme, i.e., $\theta^*(\alpha_0) = 1$. The qualified drivers' utility becomes $U_q(\tau) = \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n(\tau-\alpha_0)}\}dF(\epsilon)p\delta(\tau - \alpha_0) + p\delta\alpha_0 - \frac{1}{2}k\alpha_0^2 = \int_{-s}^{s} \min\{\tau - \alpha_0, D(\epsilon)/n\}dF(\epsilon)p\delta + p\delta\alpha_0 - \frac{1}{2}k\alpha_0^2$, we observe that $U_q(\tau)$ does not decrease in $\tau \in (\frac{p\delta}{k}, 1]$ given $\alpha_0 \in (\frac{p\delta}{k}, 1]$. Also note that the platform will increase α_0 to maximize the total matching volume, so drivers will increase their total working time and finally work binding to 1 to retain the advantages of being qualified. Therefore, we have $\tau^* = 1$.

- *Prioritization Scheme*

We keep using $\theta \in [0, 1]$ and $1 - \theta$ to denote the proportion of prioritized and unprioritized drivers respectively, and $\hat{\alpha}_p, \hat{\alpha}_{up}$ to denote the average working time during peak and off-peak periods respectively. For an unprioritized driver, given α_0, $\theta, \hat{\alpha}_p, \hat{\alpha}_{up}$, his optimization problem is $\underset{\alpha_{up}<\alpha_0}{max}\ U_{up}(\alpha_{up}) = \underset{\alpha_{up}<\alpha_0}{max}\ Pr_{up}(\hat{\alpha}_p, \hat{\alpha}_{up}) \cdot p\delta(\tau - \alpha_{up}) + p\delta\alpha_{up} - \frac{1}{2}k\alpha_{up}^2$, where $Pr_{up}(\hat{\alpha}_p, \hat{\alpha}_{up}) =$

$\int_{n\theta(\tau-\hat{\alpha}_p)-L}^{s} \min\{1, \frac{D(\epsilon)-n\theta(\tau-\hat{\alpha}_p)}{n(1-\theta)(\tau-\hat{\alpha}_{up})}\}dF(\epsilon)$. Because U_{up} is concave in α_{up}, we solve FOC and obtain $\alpha_{up}^{\dagger}(\hat{\alpha}_{up}) = \frac{p\delta}{k}(1 - Pr_{up}(\hat{\alpha}_p, \hat{\alpha}_{up}))$. Similarly, we here focus on the case where the service target $\alpha_0 > \alpha_{up}^{\dagger}(\hat{\alpha}_p, \hat{\alpha}_{up})$. The equilibrium condition requires $\alpha_{up}^{\dagger}(\hat{\alpha}_p, \hat{\alpha}_{up}) = \hat{\alpha}_{up}$, which yields α_{up}^{*} that solves $\alpha_{up}^{*} = \frac{p\delta}{k}(1 - Pr_{up}(\hat{\alpha}_p, \alpha_{up}^{*}))$ and thus $U_{up}^{*} = p\delta Pr_{up}(\hat{\alpha}_p, \alpha_{up}^{*}) + (1 - Pr_{up}(\hat{\alpha}_p, \alpha_{up}^{*}))^2 \frac{p^2\delta^2}{2k}$. For a prioritized driver, given α_0, θ, and $\hat{\alpha}_p$, his optimization problem is given by $\underset{\alpha_p \geq \alpha_0}{max}\ U_p(\alpha_p) = \underset{\alpha_p \geq \alpha_0}{max}\ Pr_p(\hat{\alpha}_p) \cdot p\delta(\tau - \alpha_p) + p\delta\alpha_p - \frac{1}{2}k\alpha_p^2$, where $Pr_p(\hat{\alpha}_p) = \int_{-s}^{s} \min\{1, \frac{D(\epsilon)}{n\theta(\tau-\hat{\alpha}_p)}\}dF(\epsilon)$. Through the same procedure as in the proof of Lemma 4, we can prove $\alpha_p^{*} = \alpha_0$. Using the same technique as in the proof of Lemma 4, we prove that given the service target $\alpha_0 \in (\alpha_{up}^{*}, 1]$, all drivers participate the prioritization scheme, i.e., $\theta^{*}(\alpha_0) = 1$. Then, drivers' problem is the same as that in qualification scheme, so similarly we can prove $\tau^{*} = 1$.

2). <u>*Open system*</u>

- *Benchmark*

  Note that a full-time driver's problem is $U_O^B(\alpha) = p\delta\alpha - \frac{1}{2}k\alpha^2$ because he does not have chance to work during off-peak. Using FOC to solve for the optimal α, we have $\alpha_O^B = \frac{p\delta}{k}$, $U_O^B = \frac{p^2\delta^2}{2k}$, and $M_O^B = \int_{-s}^{s} D(\epsilon)dF(\epsilon) + n\alpha_O^B = L + n\frac{p\delta}{k}$. Therefore, a full-time driver allocates $\frac{p\delta}{k}$ at peak and $\tau - \frac{p\delta}{k}$ at off-peak, thus the equilibrium outcomes remain the same as long as $\tau \in (\frac{p\delta}{k}, 1]$.

- *Qualification Scheme*

  The analysis of qualification scheme in open system is the same as that in closed system.

- *Prioritization Scheme*

  In an open system, unprioritized drivers face the same problem as unqualified drivers because they will not receive any service request during the off-peak period due to the infinite number of part-time drivers during the off-peak period. Through the same alaysis, given $\alpha_0(\frac{p\delta}{k}, 1]$, all full-time drivers will participate the scheme, i.e., $\theta^{*}(\alpha_0) = 1$. Then, drivers' problem is the same as that in qualification scheme, so similarly we can prove $\tau^{*} = 1$. ■

SA.D Numerical Analysis and Results

SA.D.1 Determining the Parameter Values in Section 6

In this section, we outline the methodology used to determine the values of exogenous parameters. The primary empirical evidence is drawn from Didi in the Chinese market (Bai et al. 2019, Xiong et al. 2021, Bimpikis et al. 2019, Wang and Zhang 2022). In addition, due to the absence of off-peak demand volatility information, we supplement this by using trip data from New York. We first present the baseline parameter values, followed by alternative values used for robustness checks.

- D_0, the intercept in the off-peak demand function $D(\epsilon) = D_0 - p + \epsilon$, which is normalized to 1.
- D_p, the intercept in the peak demand function. Bai et al. (2019) set the customer demand rate for individuals who may opt to use the platform to request for service during peak and off-peak periods as 200 and 100 per zone on average, respectively. Similarly, Wang and Zhang (2011) set the average customer demand rate as 4000 during the peak period and as 2000 during off-peak periods. Based on the above evidence, we use a peak-to-off-peak demand ratio of 2, leading $D_p = 2$.
- p, the transaction price per unit of service time. The ratio $\frac{D_0-p}{D_0}$ reflects the expected realized demand relative to the potential demand during the off-peak period. According to Bai et al. (2019), during the off-peak period, the average customer demand rate for individuals who may opt to use the platform to request for service per zone is set as 100, while the average realized customer request rate per zone is equal to 50 under the standard taxi price. From this, we infer that $\frac{D_0-p}{D_0} = 0.5$, resulting $p = 0.5$.
- n, the size of the group of full-time drivers. We begin by estimating the total number of drivers (including both full-time and part-time), based on Bai et al. (2019), which reported that the total number of registered drivers is equal to 390 per zone on average. Next, we determine the proportion of full-time drivers using the following two piece of empirical evidence. First, the 2017 Didi Travel Platform Employment Research Report[2] indicates that the majority of Didi drivers have other jobs, with 33.84% working more than 4 hours per day and 16.76% working more than 6 hours per day. Additionally, Xiong et al. (2021), using ride-hailing data from Xiamen, China, found that full-time drivers account for 52.80%. Based on the above evidence, we assume that one-third of the drivers are full-time. Therefore, the average number of full-time drivers per zone is $390 \times \frac{1}{3} = 130$, which is 1.3 times the potential demand in off-peak period (i.e., 100). Thus, we set $n = 1.3$.
- δ, the proportion of transaction price retained by the driver. According to Bai et al. (2019), Didi drivers received approximately 80% of the total fare as wages. Additionally, Bimpikis et al. (2019) reported that the ride-sharing industry commonly adopts a fixed commission rate, with driver compensation typically ranging from 75% to 80% of the fare paid by the rider. Based on the above evidence, we set $\delta = 0.8$.

[2] See http://www.shujuju.cn/lecture/detail/3532 (Accessed November 2024).

- s, the boundary of ϵ, which reflects the randomness in the off-peak demand. Due to the unavailability of randomness information in the Didi data, we estimate this parameter using trip data from New York City[3]. We selected trip data from Januray 2024 to June 2024 as our sample and consider it an approximation of realized demand. The average number of trips per day is 755,117, and on over 95.6% of the sampled dates, the number of trips falls within 20% above or below this average. Then we apply this to our model. The average realized off-peak demand is equal to $D_0 - p = 0.5$. The realized off-peak demand by considering the randomness $D_0 - p + s$ should be between 20% above or below this average, which is [0.4, 0.6]. Therefore, we set $\epsilon \sim [-0.1, 0.1]$, leading to $s = 0.1$.
- k, the congestion cost coefficient. We leverage the taxi data in 2008 when the ride-hailing platforms (with incentive schemes) were not introduced in China, and consider this as the benchmark case. The dataset covers the record of 10,357 taxis in Beijing during a one-week period in February (Yuan et al. 2010, Yuan et al. 2011)[4]. The taxi system can be considered as a closed system, due to the high entry cost and stringent regulations. Peak hours in Beijing are defined as 7 AM to 9 AM and 5 PM to 7 PM[5]. We find that on average, drivers allocate around 15% of their total daily workload to the peak period. According to the results of the benchmark case in a closed system, each full-time driver is willing to work α_C^B in the peak period, as shown in Lemma 1. With $p = 0.5$, $\delta = 0.8$, $n = 1.3$, $L = D_0 - p = 0.5$, we obtain $k = 1.4$. Given that the number of motor vehicles in Beijing increased from 3.504 million in 2008 to 7.85 million in 2024[6], the congestion degree and cost would also rise. Additionally, some areas are more crowded than others. Therefore, we consider k in the range of 1 to 5. For the analysis of service target and price multiplier with fixed k (see Figure 3), we choose a medium value $k = 2$.

We determine the parameter values using the approach described above, which serves as the baseline. Additionally, we vary each parameter by assigning larger or smaller values, generating multiple sets of parameters for robustness checks of the results. The parameter settings are summarized in Table SA.2.

[3] See https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page (Accessed November 2024)

[4] The data is available through the following link: https://www.microsoft.com/en-us/research/publication/t-drive-trajectory-data-sample/ (Accessed November 2024)

[5] See https://shorturl.at/IOSaN (Accessed November 2024)

[6] See https://jtgl.beijing.gov.cn/jgj/jgxx/ywsj/index.html (Accessed November 2024)

Table SA.2 Parameter Settings

	D_0	D_p	p	n	δ	s
(0) Baseline	**1**	**2**	**0.5**	**1.3**	**0.8**	**0.1**
(1) Larger D_p	1	**2.5**	0.5	1.3	0.8	0.1
(2) Smaller D_p	1	**1.5**	0.5	1.3	0.8	0.1
(3) Larger p	1	2	**0.7**	1.3	0.8	0.1
(4) Smaller p	1	2	**0.3**	1.3	0.8	0.1
(5) Larger n	1	2	0.5	**1.5**	0.8	0.1
(6) Smaller n	1	2	0.5	**1.1**	0.8	0.1
(7)Larger δ	1	2	0.5	1.3	**0.9**	0.1
(8) Smaller δ	1	2	0.5	1.3	**0.7**	0.1
(9) Larger s	1	2	0.5	1.3	0.8	**0.15**
(10) Smaller s	1	2	0.5	1.3	0.8	**0.05**

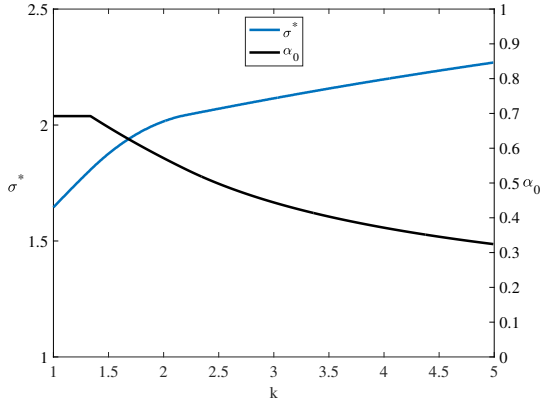
SA.D.2 Robustness checks of comparison between incentive schemes and surge pricing

In this section, we present ten sets of robustness checks for Figures 1 and 2 from Section 6.1, based on the parameter settings outlined in Table SA.2. Overall, the results remain largely consistent with Figures 1 and 2, indicating that the platform needs to raise the base price by a significant percentage (typically around 50% or more) to achieve the profit of an incentive scheme. In addition, we also find more cases where the surge pricing fails to achieve the same matching volume as an incentive scheme.

(a) Robustness checks of alternative D_p When peak period demand increases or decreases by 0.5, we observe highly consistent results regarding the minimum price multiplier required for surge pricing to achieve the same profit as the incentive scheme, as shown in Figures SA.1 and SA.3. For the comparison of maximum matching volume, we find more cases where the surge pricing might not reach the same matching volume as an incentive scheme, as shown in Figures SA.2 and SA.4.

Figure SA.1 The minimum price multiplier needed for surge pricing to achieve the same profit of an incentive scheme. ($D_0 = 1, D_p = 2.5, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)

(a) Closed system



(b) Open system

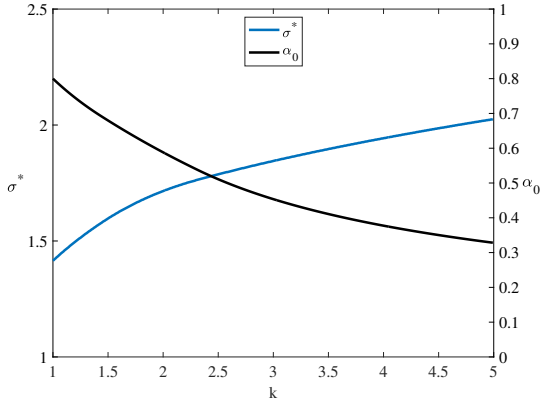
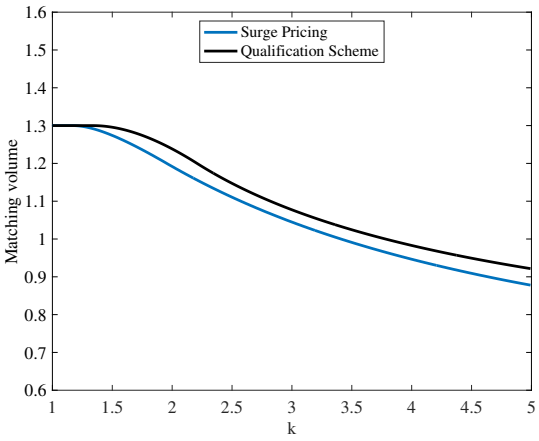
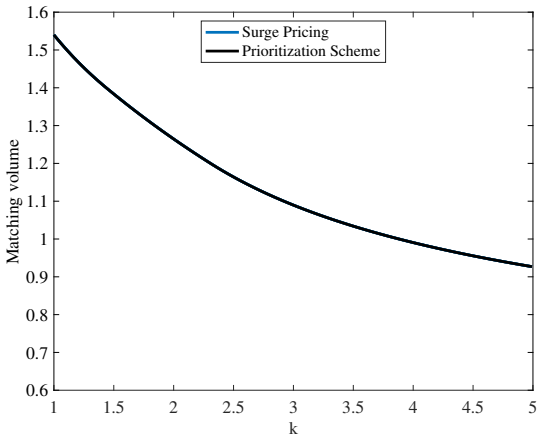


Figure SA.2 **The comparison of maximum matching volume between surge pricing and incentive schemes ($D_0 = 1, \underline{D_p = 2.5}, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



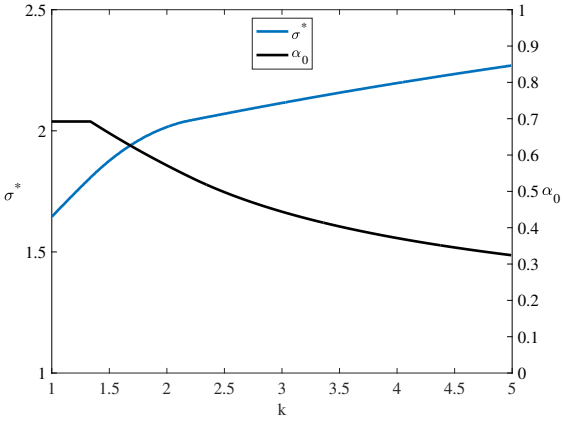
(b) Open system



Note. In figure (b), the black and blue lines overlap, indicating that surge pricing could achieve the same matching volume as the incentive scheme in an open system.

Figure SA.3 **The minimum price multiplier needed for surge pricing to achieve the same profit of an incentive scheme. ($D_0 = 1, \underline{D_p = 1.5}, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



(b) Open system

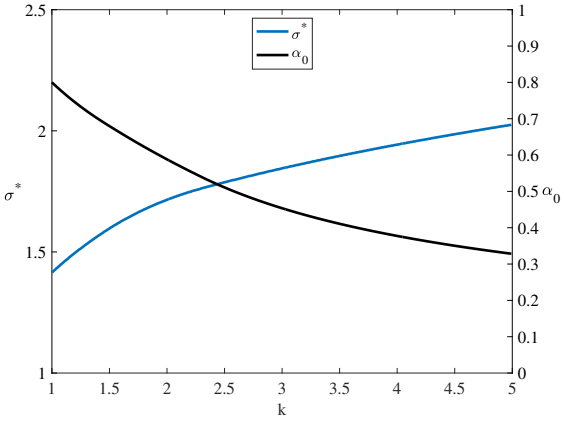
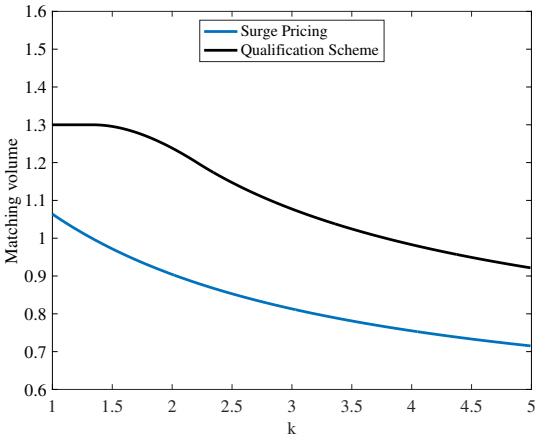
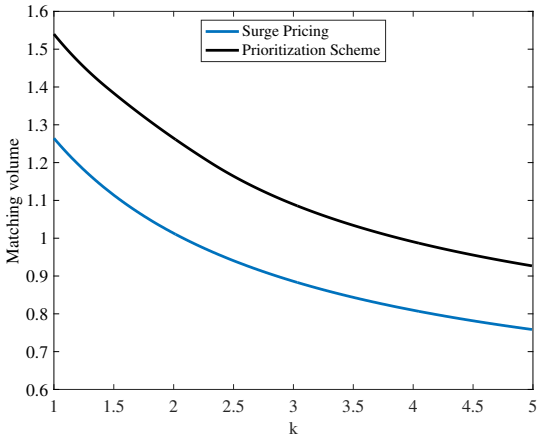


Figure SA.4 **The comparison of maximum matching volume between surge pricing and incentive schemes ($D_0 = 1, \underline{D_p = 1.5}, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



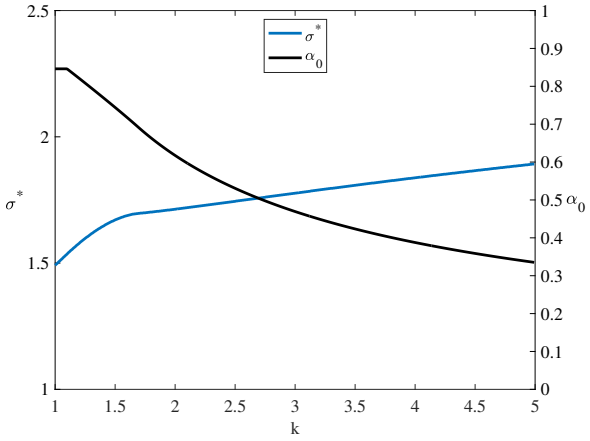
(b) Open system



(b) Robustness checks of alternative p When the base price increases or decreases by 0.2, we observe patterns that are largely consistent with those in Figures 1 and 2, as shown in Figures SA.5 to SA.8. The platform typically needs to raise its base price by more than 50% to match the profit of an incentive scheme. The exception occurs in Figure SA.5(b), where the platform needs to do so only when the congestion cost $k \geq 2$, which is because a higher base price enhances the marginal effectiveness of surge pricing. On the comparison of maximum matching volume, we observe more cases where surge pricing might not reach the same matching volume as an incentive scheme, as shown in Figures SA.6 and SA.8.

Figure SA.5 **The minimum price multiplier needed for surge pricing to achieve the same profit of an incentive scheme.** **(**$D_0 = 1, D_p = 2, \underline{p = 0.7}, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$**)**

(a) Closed system



(b) Open system

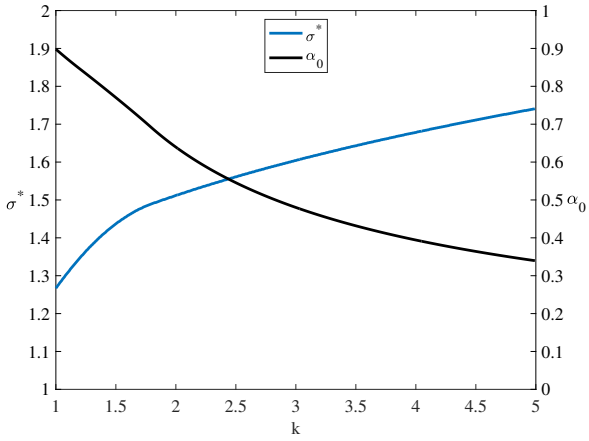
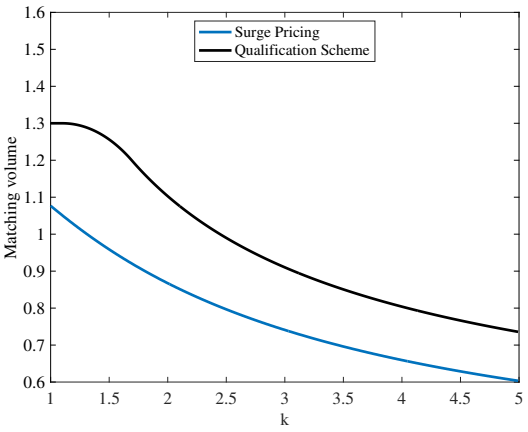


Figure SA.6 **The comparison of maximum matching volume between surge pricing and incentive schemes (**$D_0 = 1, D_p = 2, \underline{p = 0.7}, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$**)**

(a) Closed system



(b) Open system

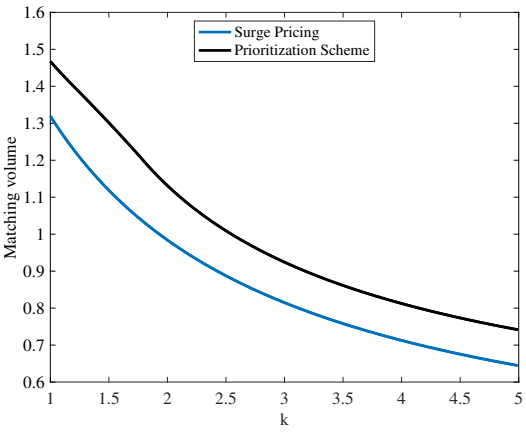


Figure SA.7 **The minimum price multiplier needed for surge pricing to achieve the same profit of an incentive scheme.** **($D_0 = 1, D_p = 2, \underline{p = 0.3}, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system (b) Open system

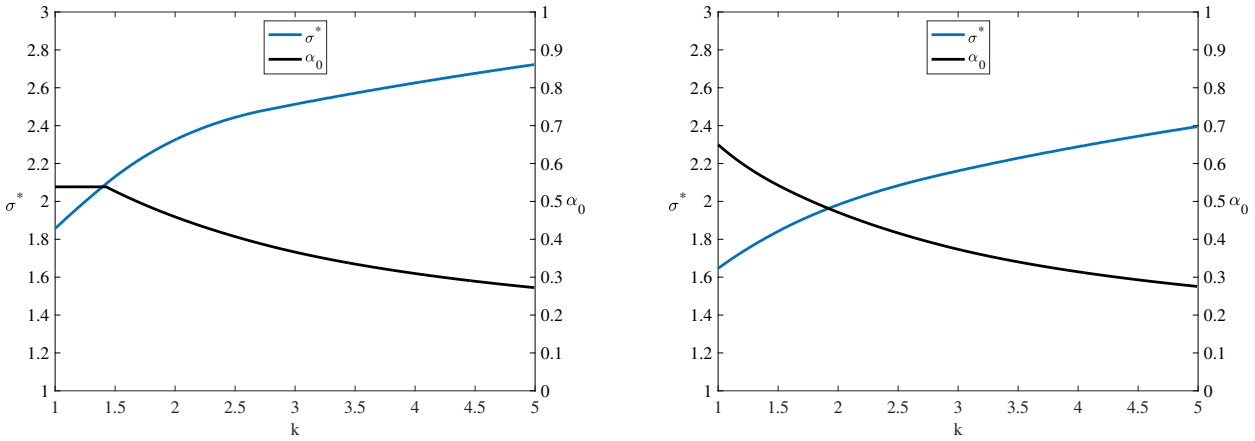
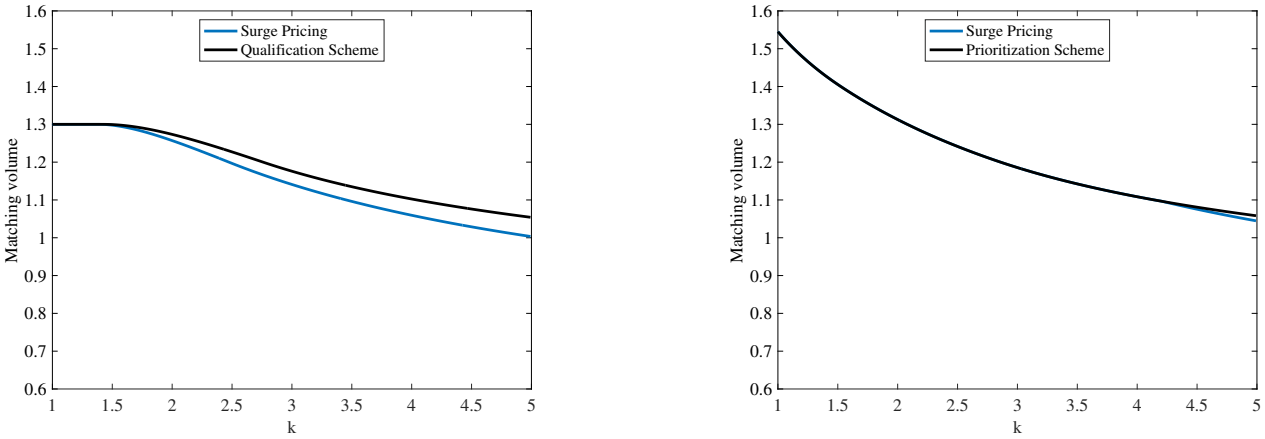


Figure SA.8 **The comparison of maximum matching volume between surge pricing and incentive schemes ($D_0 = 1, D_p = 2, \underline{p = 0.3}, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system (b) Open system



Note. In figure (b), the black and blue lines overlap with each other when k is small.

(c) Robustness checks of alternative n When the size of full-time driver increases or decreases by 0.2, the results remain highly consistent as those in Section 6.1, as shown in Figures SA.9 to SA.12.

Figure SA.9 **The minimum price multiplier needed for surge pricing to achieve the same profit of an incentive scheme.** **($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, \underline{n = 1.5}, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system (b) Open system

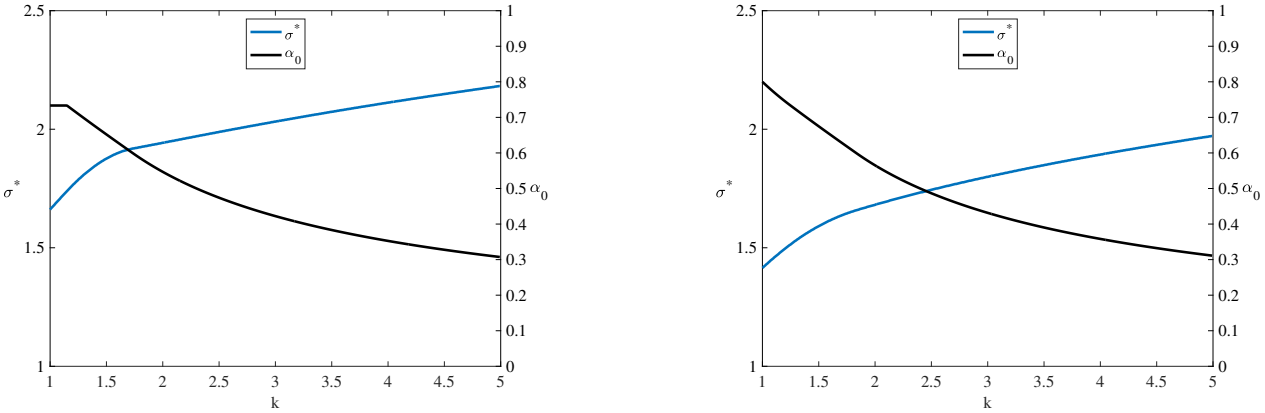
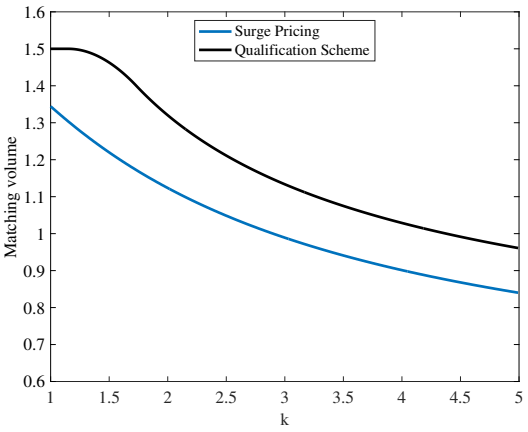


Figure SA.10 **The comparison of maximum matching volume between surge pricing and incentive schemes ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, \underline{n = 1.5}, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



(b) Open system

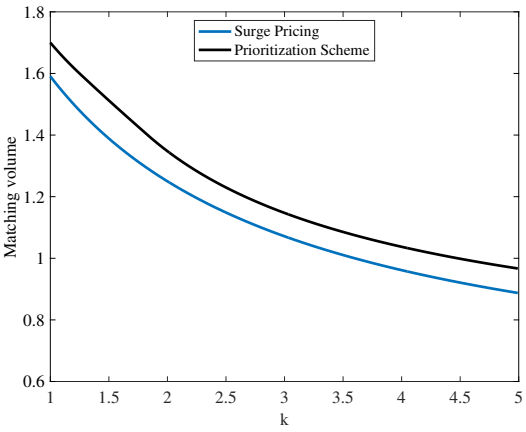
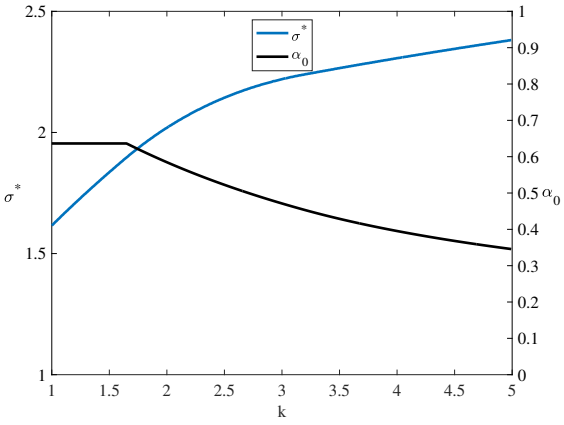


Figure SA.11 **The minimum price multiplier needed for surge pricing to achieve the same profit of an incentive scheme. ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, \underline{n = 1.1}, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



(b) Open system

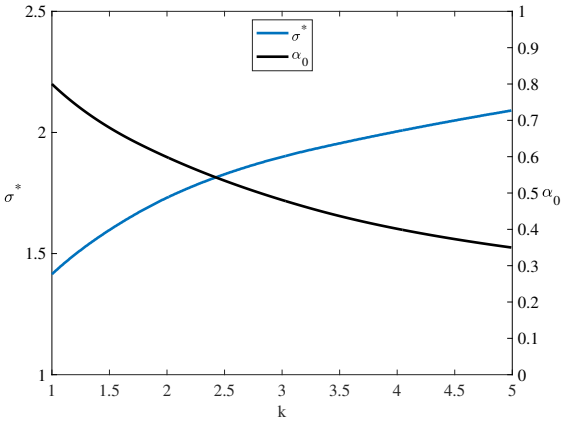
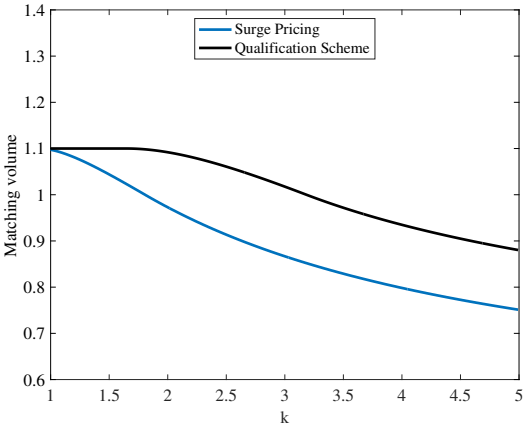
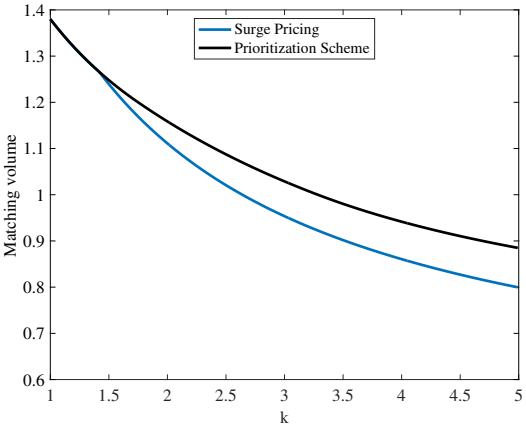


Figure SA.12 **The comparison of maximum matching volume between surge pricing and incentive schemes ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, \underline{n = 1.1}, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



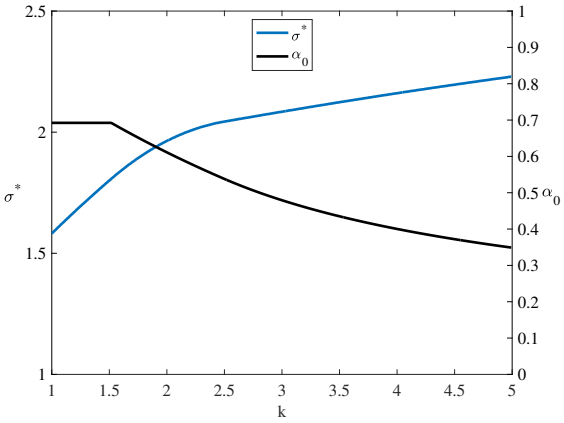
(b) Open system



(d) Robustness checks of alternative δ When the proportion of transaction price retained by the driver increases or decreases by 0.1, the results remain highly consistent with Figures 1 and 2, as shown in Figures SA.13 to SA.16.

Figure SA.13 **The minimum price multiplier needed for surge pricing to achieve the same profit of an incentive scheme. ($D_0 = 1, D_p = 2, p = 0.5, \underline{\delta = 0.9}, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



(b) Open system

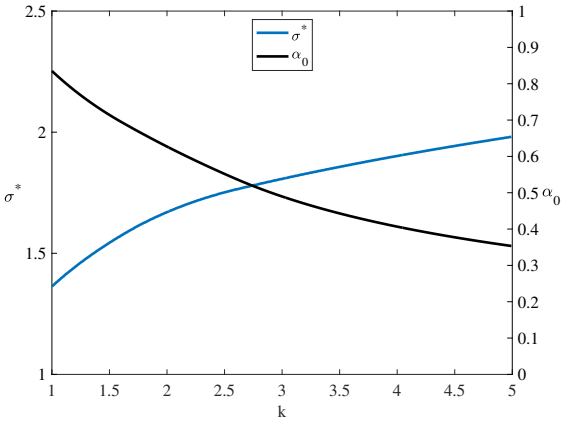
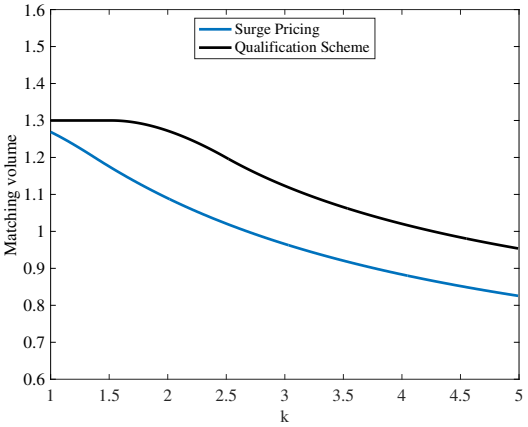


Figure SA.14 **The comparison of maximum matching volume between surge pricing and incentive schemes ($D_0 = 1, D_p = 2, p = 0.5, \underline{\delta = 0.9}, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



(b) Open system

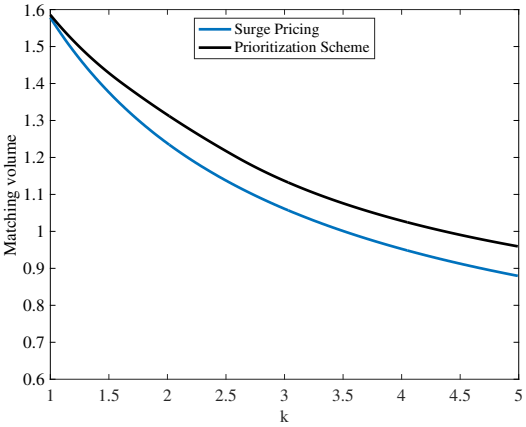
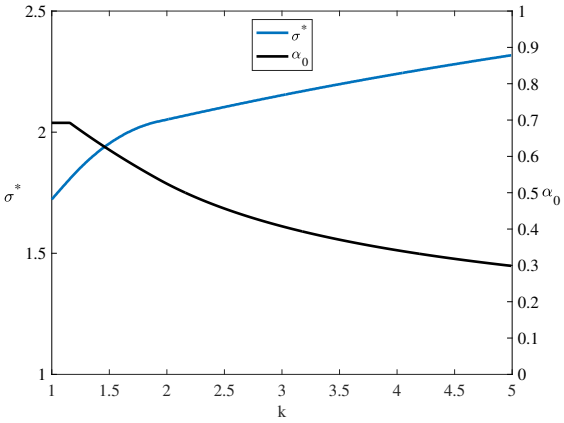


Figure SA.15 **The minimum price multiplier needed for surge pricing to achieve the same profit of an incentive scheme. ($D_0 = 1, D_p = 2, p = 0.5, \underline{\delta = 0.7}, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



(b) Open system

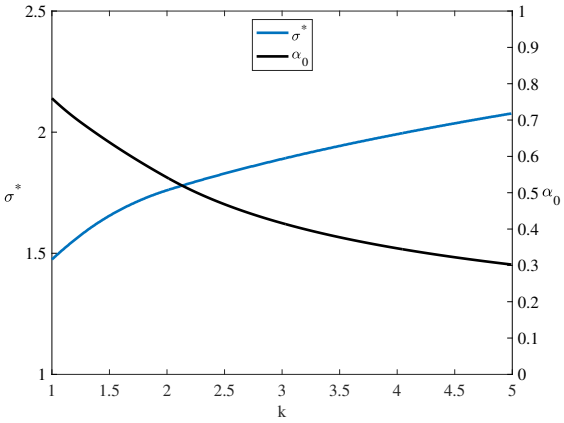
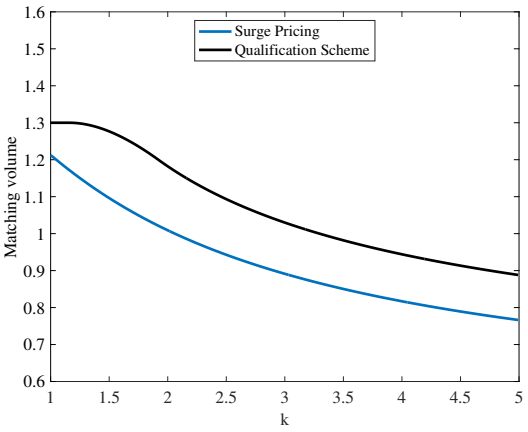
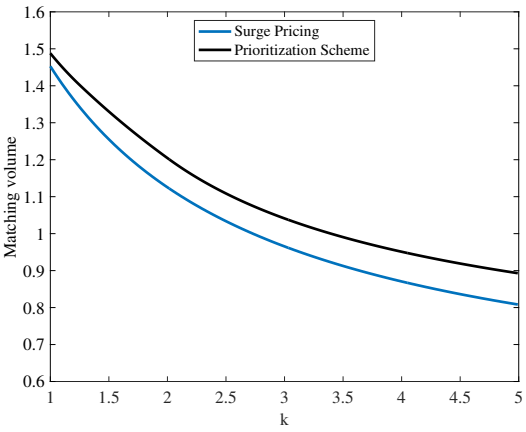


Figure SA.16 **The comparison of maximum matching volume between surge pricing and incentive schemes ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.7, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



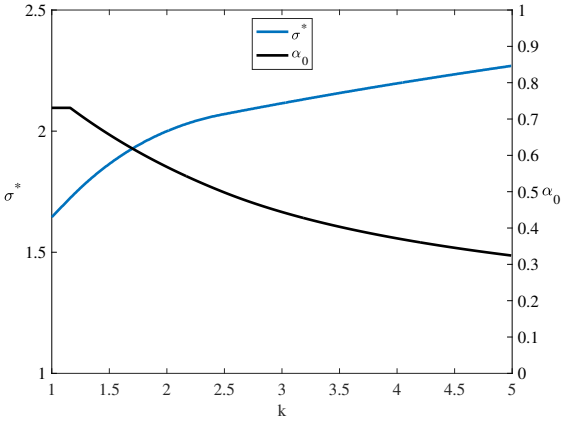
(b) Open system



(e) Robustness checks of alternative s When the randomness of off-peak demand changes, the results remain highly consistent with Figures 1 and 2, as shown in Figures SA.17 to SA.20.

Figure SA.17 **The minimum price multiplier needed for surge pricing to achieve the same profit of an incentive scheme. ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.15, 0.15]$)**

(a) Closed system



(b) Open system

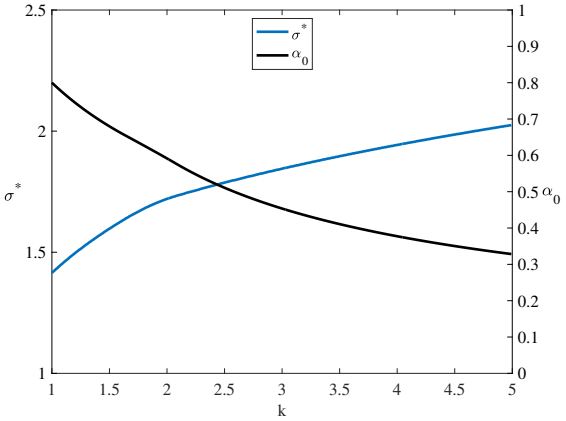
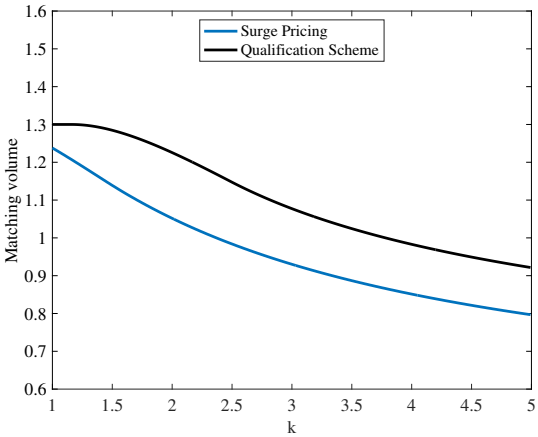


Figure SA.18 **The comparison of maximum matching volume between surge pricing and incentive schemes ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.15, 0.15]$)**

(a) Closed system



(b) Open system

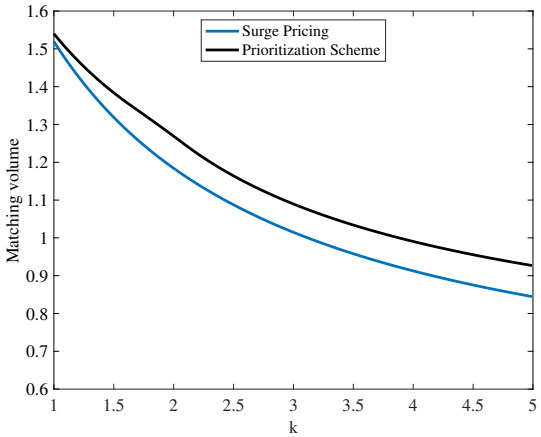
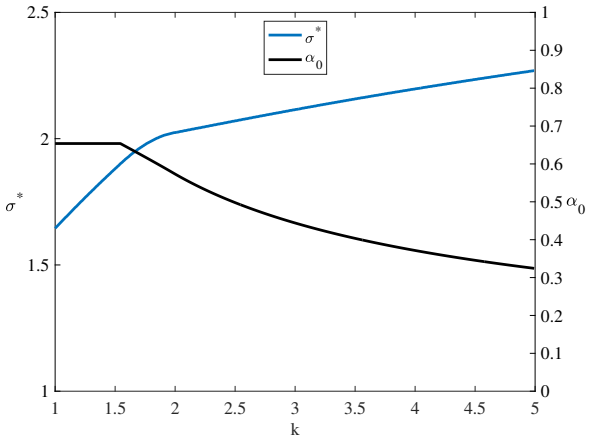


Figure SA.19 **The minimum price multiplier needed for surge pricing to achieve the same profit of an incentive scheme.** **(**$D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.05, 0.05]$**)**

(a) Same profit



(b) Same matching volume

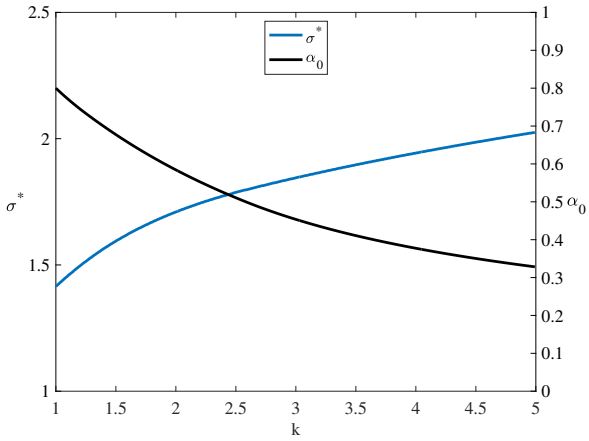
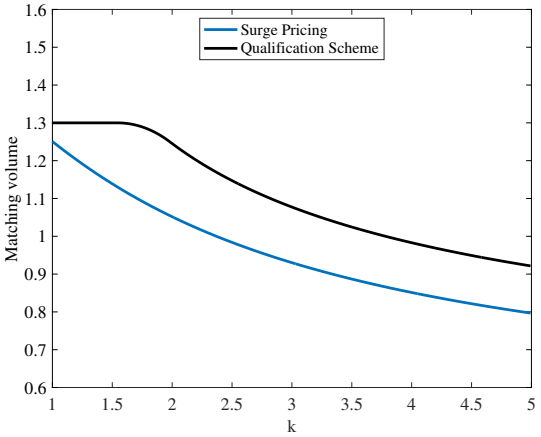
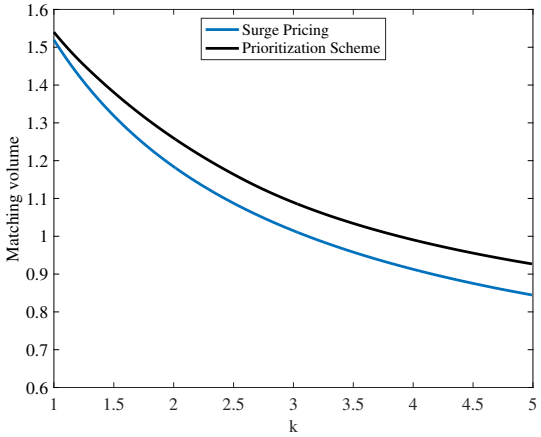


Figure SA.20 **The comparison of maximum matching volume between surge pricing and incentive schemes (**$D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.05, 0.05]$**)**

(a) Closed system



(b) Open system



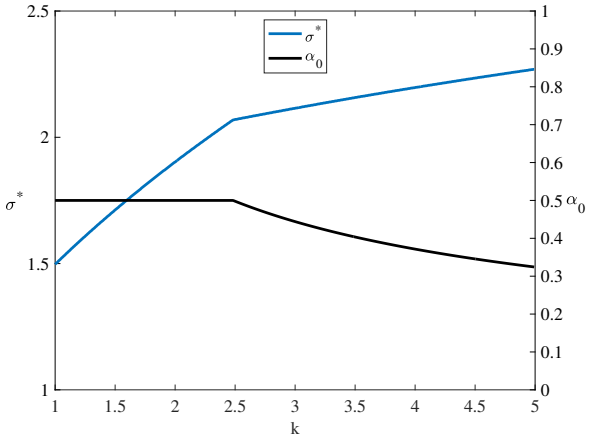
SA.D.3 Comparison between surge pricing and incentive schemes with upper bound of α_0

In this section, we examine cases where the service target has upper bounds of 0.5, 0.4 and 0.3, respectively. The results of comparison with surge pricing are robust as those in Section 6.1 as long as drivers' congestion costs are sufficiently high.

(a) Upper bound of α_0 equals 0.5

Figure SA.21 **The minimum price multiplier needed for surge pricing to achieve the performance of a qualification scheme with a service target upper bound of 0.5 in a close system. ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



(b) Open system

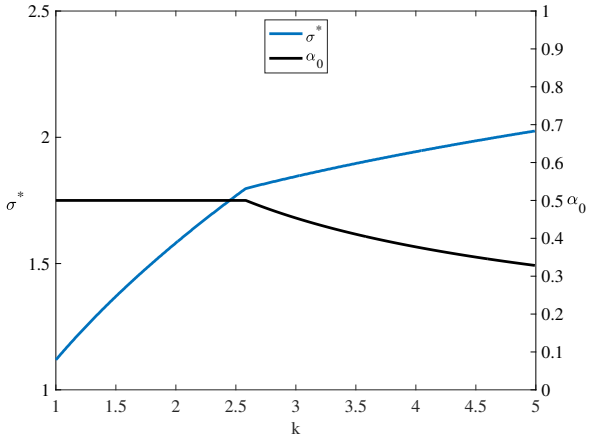
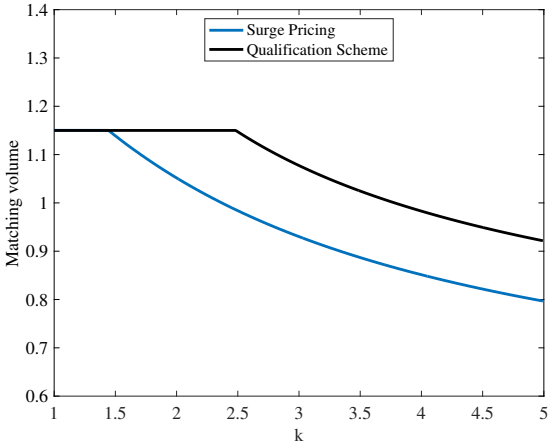
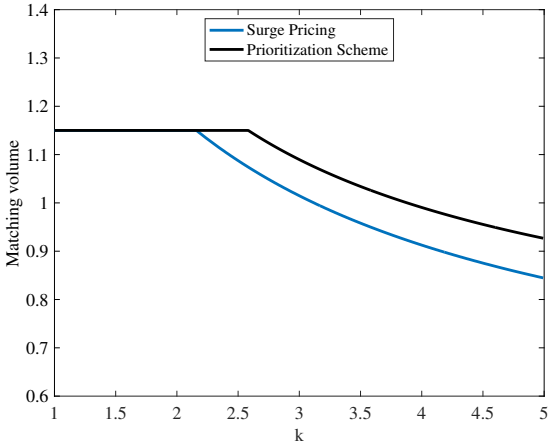


Figure SA.22 **The minimum price multiplier needed for surge pricing to achieve the performance of a prioritization scheme with a service target upper bound of 0.5 in an open system.($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



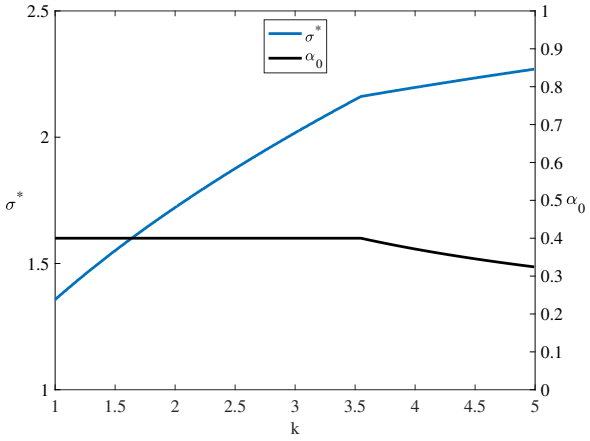
(b) Open system



(b) Upper bound of α_0 equals 0.4

Figure SA.23 **The minimum price multiplier needed for surge pricing to achieve the performance of a qualification scheme with a service target upper bound of 0.4 in a close system. ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



(b) Open system

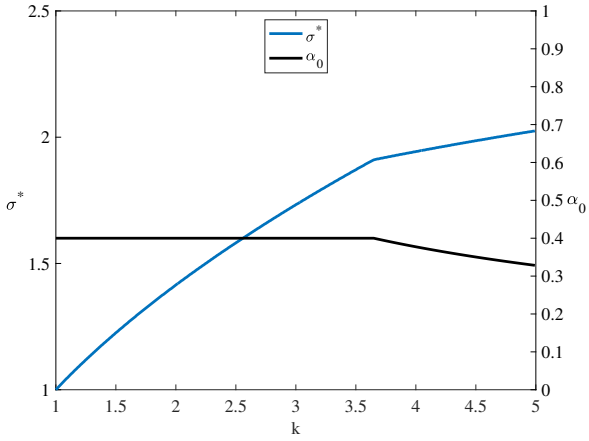
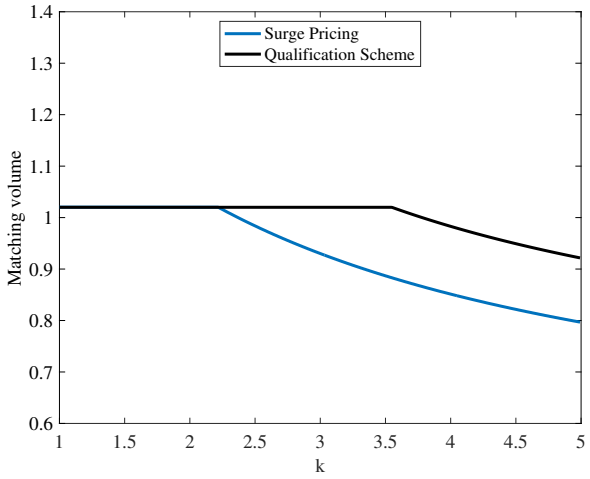
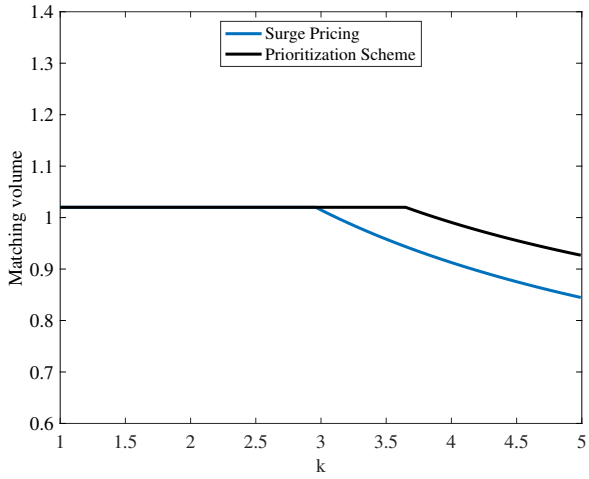


Figure SA.24 **The minimum price multiplier needed for surge pricing to achieve the performance of a prioritization scheme with a service target upper bound of 0.4 in an open system. ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)**

(a) Closed system



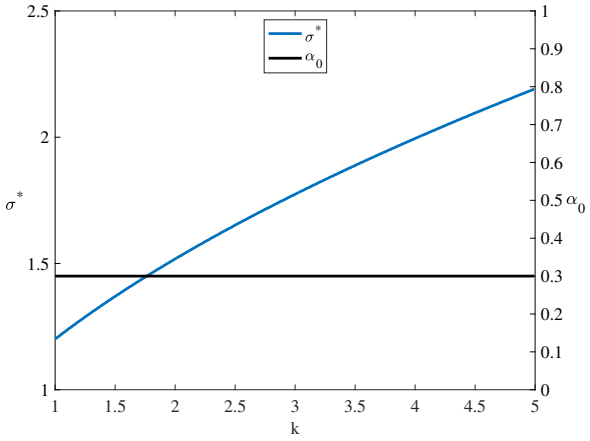
(b) Open system



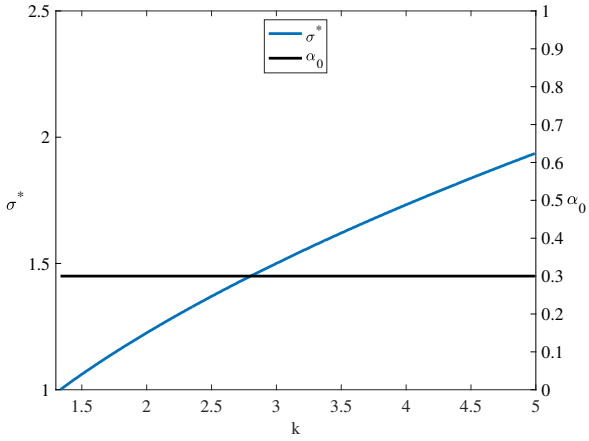
(c) Upper bound of α_0 equals 0.3

Figure SA.25 **The minimum price multiplier needed for surge pricing to achieve the performance of a qualification scheme with a service target upper bound of 0.3 in a close system. (**$D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$**)**

(a) Closed system



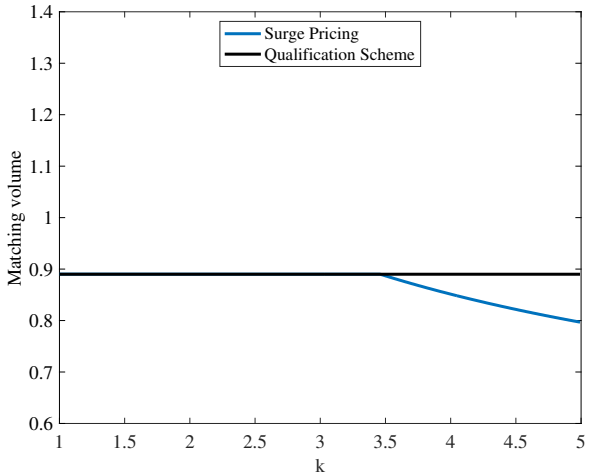
(b) Open system



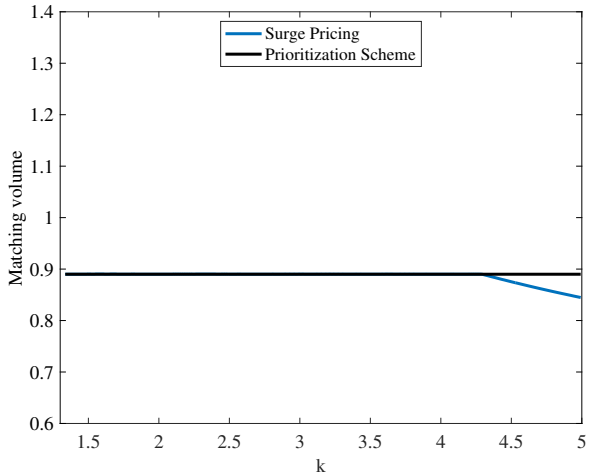
Note. In figure (b), k begins at the level where each full-time driver would choose to work 0.3 in peak period in the absence of an incentive scheme.

Figure SA.26 **The minimum price multiplier needed for surge pricing to achieve the performance of a prioritization scheme with a service target upper bound of 0.3 in an open system.(**$D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$**)**

(a) Closed system



(b) Open system

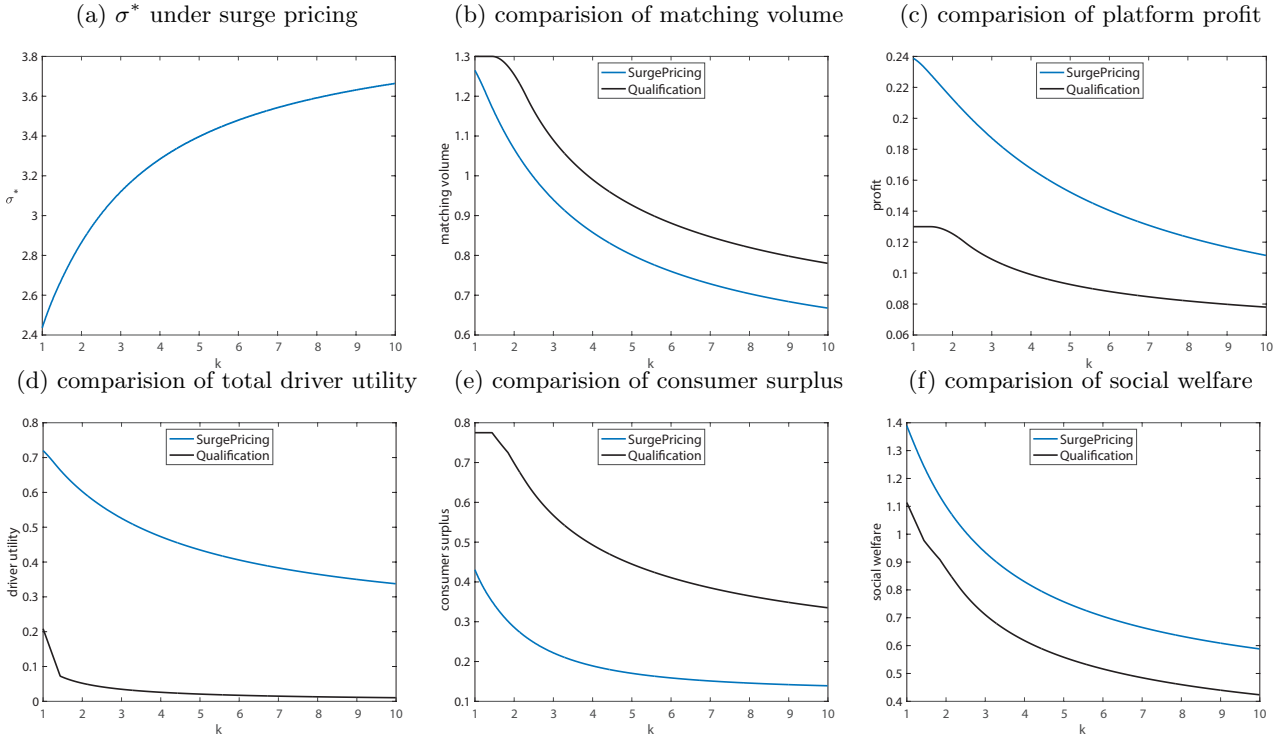


Note. In figure (b), k begins at the level where each full-time driver would choose to work 0.3 in peak period in the absence of an incentive scheme.

SA.D.4 Comparison between surge pricing and incentive schemes when the price multiplier is endogenous

In the following, we present comparisons between surge pricing and incentive schemes when the price multiplier is optimally determined. Due to the tractability challenges, we utilize numerical analysis. Figure SA.27 and Figure SA.28 display the results for a closed system and an open system, respectively.

Figure SA.27 The comparision between surge pricing and qualification scheme when the price multiplier is endogeneous in a closed system . ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)

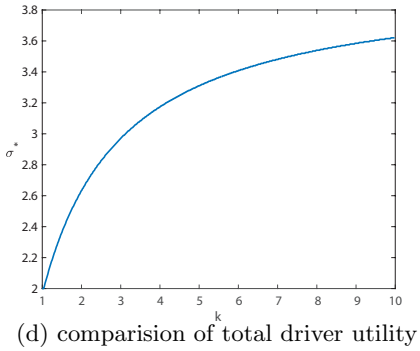


We observe similar patterns in both figures. To start with, the optimal price multiplier under surge pricing is high. It requires the platform to raise the base price by more than 140% in a closed system and over 100% in an open system. This could be challenging to implement in practice, especially in markets with price regulations. Regarding the comparison between surge pricing and incentive schemes, we find that surge pricing results in higher platform profit, driver utility, and social welfare, while the incentive scheme yields higher matching volume and consumer surplus.

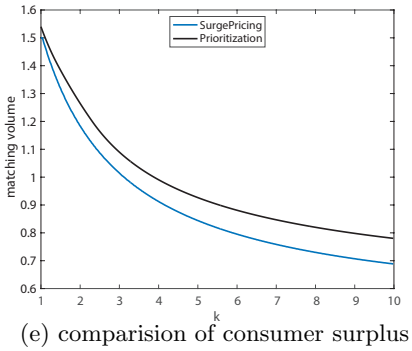
Specifically, as shown in Figure SA.27(b) and Figure SA.28(b), the incentive scheme achieves a higher matching volume. Although surge pricing allows the platform to increase prices in order to attract drivers during the peak period, this reduces the demand at the same time, which leads to a lower matching volume. Furthermore, as illustrated in Figure SA.27(c)(d) and Figure SA.28(c)(d), surge pricing generates higher platform profit and driver utility, which is primarily driven by the elevated peak-period prices. Next, as shown in Figure SA.27(e) and Figure SA.28(e), incentive scheme leads to higher consumer surplus, due to the lower prices and a larger matching volume. Finally, as indicated in Figure SA.27(f) and Figure SA.28(f), surge pricing results in higher social welfare under the optimal price multiplier. The

Figure SA.28 The comparision between surge pricing and prioritization scheme when the price multiplier is endogeneous in an open system . ($D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$)

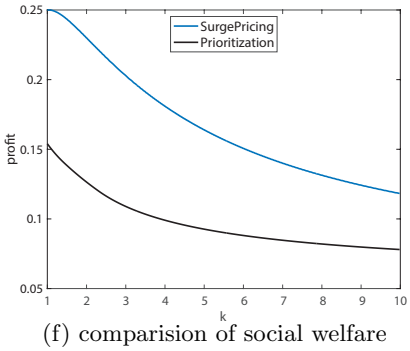
(a) σ^* under surge pricing



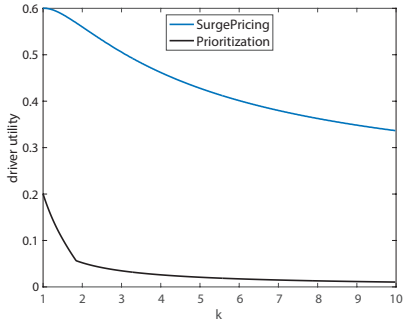
(b) comparision of matching volume



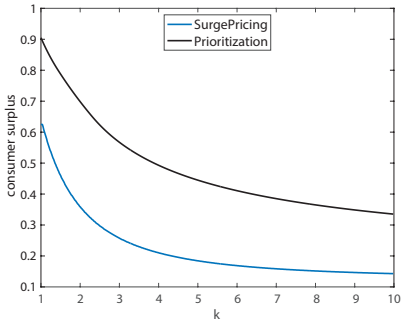
(c) comparision of platform profit



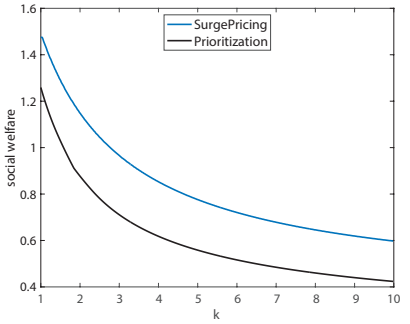
(d) comparision of total driver utility



(e) comparision of consumer surplus



(f) comparision of social welfare



benefits of surge pricing, in terms of increased platform profit and driver utility, outweigh the advantage of the incentive scheme in boosting consumer surplus.

The above results suggest that the optimal mechanism depends on the platform's priorities. The incentive scheme is more effective in scenarios where maximizing matching volume and consumer surplus is a priority. It increases the supply during the peak period without increasing the price, leading to better service availability and affordability for consumers. In contrast, surge pricing is advantageous in contexts where maximizing platform profit and driver earnings is critical or maximizing the social welfare is the goal. Finally, in markets where surge pricing is difficult to implement, the incentive schemes studied in the paper offer an alternative strategy to address the demand-supply imbalance.

SA.D.5 Robustness checks of heterogeneous congestion cost

In this section, we present ten sets of robustness checks for Figure A1 from the Online Appendix. Eight of these follow the parameter settings outlined in Table SA.2 of Supplementary Analysis D.1 (from (3) to (10)), with no changes made to D_p as it is not relevant in the context of heterogeneous drivers. In the remaining two robustness checks, we varied the congestion cost for low-type drivers (k_l). The figures consistently display similar patterns with Figure A1, confirming the robustness of our results regarding heterogeneous drivers.

(a) Robustness check of alternative p

Figure SA.29 Numerical Analysis with Heterogeneous Drivers. ($D_0 = 1, \underline{p = 0.7}, \delta = 0.8, n = 1.3, k_l = 1, \epsilon \sim U[-0.1, 0.1]$)

(a) The optimal composition of drivers (b) The optimal implementation of incentive schemes

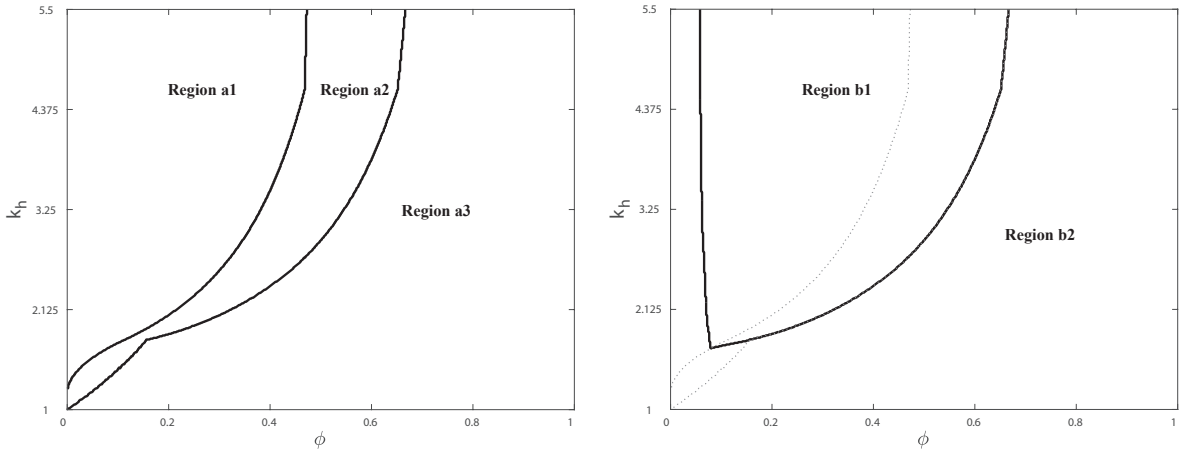
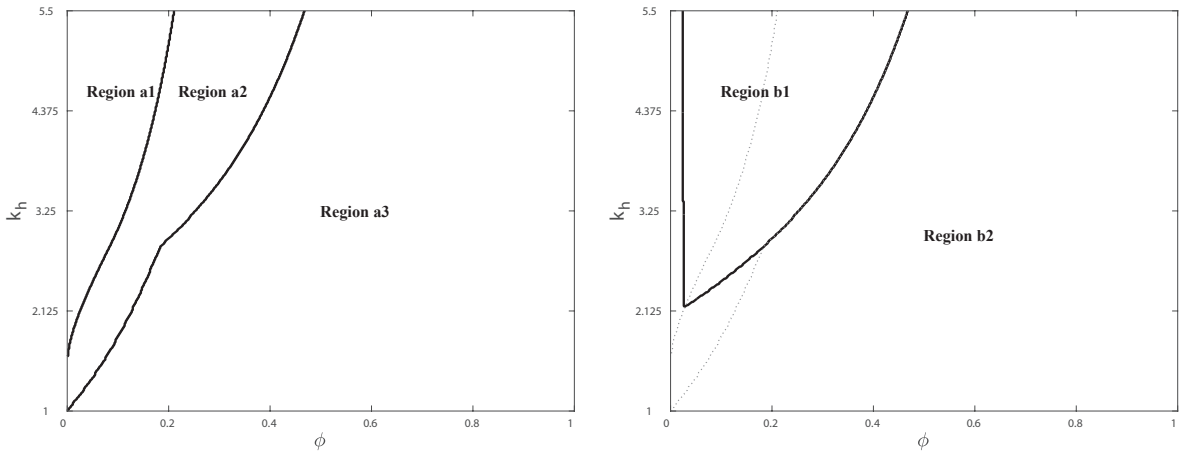


Figure SA.30 Numerical Analysis with Heterogeneous Drivers. ($D_0 = 1, \underline{p = 0.3}, \delta = 0.8, n = 1.3, k_l = 1, \epsilon \sim U[-0.1, 0.1]$)

(a) The optimal composition of drivers (b) The optimal implementation of incentive schemes



(b) Robustness check of alternative n

Figure SA.31 Numerical Analysis with Heterogeneous Drivers. ($D_0 = 1, p = 0.5, \delta = 0.8, \underline{n = 1.5}, k_l = 1, \epsilon \sim U[-0.1, 0.1]$)

(a) The optimal composition of drivers (b) The optimal implementation of incentive schemes

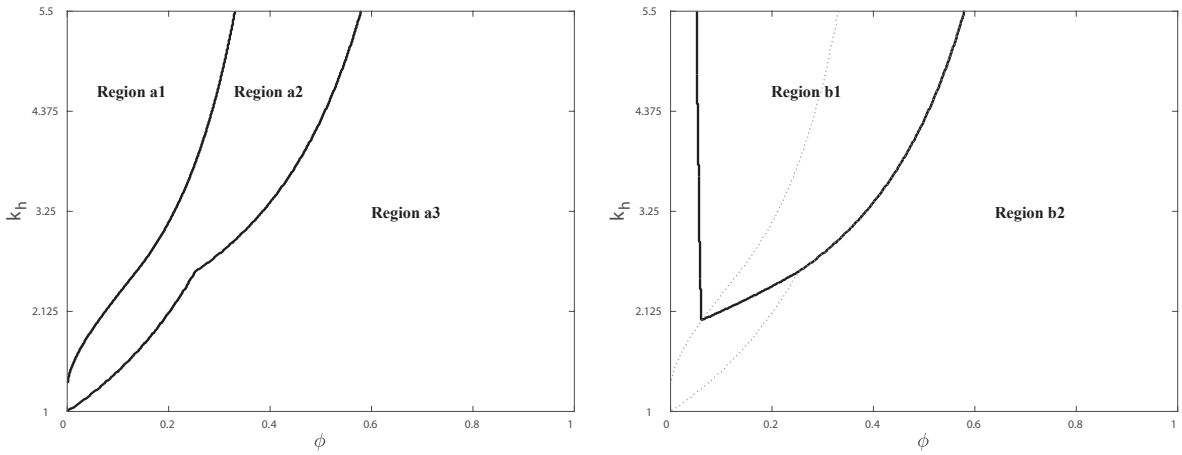
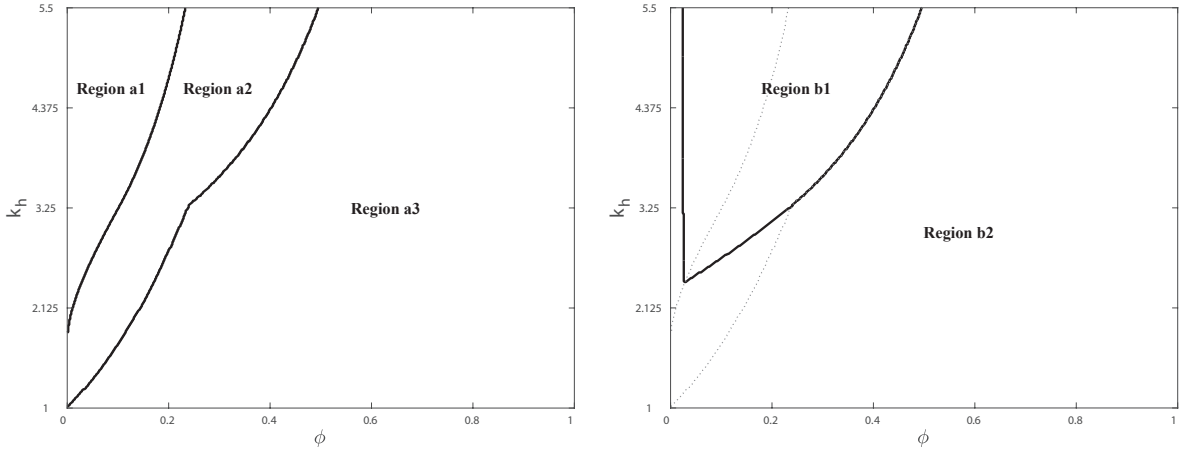


Figure SA.32 Numerical Analysis with Heterogeneous Drivers. ($D_0 = 1, p = 0.5, \delta = 0.8, \underline{n = 1.1}, k_l = 1, \epsilon \sim U[-0.1, 0.1]$)

(a) The optimal composition of drivers (b) The optimal implementation of incentive schemes



(c) Robustness check of alternative δ

Figure SA.33 Numerical Analysis with Heterogeneous Drivers. ($D_0 = 1, p = 0.5, \underline{\delta = 0.9}, n = 1.3, k_l = 1, \epsilon \sim U[-0.1, 0.1]$)

(a) The optimal composition of drivers (b) The optimal implementation of incentive schemes

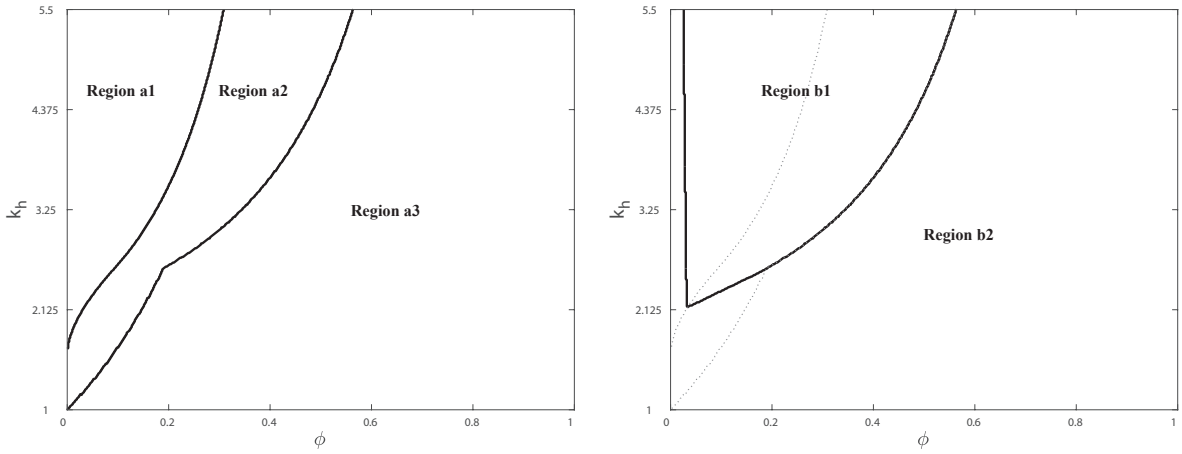
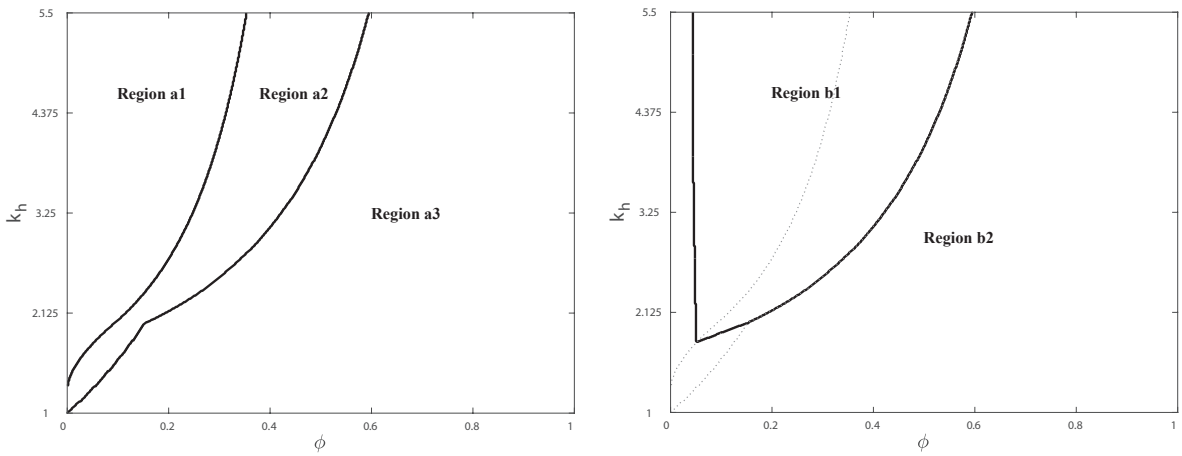


Figure SA.34 Numerical Analysis with Heterogeneous Drivers. ($D_0 = 1, p = 0.5, \underline{\delta = 0.7}, n = 1.3, k_l = 1, \epsilon \sim U[-0.1, 0.1]$)

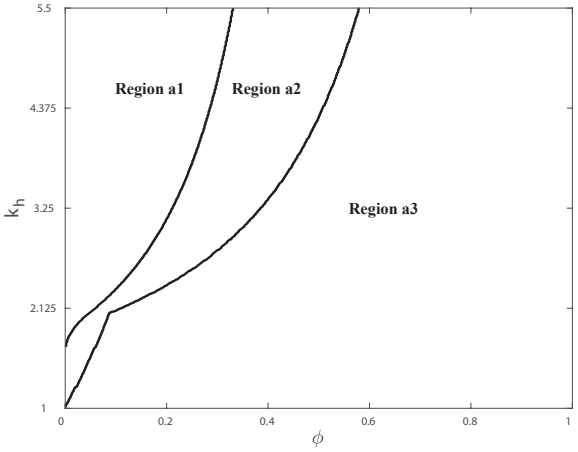
(a) The optimal composition of drivers (b) The optimal implementation of incentive schemes



(d) Robustness check of alternative s

Figure SA.35 Numerical Analysis with Heterogeneous Drivers. (**$D_0 = 1, p = 0.5, \delta = 0.8, n = 1.3, k_l = 1, \epsilon \sim U[-0.05, 0.05]$**)

(a) The optimal composition of drivers



(b) The optimal implementation of incentive schemes

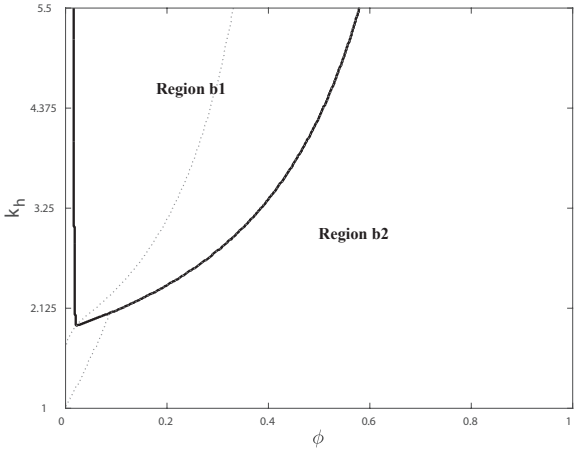
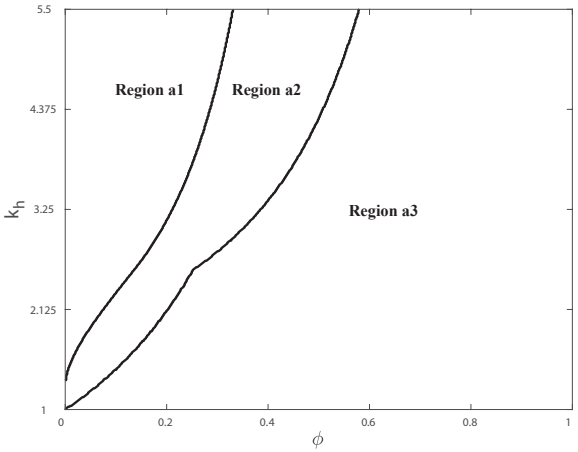
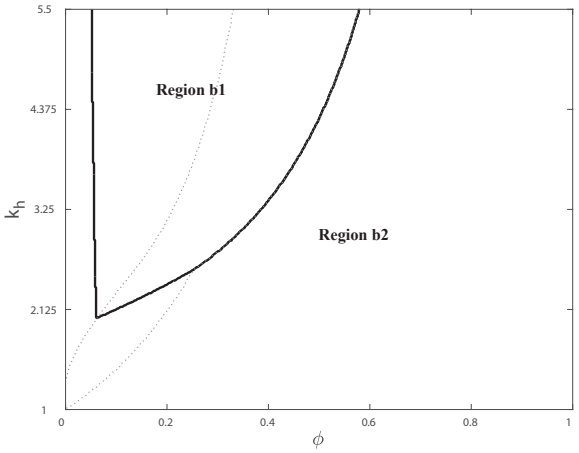


Figure SA.36 Numerical Analysis with Heterogeneous Drivers. (**$D_0 = 1, p = 0.5, \delta = 0.8, n = 1.3, k_l = 1, \epsilon \sim U[-0.15, 0.15]$**)

(a) The optimal composition of drivers



(b) The optimal implementation of incentive schemes



(e) Robustness check of alternative k_l

Figure SA.37 Numerical Analysis with Heterogeneous Drivers. ($D_0 = 1, p = 0.5, \delta = 0.8, n = 1.3, \underline{k_l = 0.7}, \epsilon \sim U[-0.1, 0.1]$)

(a) The optimal composition of drivers (b) The optimal implementation of incentive schemes

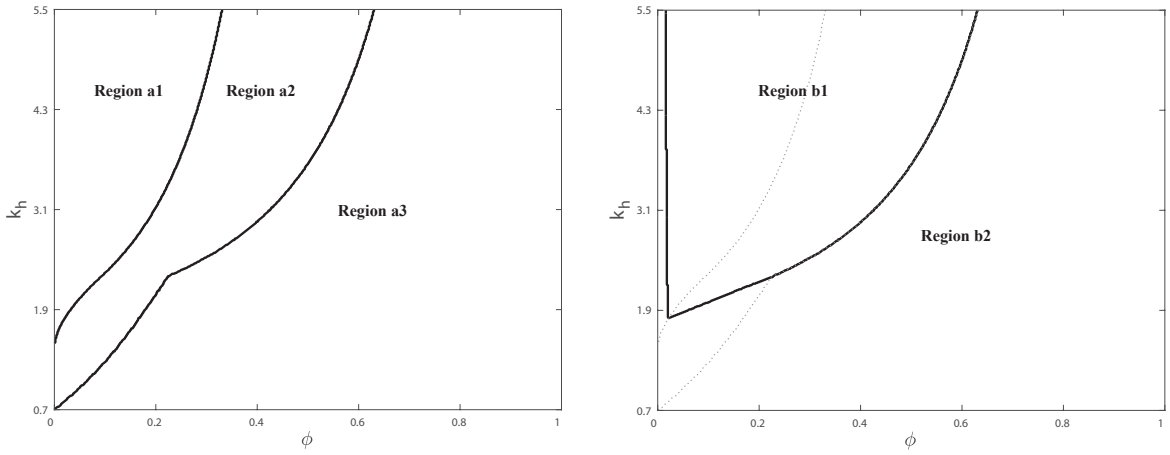


Figure SA.38 Numerical Analysis with Heterogeneous Drivers. ($D_0 = 1, p = 0.5, \delta = 0.8, n = 1.3, \underline{k_l = 1.3}, \epsilon \sim U[-0.1, 0.1]$)

(a) The optimal composition of drivers (b) The optimal implementation of incentive schemes

